# SILENT SCALE SHIFT: WHEN SHARED UPDATES BREAK EXISTING READOUTS

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## ABSTRACT

Shared encoders keep learning long after the readouts that consume them were fitted. Auxiliary-task updates can preserve what a task needs while shrinking the features, so a frozen readout loses accuracy although a refitted probe does not. We isolate this silent scale shift in matched continuations. On HAR, frozen-readout accuracy falls from $0.946$ to $0.453$ across ten seeds while the jointly trained head reaches $0.948$, and one global input-scale correction fitted without labels restores $0.938$. With a real reconstruction auxiliary, error falls to $0.765$ of detachment while frozen accuracy drops to $0.826$; a paired linear map restores $0.947$ without touching either model. Exact scale boundaries of an affine readout locate the first class change, and a repair hierarchy separates scale from the richer drift that follows stronger contraction. A signed chain-rule audit explains why task-gradient conflict and unsigned variance probes misjudge the direction of variance change. On SNR, ten confirmation seeds keep a $0.89$-point auxiliary gain and the old logits under a reference-anchored constraint, at $2.37\times$ CPU inference cost; folding the constraint into the encoder removes the cost and the gain. A coordinate map fitted after training keeps the gain at native cost and improves old-reader accuracy, yet changes $44.07\%$ of legacy decisions. Penalizing those changes during training halves them at native cost while keeping most of the gain. No single-encoder arm keeps the auxiliary gain and every seed inside a five-point disagreement budget. Accuracy and compatibility are separate acceptance criteria.

## 1 INTRODUCTION

An encoder and its readout are fitted together and deployed together. Further training changes the features, while the deployed classifier keeps the weights and input statistics it was shipped with. The current head adapts and stays accurate. What happens to the readout already in use?

We isolate the failure by cloning a healthy HAR model with its Adam state, retaining the fitted readout, and continuing matched branches on identical minibatches. Across ten seeds at variance-auxiliary dose 3, frozen-readout accuracy falls from $0.946$ to $0.453$ while the jointly trained head reaches $0.948$. One global input-scale correction, fitted without new labels, restores $0.938$ (Figure 1b). We call this component *silent scale shift*: the task information survives, and the existing readout no longer fits its input.

The failure does not need a synthetic auxiliary. HAR reconstruction lowers held-out MSE to $0.765$ of the dose-matched detached decoder while frozen-readout accuracy falls to $0.826$; a linear map fitted to paired old and new training features restores $0.947$, with the encoder and reconstruction head unchanged (Figure 1c). Auxiliary progress and readout failure coexist, and the scatter criterion meant to catch representation collapse never fires.

The rest of the paper asks how much of the damage is scale, which updates cause it, and whether useful auxiliary learning can coexist with an unchanged readout. Exact boundaries for an affine readout give the scale answer: a single scalar predicts where a frozen prediction changes class, and a fixed hierarchy of repairs, from mean and scale to per-feature and paired alignment, places the damage that remains. The diagnostic starts from the variance gradient. Task-gradient relations, the object of PCGrad (Yu et al., 2020) and CAGrad (Liu et al., 2021a), do not determine the sign of variance change; the update's projection onto the variance gradient does, and for Adam the applied increment matters as well.

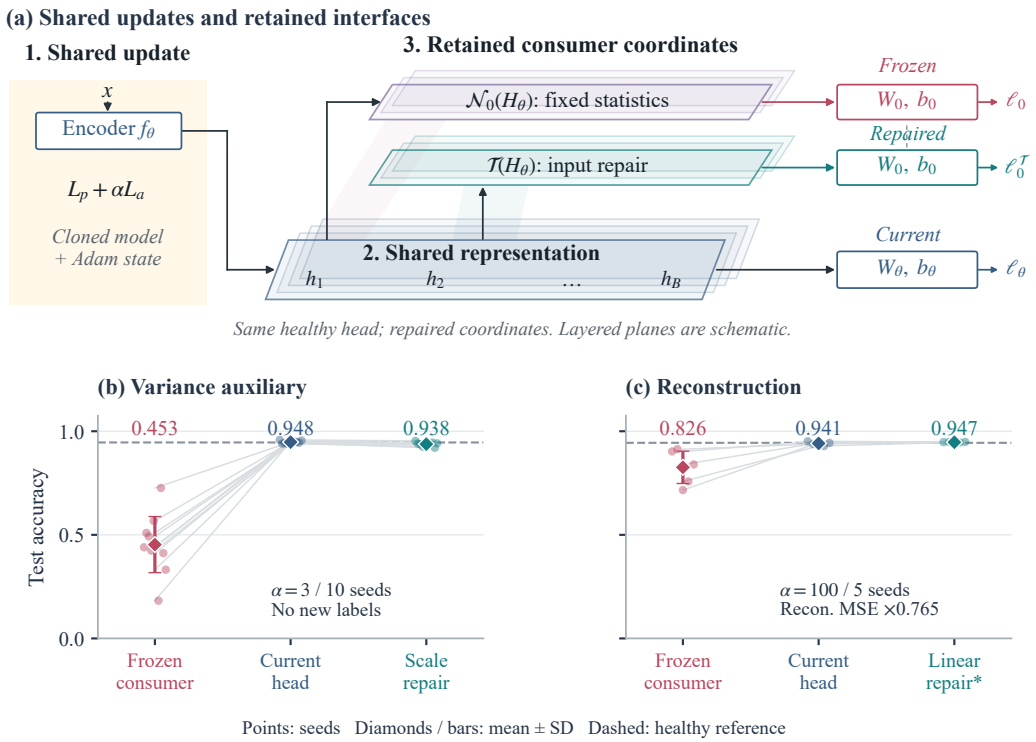



Figure 1: An updated representation can support the task while breaking an existing readout. (a) Three paths through shared features: the current head, the frozen readout, and the same readout after input repair. $\mathcal{N}_0$ is the healthy normalizer; $\mathcal{T}$ repairs normalization or alignment while retaining $(W_0, b_0)$. (b) HAR variance continuation: global scale correction restores much of the lost frozen accuracy. (c) HAR reconstruction: auxiliary MSE falls to $0.765$ of detachment; paired linear alignment restores readout accuracy. Repairs use training features without new labels and leave both learned models fixed. *Linear repair requires paired old/new anchors. Accuracies use the full test set; bars show population seed SD.

For coexistence, a reference-anchored readout-nullspace control (Shen et al., 2020; Farajtabar et al., 2020) preserves healthy logits while leaving $53$ of $64$ feature directions free. Ten new seeds confirm a $0.89$-point auxiliary gain at $2.37\times$ CPU inference cost. Freezing the nonlinear trunk removes the cost and loses the gain. A post-hoc single-encoder chart retains the gain and improves old-reader accuracy, yet changes $44.07\%$ of old predictions. A penalty on the frozen consumer's margin during training halves that change at native cost while keeping most of the gain. Each arm buys one property with another. Decodability, aggregate utility, and decision compatibility are separate measurements; a deployed consumer also carries cached decisions, calibrated thresholds, and human review. Compatibility with those is an acceptance criterion in its own right.

## 2 When Existing Readouts Fail

Let $h_\theta(x) \in \mathbb{R}^m$ be the shared features. An existing readout applies the fixed map $W_0 D_0^{-1}(h_\theta(x) - \mu_0) + b_0$, with healthy means $\mu_0$, positive diagonal scales $D_0$, weights $W_0$, and bias $b_0$. We call this complete input-to-output map the *consumer*. During continuation, the current task head learns with the encoder, while the old consumer stays fixed. A refitted probe tests what the new features can support; the old consumer tests whether its interface still works.

Pure contraction separates the two questions. For $H_c = cH$, an affine readout has logits $cWh + b$, which approach $b$ as $c \to 0$. A unique bias winner then makes the prediction constant, although every $c > 0$ preserves linear decodability.

**Proposition 1** (Two scale boundaries of a frozen affine readout)**.** *Write the logits after fixed input statistics as* $z_i(c) = z_0 + c\, s_i$*. Assume unique winners* $t = \arg\max_j z_0[j]$ *and* $p_i = \arg\max_j z_i(1)[j]$*. The bias-only class first loses as* $c$ *increases from zero at*

$$c_i^{\text{bias}} = \min_{j:\, s_i[j] > s_i[t]} \frac{z_0[t] - z_0[j]}{s_i[j] - s_i[t]}, \tag{1}$$

*with the minimum of an empty set taken as* $\infty$*. The first class change while shrinking from* $c = 1$ *instead occurs at*

$$c_i^{\text{first}} = \max_{\substack{j:\, s_i[p_i] > s_i[j] \\ 0 < \xi_{ij} < 1}} \xi_{ij}, \qquad \xi_{ij} = \frac{z_0[j] - z_0[p_i]}{s_i[p_i] - s_i[j]}, \tag{2}$$

*with zero denoting no positive-scale flip. The two boundaries differ in general. With nonzero feature variances and exact statistics, refitting the input statistics makes the readout invariant to every* $c > 0$*.*

*Proof.* Each pairwise margin is affine in $c$. Increasing $c$ from zero reaches the smallest positive crossing of the bias winner; decreasing $c$ from one reaches the largest crossing of the healthy winner in $(0, 1)$. Standardising $cH$ cancels $c$ from numerator and denominator. For example, logits $(1, 10c, -1 + 12c)$ give $c^{\text{bias}} = 0.1$ but $c^{\text{first}} = 0.5$. □

The imposed-scale sweep in Figure 10 isolates this mechanism from the drift that training adds (Appendix H.6).

The matched HAR trajectory in Figure 11a starts from one healthy checkpoint and keeps calibration examples out of encoder and readout fitting. At $\alpha = 3$ the current head recovers while the frozen consumer stays inaccurate. Repairs fitted to training features without labels separate mean from scale: mean correction reaches $0.813$, global scale correction $0.938$, and their combination $0.944$, close to per-feature recalibration ($0.947$). At $\alpha = 100$, per-feature repair reaches $0.829$ against $0.935$ for the refitted probe. Scale accounts for most of the moderate-dose failure; stronger or projected contraction needs richer alignment (Appendix H.8).

Repairs act upstream of the frozen weights, through a versioned encoder output or a consumer-input adapter. Mean and scale repairs need cached healthy statistics and unlabeled new features; paired alignment also needs old features for the same anchors. A supervised refit reuses existing labels and is the stronger-access baseline. We report NLL and 15-bin ECE with accuracy.

## 3 Auditing Shared Updates

### 3.1 Signed variance change

Let $H(\theta) \in \mathbb{R}^{B \times m}$ be the outputs of a shared encoder on a fixed diagnostic batch; a linear primary head and an auxiliary head consume $H$. Training minimizes $L_p + \alpha L_a$, where $L_p$ is the primary loss, $L_a$ the auxiliary loss, and $\alpha \geq 0$ the dose; multiple auxiliaries can be absorbed into their weighted sum. Write $G_p = \nabla_\theta L_p$ and $G_a = \nabla_\theta L_a$. Define

$$V(\theta) = \frac{1}{2B} \|C_B H(\theta)\|_F^2, \qquad C_B = I - \mathbf{1}\mathbf{1}^\top / B, \qquad v = \nabla_\theta V. \tag{3}$$

Here $2V$ is the covariance trace with divisor $B$: a rotation-invariant measure of variation across examples. Under gradient flow,

$$\dot{V} = r - \alpha c, \qquad r = -v^\top G_p, \qquad c = v^\top G_a. \tag{4}$$

We call $c > 0$ *auxiliary contraction*, and $r > 0$ *primary restoration*. When both are positive, $\alpha c / r > 1$ defines the instantaneous contraction-dominated regime. The pair $(r, c)$ also covers a primary update that reduces variance on its own. The identity keeps the encoder Jacobian, so the diagnostic needs no reduced dynamical model.

**Proposition 2** (Conflict information does not identify contraction)**.** *The task-gradient Gram matrix does not determine the sign of* $\dot{V}$*. In particular, there are two auxiliary updates with identical task-gradient norms and conflict, but opposite effects on variance under the same primary update.*

*Proof.* Take $G_p = e_1$, $G_a^{\pm} = \pm e_2$, and $v = -e_1 + \beta e_2$ for $\beta > 0$. Both pairs have identity Gram matrix and zero conflict, whereas $r = 1$ and $c^{\pm} = \pm\beta$. For $\alpha\beta > 1$, $\dot{V}^+ < 0 < \dot{V}^-$. These gradients are locally realizable by a smooth encoder and losses (Appendix A). □

Task-gradient relations leave the sign of variance change open. A combiner can preserve variance by lowering the auxiliary dose; preserving a particular auxiliary update requires its projection onto $v$.

The local balance also differs from eventual collapse. In the symmetric linear model $\dot{z} = (R - \alpha A)z$, $R \succ 0$, $A \succeq 0$, some mode contracts when $\alpha\lambda_{\max}(R^{-1/2}AR^{-1/2}) > 1$; all modes contract when $\alpha\lambda_{\min}(R^{-1/2}AR^{-1/2}) > 1$ (Theorem 3). A head can exploit the surviving modes. In HAR, a $4{,}600\times$ trace contraction leaves standardized-probe accuracy unchanged (Appendix H.4), so variance, effective rank, task decodability, and old-readout utility are measured separately.

### 3.2 What an unsigned probe leaves out

Let $h$ and $u$ stack the rows of $H$ and $C_B H$, respectively, and let $g_p = \nabla_h L_p$, $g_a = \nabla_h L_a$. A cheap feature-space probe uses the centered radial direction:

**Definition 1** (Unsigned radial toxicity proxy)**.** *For $u \neq 0$ and $\|g_p\| > 0$, define*

$$\rho_b = \frac{\|\Pi_u g_a\|}{\|g_p\|}, \qquad \Pi_u g = \frac{u^\top g}{\|u\|^2} u. \tag{5}$$

*The experimental estimator $\hat{\rho}$ instead averages regularized per-example ratios. The frozen decision uses a recorded value $q = \hat{\rho}_{t_0}$ and the nominal dose $\alpha_{\text{probe}} = 1/q$.*

The proxy drops the sign of the auxiliary direction and takes the whole primary-gradient norm as its reference. Expansion and contraction can then give the same $\rho_b$, and a large primary gradient need not restore centered variation. The two drivers also measure different objects: the controlled one probes the isotropic surrogate $\|C_B H\|_F^2/(Bm)$ while training with fixed projected-variance auxiliaries, and the real-head one probes the actual auxiliary.

### 3.3 A local validity boundary

The discrepancy can be exposed without assuming radial training dynamics. Let $J = \partial h/\partial\theta$, $K = JJ^\top$, $e = u/\|u\|$, and $n_p = \|g_p\| > 0$. At an audited checkpoint choose any $k > 0$ and write $K = k(I + E)$. Define

$$a = \frac{e^\top g_a}{n_p}, \quad d = -\frac{e^\top g_p}{n_p}, \quad \epsilon_a = \frac{e^\top E g_a}{n_p}, \quad \epsilon_p = -\frac{e^\top E g_p}{n_p}. \tag{6}$$

Here $a$ is signed radial loading relative to the primary norm, $d$ measures restoration alignment, and the $\epsilon$ terms measure the effect of encoder geometry. For the actual auxiliary, $|a| = \rho_b$. Products with $K$ can be evaluated by Jacobian–vector and vector–Jacobian products, without constructing the full matrix.

**Theorem 1** (Local validity boundary of a frozen probe)**.** *For any frozen probe value $q \geq 0$, the variance dynamics satisfy*

$$\frac{\dot{V}}{N} = 1 - \alpha q + \mathcal{E}, \qquad N = \frac{k\|u\| n_p}{B} > 0, \qquad \mathcal{E} = (d-1) + \epsilon_p - \alpha(a - q + \epsilon_a). \tag{7}$$

*Consequently, with*

$$\Delta = |d-1| + |\epsilon_p| + \alpha\big(|a - q| + |\epsilon_a|\big), \qquad |1 - \alpha q| > \Delta, \tag{8}$$

*the probe and the exact variance derivative agree in sign. If $1 - \alpha q > \Delta$, variance increases; if $1 - \alpha q < -\Delta$, it decreases. The remaining region requires the signed audit.*

*Proof.* The chain rule gives $\dot{V} = -u^\top K(g_p + \alpha g_a)/B$. Substitute $K = k(I + E)$, divide by $N$, and add and subtract $1 - \alpha q$. The triangle inequality bounds $|\mathcal{E}|$ by $\Delta$. □

The margin is sufficient and checkpoint-local. Computing it requires the exact flux, so it serves as an offline explanation of the cheap probe. On $405$ HAR checkpoints, the frozen certificate fires on $2$; the signed balance $\psi = d - \alpha a$ with a combined residual certifies $10$ at the same scale. Per-checkpoint scale optimization raises the counts to $15$ and $154$ (Table 2). Every certified sign is correct. The corrected balance alone agrees in sign on $58\%$ of checkpoints, leaving substantial encoder-geometry mismatch.

In the identity metric ($E = 0$), $\dot{V}/N = d - \alpha a$. If $a = q$ and $0 < d < \alpha q < 1$, the nominal gate stays closed while variance contracts: the primary norm overstates restoration. If $a = -q$, the same unsigned gate can open while the auxiliary expands variance. Drift after calibration contributes through $a - q$. These examples identify concrete sources of a wrong sign.

The mismatch term also separates into actionable components. If $a_0$ is signed loading at calibration for the actual auxiliary, then

$$|a_t - q| \le |a_t - a_0| + |a_0 - |a_0|| + \big||a_0| - q\big|. \tag{9}$$

The terms separate temporal drift, discarded sign, and surrogate or aggregation mismatch. A radial approximation is informative when the auxiliary contracts, the primary gradient restores centered variation, the encoder metric is compatible with the approximation, and calibration transfers to the audited checkpoint.

A scalar reduction, $\dot{\sigma} = -\alpha\kappa\sigma + b(\sigma)$, yields a saddle-node threshold $\alpha_c = \lambda_\star/\kappa$ under the explicit shape and endpoint assumptions of Theorem 2. Equating $1/q$ with this threshold additionally requires calibration to $\kappa/\lambda_\star$. Empirically, we compare $1/q$ with the half-height crossing $\alpha_{1/2}$ and report the operational collapse threshold separately.

# 4 Evidence Across Tasks and Protocols

## 4.1 Protocols and observables

The controlled benchmark spans HAR (Anguita et al., 2012), RadioML (O'Shea et al., 2016), battery health (Saha & Goebel, 2007), and NYUv2. A three-layer MLP with a fixed projected-variance auxiliary isolates contraction; Transformer, CNN, real-head, CelebA, and financial studies test its scope. Appendices F–H give the protocols and optimizer grids. Error bars show population seed SD unless stated otherwise.

The operational label is relative scatter $s = \mathcal{S}_h/\mathcal{S}_{\text{single}} < 0.3$, using the same-seed single-task reference and

$$\mathcal{S}_h = \mathbb{E}_b\Big[\mathrm{std}_j\Big(h_j^{(b)} - \mathbb{E}_{b'} h_j^{(b')}\Big)\Big]. \tag{10}$$

The proxy is coordinate-dependent and blind to the all-ones direction. "Collapse" therefore refers to the declared $0.3$ criterion; information is measured separately. The log-dose interpolated crossing at $s = 0.5$, $\alpha_{1/2}$, summarizes the dose curve.

## 4.2 Calibration is task- and width-dependent

Across sixteen domain–width cells, the ratio of predicted dose to observed half-height crossing is $0.95$–$3.01$ for battery and 30–$116$ for RadioML; HAR reverses bias with width (Figure 7a). All cells train projected auxiliaries and probe the isotropic surrogate.

## 4.3 Calibration and drift change the decision

On HAR at $m = 64$, detached warm-up moves $1/\overline{\hat{\rho}}$ from $1.34$ at $W = 1$ to $0.00907$ at $W = 4$, against a half-height crossing of $0.00141$. At dose $0.03$, $W = 1, 2$ leave all gates closed and all ten seeds collapse; $W \ge 4$ opens every gate and prevents collapse. At dose $100$, all tested warm-ups protect scatter (Figure 7b,d). An extreme dose can therefore hide a calibration error.

Recomputing the probe during training can produce false alarms. HAR ratios grow even at zero auxiliary dose, and running gates activate in every seed of ten configurations with final scatter $0.50$–$1.04$. Battery has one activation among 35 seeds ending above half-height (Appendix H.2). A shrinking

Table 1: Independent confirmation of the validation-selected SNR auxiliary: 20% of fitting primary labels, dose 1, ten new seeds. D/J denote detachment/joint training; G denotes the reference-anchored readout-nullspace constraint. Own accuracy uses the current task head. The same healthy consumer is retained for frozen accuracy. Means with population seed SD; bounds use a separate fixed 40-consumer bank and a five-point relative-error budget. The guarded representation requires both encoder evaluations at inference.

| Arm | own accuracy | frozen accuracy | frozen NLL | ECE | certified |
|---|---|---|---|---|---|
| D | $0.275 \pm 0.010$ | $0.259 \pm 0.016$ | 2.176 | 0.153 | 0/10 |
| J | $0.285 \pm 0.009$ | $0.258 \pm 0.019$ | 2.135 | 0.155 | 0/10 |
| GD | $0.279 \pm 0.010$ | $0.259 \pm 0.013$ | 2.031 | 0.048 | 10/10 |
| GJ | $0.288 \pm 0.009$ | $0.259 \pm 0.013$ | 2.031 | 0.048 | 10/10 |

primary-gradient denominator can inflate the ratio. On $405$ HAR checkpoints, the unsigned predictor agrees with the exact variance derivative on $51\%$, with median restoration alignment $0.065$.

### 4.4 Real auxiliaries and stronger encoders

Four real auxiliary heads separate loading from contraction (Figure 5a). Mean per-example task-gradient cosines are at most $0.06$ in magnitude, while SNR radial loading is $88\%$. Yet the endpoint signed radial cosine is negative on SNR, and $c_t = v^\top G_a < 0$ at $74/75$ historical epoch-end checkpoints, so sampled flow contraction does not explain its collapse. Flow and Adam signs also disagree on $33/75$ sampled paired updates; we record the applied increment and a primary-only Adam counterfactual (Appendix H.7).

Protocol changes alter the observed failure. Historical SNR scatter onset lies between doses 3 and 10, whereas I/Q reconstruction stays above the criterion through $10000$. Under healthy-checkpoint continuation, I/Q at $3000$ gives frozen accuracy $0.301$ versus $0.304$ detached, with trace $1.061$; SNR at $30$ gives $0.243$ versus $0.304$, with scatter $0.870$. Resetting auxiliary weights or their Adam state also fails to reproduce the historical severe collapse (Appendix H.11). Warm-up, split, architecture, and horizon define the operating regime.

A waveform CNN on the same I/Q split reaches $0.424$ healthy-readout accuracy. At reconstruction dose $3000$, frozen accuracy is $0.415$ versus $0.418$ detached, while the current head reaches $0.437$. This stronger encoder leaves the consumer largely intact on the fixed real-auxiliary grid (Appendix H.10).

## 5 Repairing and Preserving Compatibility

Real reconstruction tests repair while the auxiliary makes progress (Table 34, Appendix H.9). At HAR dose $100$, frozen accuracy falls from $0.941$ under detachment to $0.826$, while the current head reaches $0.941$. Test reconstruction MSE falls to $0.765$ of detachment, with $0/5$ scatter collapses. Per-feature recalibration reaches $0.863$; paired linear alignment reaches $0.947$ (Figure 1c). Repair changes only the consumer input, retaining the auxiliary improvement.

Let $A = W_0 \operatorname{diag}(1/s_0)$ and let $P_\perp$ project onto $\ker A$. The reference-anchored control $H_0 + (H_\theta - H_0)P_\perp$ preserves all healthy logits while emitting one shared $64$-vector (Proposition 10, Figure 2). Validation selects SNR dose $1$ from six fixed candidates using $20\%$ of fitting primary labels. Ten new seeds confirm an ordinary auxiliary gain J–D of $0.97$ percentage points, paired $95\%$ interval $[0.63, 1.31]$, and a guarded gain GJ–GD of $0.89$, $[0.60, 1.18]$ (Table 1). The guarded model clears the declared $0.5$-point noninferiority margin against J, and its predictions, NLL, and ECE match the reference to a maximum logit error of $4.6 \times 10^{-14}$; its ten endpoints also satisfy a separate five-point disagreement budget. Frozen accuracy barely moves, while ECE rises from $0.048$ to $0.155$. Both encoders remain at inference, at a measured $2.37\times$ CPU cost (Appendix H.12).

With a frozen nonlinear trunk, the last-affine constraint folds into one encoder but loses the SNR gain; freezing accounts for $-3.53$ points, while the constraint adds $-0.003$ (Appendix H.13). A nonlinear-side grid also fails its benefit screen. Fitting and folding a chart *after* unrestricted learning instead retains a $1.16$-point gain at $0.990\times$ native CPU cost. Old-reader accuracy improves by $1.36$

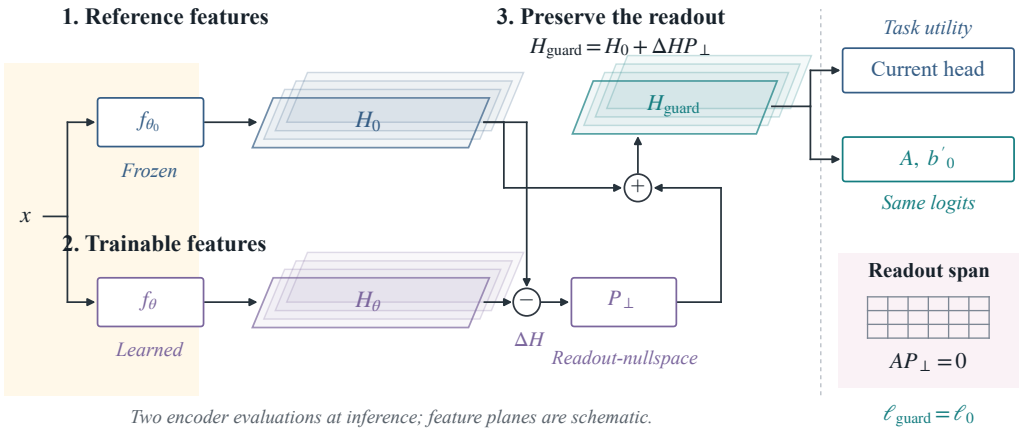



Figure 2: A fixed readout leaves a nullspace for learning. Projecting $\Delta H = H_\theta - H_0$ into $\ker A$ and adding $H_0$ gives one shared 64-vector with unchanged reference logits in real arithmetic. The normalized head is $A = W_0 D_0^{-1}$ and its effective bias is $b'_0 = b_0 - A\mu_0$. The inset shows the annihilation condition; planes are schematic. Both encoders are evaluated at inference.

points and its NLL and ECE fall, yet $44.07\%$ of its predictions change, and none of ten repaired consumers meets the fixed five-point disagreement budget (Appendix H.15). Every aggregate check can pass while the deployed interface changes. A training-time hinge on the frozen consumer's margin keeps one encoder and a $0.72$-point gain, against $0.97$ for joint training, while halving decision changes to $29.2\%$; it still misses the budget, and on a second corpus whose auxiliary barely moves the readout the same rule meets it with eight of ten certificates (Appendix H.16). On a pretrained vision backbone, three epochs of primary-only continuation raise the frozen readout's accuracy by $8.5$ points and still change $42.1\%$ of its decisions (Appendix H.17).

Detachment blocks auxiliary gradients to the encoder and provides the reference when the auxiliary brings no primary-task value. The spectral toxicity filter (STF) instead removes a chosen component of the auxiliary channel. Its idealized backward form is

$$F_w[g_a] = g_a - w\Pi_u g_a, \qquad w \in [0, 1], \tag{11}$$

The implemented selective arms transform the auxiliary input, $\widetilde{H} = H - wC_B HP$, using a frozen projector $P$ or a centered radial direction. The transformation changes the forward input as well as its gradient, while the auxiliary head continues training (Appendix E).

The calibrated arm freezes the warm-up probe and opens the gate when $\alpha\hat{\rho} \geq 1$. The rank variant then selects

$$\text{rank}_{\text{sel}} = \min\{\text{rank} \leq \text{rank}_{\max} : \alpha\hat{\rho}_{\text{res}}[\text{rank}] < 1\}, \tag{12}$$

from nested candidate subspaces. If none passes, it uses the cap and records the unresolved residual. The radial arm is a separate intervention. Theorem 5 gives the minimum deletion rank for a specified PSD operator; using an empirical PCA basis adds a basis-identification question. Gate calibration and subspace choice therefore need separate evaluation.

The low-label SNR dose scan shows both the value and limits of filtering (Figure 8a,b). With $20\%$ primary labels, joint training is healthy at low dose but collapses in every seed from $\alpha = 30$. Calibrated STF avoids collapse at every sampled dose and exceeds detachment by $0.5$–$1.0$ percentage points over $\alpha \in [15, 100]$; the advantage disappears at extreme doses. This five-seed scan uses a different label budget from the fully labeled study in Appendix H.20.

The fully labeled SNR comparison shows why clearing the scatter threshold is insufficient (Figure 8c). Detachment reaches $0.307$ accuracy with $0/10$ collapses. The radial arm reaches $0.199$ with $1/10$, and the rank-3 arm $0.211$ with $0/10$. Basis choice also changes how much auxiliary learning is removed: in HAR reconstruction, an auxiliary-gradient basis uses $30.75$ directions versus PCA's $62.0$ among the same four gate-open seeds, at similar accuracy (Table 7).

**Algorithm 1** Auditing representation survival and probe validity

**Require:** Diagnostic batch, healthy reference, checkpoint $\theta$, operating losses and dose $\alpha$, recorded calibration value $q$.

1: Report task metrics, covariance trace $2V$, effective rank, standardized linear probe, and the declared operational scatter criterion.
2: Audit healthy-readout accuracy/NLL/ECE, mean/global/per-feature repairs, and any paired alignment; compare with a refitted probe.
3: With unlabeled monitor inputs and healthy reference predictions, bound consumer disagreement; select a certified candidate or abstain.
4: Report gradient-flow fluxes $(r, c)$ from (4). For Adam, audit $v^\top s_\theta$ and the actual post-step $V$ change; specify any auxiliary counterfactual.
5: Record $1-\alpha q$, the signed balance $\psi = d-\alpha a$, and the corresponding residuals. Form $q^\star = a/d$ only when $d \neq 0$.
6: **if** $|1 - \alpha q| > \Delta$ **then**
7: Mark the gradient-flow sign locally certified for this batch.
8: **else**
9: Use the signed audit; report restoration misalignment, geometry, and probe mismatch separately.
10: **end if**
11: Compare joint training, detachment, and any selective intervention using both survival and task utility at matched doses and seeds.

At the controlled high doses, calibrated filtering reduces collapse from $10/10$ to $0/10$ on HAR, battery, and RadioML, and from $5/5$ to $0/5$ on CelebA Smiling (Liu et al., 2015). With a Transformer at RadioML dose $3000$, joint training collapses in $10/10$ seeds at accuracy $0.156$; filtering gives $0/10$ collapses and relative scatter $0.970$. Reliable gate activation at these extreme doses coexists with the calibration errors in Figure 7.

## 6 Evaluating a Shared Representation

The evidence suggests three questions for evaluation: how did the features change, what task information remains decodable, and does the existing consumer still work? Algorithm 1 combines these checks with the signed update audit.

When $\Delta$ exceeds the probe margin, the unsigned decision remains unresolved. The signed audit attributes its current error to restoration alignment, encoder geometry, discarded sign, or calibration mismatch. Repeating the same unsigned ratio cannot separate these effects.

The intervention comparison also needs both compatibility and task utility. A rank-3 SNR filter can pass the scatter criterion while underperforming detachment; the low-label radial filter can lose its advantage at extreme doses. An open gate alone says little about the quality of the selected subspace.

For a present checkpoint, unlabeled inputs can support a conservative consumer decision. On a fixed corpus, the increase in classification error is bounded by disagreement with healthy predictions. We sample $1{,}024$ monitor inputs, select the first candidate in a fixed repair hierarchy whose disagreement bound meets the budget, and otherwise abstain. Across $105$ joint endpoints, a $10$-percentage-point budget certifies $49$, with $0$ observed violations and $56$ abstentions; a healthy-reference scatter heuristic makes $11$ false-safe declarations (Appendix H.18).

Each check has a scope. The signed flux describes the current gradient-flow direction; Adam needs its applied increment. The STF gate is an empirical calibration rule. The disagreement bound covers a fixed checkpoint family and corpus remainder, given healthy predictions; changed populations and later updates fall outside it. The CNN and single-encoder controls bound how far the protection extends.

## 7 Related Work

Multi-task learning (Caruana, 1997) and task grouping (Standley et al., 2020) ask when shared training helps. PCGrad and CAGrad address gradient conflict (Yu et al., 2020; Liu et al., 2021a); MGDA, Nash-MTL, and FAMO optimize Pareto, bargaining, or progress-balancing criteria (Sener & Koltun, 2018; Navon et al., 2022; Liu et al., 2023). Other methods adapt task-gradient geometry or loss weights (Senushkin et al., 2023; Liu et al., 2021b; Chen et al., 2018; Kendall et al., 2018; Wang et al., 2021b; Limarenko et al., 2026; Zhou et al., 2025). We add the variance gradient to this picture: task-gradient relations alone leave the sign of variance change unidentified.

Backward-compatible learning studies model-version compatibility (Shen et al., 2020; Hu et al., 2022b). LDC (Gomez-Villa et al., 2024) shows retained discrimination behind drifting class prototypes and learns label-agnostic feature maps. Two arms we measure are these controls: recomputing per-feature input statistics is the AdaBN operation, and the paired linear map is a label-free old-to-new feature map of the LDC family. Our question is narrower: fixed affine consumers under auxiliary updates, with scale, useful learning, and preserved decisions separated. CORAL (Sun et al., 2016), AdaBN (Li et al., 2016), and Tent (Wang et al., 2021a) supply alignment and unlabeled-adaptation baselines; our anchor-fitted repairs test model drift without adapting on test inputs. Shift and model-selection assumptions matter (Zhao et al., 2023). Calibration (Guo et al., 2017; Ovadia et al., 2019), probes (Alain & Bengio, 2016), and unlabeled performance estimation (Garg et al., 2022) complement the fixed-corpus disagreement check.

Self-supervised learning studies collapse through stop-gradient mechanisms (Grill et al., 2020; Chen & He, 2021; Caron et al., 2021), variance and covariance regularization (Zbontar et al., 2021; Bardes et al., 2022; Ermolov et al., 2021), and spectral dynamics (Jing et al., 2022; Dong et al., 2021; Kim et al., 2025). RankMe evaluates representations by effective rank (Garrido et al., 2023). Our observable is variation across examples, distinct from terminal classifier geometry in neural collapse (Papyan et al., 2020; Wang et al., 2026). Input repair shows how reduced variation can coexist with retained decodability and lost readout compatibility.

Auxiliary-gradient projection (Dery et al., 2021; Zhu et al., 2025), auxiliary selection (Navon et al., 2021; Jiang et al., 2023; Shin et al., 2024), weight-space nullspace adaptation (Pham et al., 2026), and mergeable frozen-backbone updates (Hu et al., 2022a) provide the context for our intervention controls. Activity recognition (Anguita et al., 2012), modulation recognition (O'Shea et al., 2016), and battery health (Saha & Goebel, 2007; Severson et al., 2019; Attia et al., 2020; Pradhan & Chakraborty, 2022) provide the test domains.

## 8 Conclusion

An accurate current head can conceal a broken existing readout. Matched continuation and fixed input repairs expose the gap; exact affine boundaries account for its scale component; signed audits identify what gradient alignment and unsigned probes miss. The SNR confirmation shows that auxiliary benefit and unchanged old logits can coexist at the price of a second encoder, and the single-encoder controls show what preserving both costs. Shared-representation evaluation should measure what new features can predict and whether existing readouts can still use them.

## AI USE STATEMENT

Generative AI tools assisted with drafting, language editing, code, literature retrieval, proof development, and figure design. The new diagnostic identities were checked with executable numerical examples; assumptions are stated explicitly with each claim and in Appendix D. Figures are drawn from recorded experimental results. The authors remain responsible for verifying the final text, claims, code, and artifacts.

## ETHICS STATEMENT

We use existing datasets: UCI-HAR, RadioML2016.10A, the NASA PCoE and MATR battery archives, public A-share market data, NYUv2, and CelebA. We collected no new human-subject data. The HAR identity auxiliary is evaluated as a training-head probe, not a held-out privacy attack. Battery prediction is safety-relevant; Section 4.3 describes the observed limitations, and deployment requires task-specific validation.

## REPRODUCIBILITY STATEMENT

The artifact contains theoretical assumptions, code, per-seed records, the fixed development/confirmation plan, checkpoint and consumer replays, scale-boundary checks, actual-step audits, and the corrected battery protocol. Historical battery checkpoint selection and readout calibration are exploratory. The anonymized repository supplies released code and earlier records: `https://github.com/code-2026l/RepFix`.

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# A Proofs and Audit Details for the Diagnostic Boundary

Let $\theta \in \mathbb{R}^2$, $v_0 = -e_1 + \beta e_2$, and $z(\theta) = v_0 + \theta$. Choose a two-example encoder $H(\theta) = (z^\top, -z^\top)^\top$. Then $V = \|z\|^2/2$ and $\nabla_\theta V = v_0$ at $\theta = 0$. The losses $L_p = \|z - (v_0 - e_1)\|^2/2$ and $L_a^\pm = \|z - (v_0 \mp e_2)\|^2/2$ are nonnegative smooth functions of the representation. Their gradients at zero are $e_1$ and $\pm e_2$. Thus Proposition 2 concerns realizable encoder/loss configurations, not arbitrary incompatible gradients. Both task pairs have the same Gram matrix, while the signed variance derivatives differ.

Vectorization of the centered batch gives $u = (C_B \otimes I_m)h$ and $\nabla_\theta V = J^\top u/B$. Hence $\dot{V} = -u^\top J J^\top (g_p + \alpha g_a)/B$. For any fixed positive scale $k$, write $JJ^\top = k(I + E)$. This is a decomposition, not an assumption of isotropy. With the definitions in (6), the normalized derivative is $d + \epsilon_p - \alpha(a + \epsilon_a)$, which proves (7). Taking absolute values term by term proves (8). The exact discrepancy $\mathcal{E}$ can be smaller than $\Delta$ because signed terms cancel. Failure of the sufficient margin condition therefore leaves the probe undecided; it does not prove the probe wrong. Choosing $k$ must be reported, since it changes the decomposition and the tightness of this conservative envelope, although it cannot change $\dot{V}$.

Record the batch, checkpoint, losses, auxiliary dose, probe value $q$, and scale $k$. Compute $g_p$, $g_a$ for the actual operating losses and $u$ from the unfiltered representation. Evaluate $Kg$ by first computing $J^\top g$, then a Jacobian–vector product with $J$; a dense $K$ is unnecessary. This yields $a, d, \epsilon_a, \epsilon_p$, the exact signed flux, and the validity margin. When $u = 0$ or $\|g_p\| = 0$, the normalized decomposition is undefined and the unnormalized identity (4) remains the appropriate diagnostic. This audit costs more than a frozen feature-space probe: at $B = 256$ and $m = 64$ on CPU the exact flux costs $0.30\,\mathrm{s}$ while the exact $k = \mathrm{tr}(K)/(Bm)$ certificate costs $20.7\,\mathrm{s}$, a factor of $69.6$ (`results/CERTCOST/`). The trace enters only through the canonical scale: an envelope at any other $k$, including the per-checkpoint minimizer of Table 2, needs the exact flux and the head-level gradients but no trace of $K$. The certificate is therefore an offline audit, not a cheap substitute for measuring the flux; the cheap object in the deployed rule is the probe $q$. Its purpose is to identify why the cheap decision can or cannot be transported to an operating checkpoint. The archive contains cheap proxies and intervention outcomes for $405$ HAR checkpoints. The frozen predictor agrees in sign with the exact variance derivative on $51\%$. Its sufficient certificate fires on $2$ checkpoints at the trace-normalized scale $k = \mathrm{tr}(K)/(Bm)$ and $15$ under per-checkpoint scale optimization, with no incorrect firing.

The signed balance $\psi = d - \alpha a$ replaces the unsigned loading and primary-norm reference with directional quantities. When $d \neq 0$, it equals $d(1 - \alpha q^\star)$ with $q^\star = a/d$. Sign agreement rises to $58\%$ overall and $61\%$ at positive dose. Computing the balance needs the same two head-level backward passes as the original probe; certifying it additionally needs the exact flux.

Table 2 varies the predictor, residual envelope, and scale separately. At canonical $k$, replacing the frozen predictor with $\psi$ raises coverage from 2 to 3 with separate absolute geometry terms and to $10$ when their signed cancellation is retained. Per-checkpoint scale optimization gives $15$, $121$, and $154$. All rows have zero false firings, as required by the sufficient condition. For comparison, the frozen predictor with directly measured $|\mathcal{E}|$ certifies $128$ at canonical $k$. Every arm uses the common normalization $N$, so these counts compare the full retrospective audits on the same prediction scale.

The optimized combined certificate covers $57/226$ contracting checkpoints ($25\%$) and $97/179$ expanding checkpoints ($54\%$). Restricting to $|d| \geq 0.1$ leaves $168$ checkpoints, of which $47/97$ contracting and $52/71$ expanding points certify. The archive also contains $51$ points with $|d| \leq 0.01$. Using $\psi$ directly avoids division by weak restoration and remains defined at $d = 0$. These subsets describe the same audit archive.

Figure 3 ranks observed sign agreement using margins that already consume the exact derivative. AUC is $0.60$ for the canonical corrected combined arm, $0.80$ after scale optimization, and $0.63$ for the canonical frozen arm. Scale optimization can itself improve these retrospective scores: whenever $\psi$ and $r - \alpha c$ have matching nonzero signs, choosing $k = (r - \alpha c)/(\|u\| n_p \psi/B) > 0$ makes the combined residual zero. Unrestricted optimization would therefore certify all $236$ matching-sign points; the recorded $154$ uses a finite $241$-point grid. Script `certificate_matched_audit.py` regenerates the arms from stored primitives.

Table 2: Matched retrospective certificate ablations on 405 HAR checkpoints. "Contracting" counts certified points among the 226 negative exact derivatives. All rows have zero false firings. Canonical $k = \mathrm{tr}(K)/(Bm)$ is fixed; optimized $k$ is selected per checkpoint on the same 241-point grid. Triangle and combined geometry residuals are $|\epsilon_p| + \alpha|\epsilon_a|$ and $|\epsilon_p - \alpha\epsilon_a|$. Every envelope uses the exact flux.

| Predictor | scale | residual | fires | contracting |
|---|---|---|---|---|
| frozen $1 - \alpha q$ | canonical | original triangle | 2 | 2 |
| frozen $1 - \alpha q$ | optimized | original triangle | 15 | 10 |
| corrected $\psi$ | canonical | geometry triangle | 3 | 3 |
| corrected $\psi$ | optimized | geometry triangle | 121 | 41 |
| corrected $\psi$ | canonical | geometry combined | 10 | 10 |
| corrected $\psi$ | optimized | geometry combined | 154 | 57 |
| frozen $1 - \alpha q$ | canonical | exact $\lvert\mathcal{E}\rvert$ | 128 | 115 |

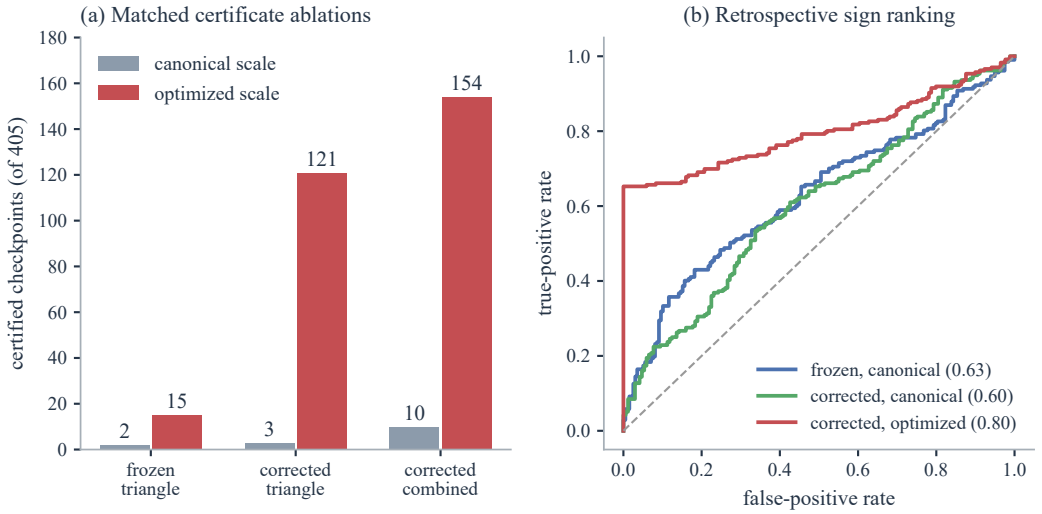



Figure 3: Certificate coverage depends on the predictor, residual, and scale. (a) Frozen and signed predictors under canonical and optimized scales, with separate and combined geometry residuals. (b) ROC curves for observed sign agreement. Every margin consumes the exact flux, so the curves describe retrospective audit discrimination.

The operational gate and the rank rule rely on different assumptions. For the gate, Theorem 1 requires the actual operating losses and encoder Jacobian, a recorded calibration value $q$, and discrepancy smaller than the sign margin at the audited checkpoint. For the minimum-rank claim, Theorem 5 requires a specified PSD operator; an empirical PCA basis needs separate identification. Deployed STFCAL evaluates neither certificate online. Table 3 records the scope of each arm.

Table 3: Assumptions behind the gate, rank rule, and deployed filters. The sign certificate and minimum-rank theorem concern different mathematical objects; the empirical arms retain their own calibration and basis assumptions.

| Arm | Claim | Required assumption |
|---|---|---|
| gate $\alpha\hat{\rho} \geq 1$ | local sign of $\dot{V}$ (Theorem 1) | frozen $q$; actual operating losses and encoder Jacobian; drift inside $\Delta$ |
| rank rule $\mathrm{rank}_{\mathrm{sel}}$ | minimum deletion rank (Theorem 5) | specified PSD operator; nested candidate subspaces |
| PCA basis | empirical surrogate | identified separately; not assumed to be the operator's eigenbasis |
| deployed STFCAL | local heuristic | no online certificate; calibration audit only |

**Corollary 1** (Transport to a later audited checkpoint)**.** *Let $q_0$ be the probe frozen at calibration, and let $q_t$ and $\Delta_t$ be the live probe and the envelope of Theorem 1 at a later checkpoint. If*

$$|1 - \alpha q_0| > \Delta_t + \alpha|q_t - q_0|,$$

*then the frozen decision has the same sign as the exact variance derivative at that checkpoint.*

*Proof.* $|(1 - \alpha q_0) - (1 - \alpha q_t)| = \alpha|q_t - q_0|$. If the displayed inequality holds, then $|1 - \alpha q_t| \geq |1 - \alpha q_0| - \alpha|q_t - q_0| > \Delta_t$, so Theorem 1 applies at $t$ with the live terms. □

**Corollary 2** (Signed directional balance)**.** *Let $\psi = d - \alpha a$. With the decomposition of Theorem 1,*

$$\frac{\dot{V}}{N} = \psi + \epsilon_p - \alpha\epsilon_a.$$

*If $|\psi| > |\epsilon_p - \alpha\epsilon_a|$ at a specified $k > 0$, then $\dot{V}$ has the sign of $\psi$, including when $d = 0$. When $d \neq 0$, define $q^\star = a/d$ and $\Delta^\star = |\epsilon_p - \alpha\epsilon_a|/|d|$; the same condition is $|1 - \alpha q^\star| > \Delta^\star$, with predicted sign* $\operatorname{sign}[d(1 - \alpha q^\star)]$*. The $|d - 1|$ discrepancy is absent.*

*Proof.* Regroup (7) as $d - \alpha a + \epsilon_p - \alpha\epsilon_a$. A perturbation smaller in magnitude than $\psi$ cannot change its sign. Substitution of $q^\star$ gives the ratio form when $d \neq 0$. □

The quantities $a \propto \sum_i (h_i - \bar{h})^\top g_{a,i}$ and $d \propto -\sum_i (h_i - \bar{h})^\top g_{p,i}$ are the centered auxiliary and primary radial derivatives. Their signed ratio $q^\star$ allows cancellation across examples. The archived probe instead averages per-example magnitudes divided by per-example primary norms. Replacing that primary reference by $d\, n_p$ removes the $|d - 1|$ discrepancy.

Corollary 1 requires the frozen margin to exceed the current envelope plus probe drift. It certifies $2/405$ checkpoints at the canonical trace-normalized scale and $14/405$ under scale optimization, with no incorrect firing. The corresponding frozen certificate without the added drift term fires on $15/405$. Outside the certified checkpoints, calibration remains an empirical decision rule.

Figure 4 shows how closely the margin approaches the transport envelope. The $14$ certified checkpoints in (a) lie just above it; the margin/envelope ratio in (b) has median $0.86$ and upper quartile $0.996$. Across the $27$ width–dose cells in (c), sign agreement is $0.56$ above median restoration alignment and $0.47$ below it. Signed mismatch $|a - q|$ varies largely independently of $d$ (log-scale correlation $-0.05$). Panel (d) compares $|\mathcal{E}|$ and the minimum envelope at the same per-checkpoint scale, with median ratio $0.885$. Matching the normalization is essential to interpreting envelope tightness. These archive associations describe where the offline audit is informative; they leave prospective performance to a separate evaluation.

For $K = I$, the local condition is $d - \alpha a < 0$. Take $d = 0.1$, $a = q = 0.2$, and $\alpha = 1$: the nominal gate is closed while the exact derivative is negative. This is a false negative caused by treating the primary norm as radial restoration. Conversely, with $d = 1$, $a = -q = -2$, and $\alpha = 1$, the gate is open but the exact derivative is positive. For a fixed $q$, the theorem applies at each time with time-indexed components. If $\sup_{t \in I} \Delta_t < |1 - \alpha q|$, its sign conclusion holds throughout the interval $I$. Obtaining this uniform bound is an additional task; a single frozen observation does not supply it. Even a negative variance derivative on a finite interval does not by itself imply convergence to zero.

Let $s_\theta$ be the actual proposed parameter increment, including any optimizer preconditioning, momentum, and weight decay. If $\nabla V$ is $L$-Lipschitz on the segment from $\theta$ to $\theta + s_\theta$, then

$$\left|V(\theta + s_\theta) - V(\theta) - v^\top s_\theta\right| \leq \tfrac{L}{2}\|s_\theta\|^2. \tag{13}$$

This follows by integrating the gradient along the segment. A first-order margin exceeding this remainder certifies the sign of the finite-step change. The gradient-flow decomposition applies directly to plain gradient updates; the actual-step audit is the corresponding observable for Adam. Separating primary and auxiliary contributions through a nonlinear optimizer requires specifying the counterfactual update and its state. A feature-gradient norm ratio is not an Adam-step certificate.

The controlled driver computes the probe in `rho_probe` within `cross_domain_mtl.py`:

$$\hat{\rho} = \frac{1}{B}\sum_{i=1}^{B} \frac{\|\Pi_{u_i} \widetilde{g}_{a,i}\|}{\|g_{p,i}\| + 10^{-8}}, \qquad \widetilde{g}_a = \nabla_H \frac{\|C_B H\|_F^2}{Bm}. \tag{14}$$

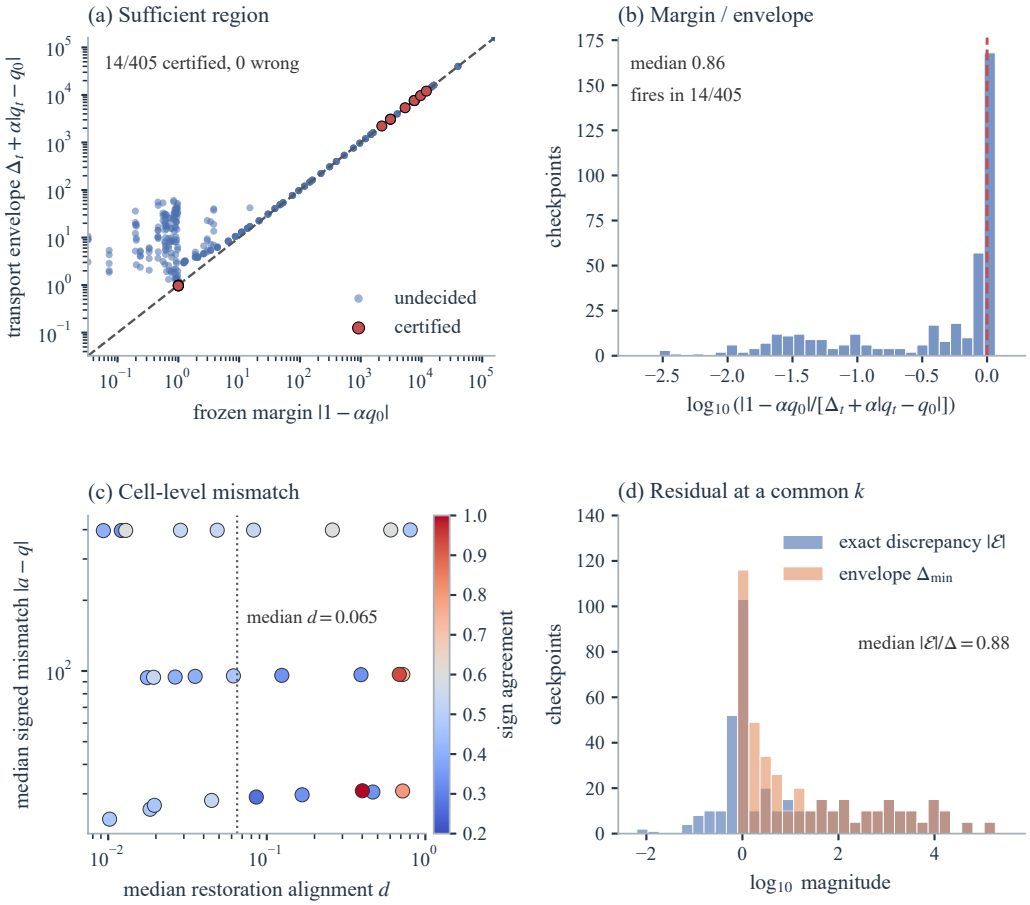



Figure 4: Most audited checkpoints fall outside the sufficient transport margin. Audit of 405 HAR checkpoints. (a) Frozen margin versus the optimized transport envelope; 14 checkpoints certify, all correctly. (b) Margin/envelope ratio: median 0.86, upper quartile 0.996. (c) Restoration alignment and signed mismatch across 27 width–dose cells. (d) Exact discrepancy and envelope at the same optimized scale, with median $|\mathcal{E}|/\Delta_{\min} = 0.885$.

Training instead uses the model's fixed random projections. Thus a contracting surrogate, per-example averaging, and regularization all enter $q$. The real-head driver differentiates the specified reconstruction, identity, or SNR loss. The A1 alignment records are probes *after* joint training; their mean cosine is an average of examplewise cosines, not the cosine of the aggregate parameter gradients. Their unsigned loading cannot establish the sign of $v^\top G_a$. The complete main-figure audit is generated by `paper/revision/build_figures.py`, with source hashes and reconstructed statistics. Duplicate low-label dose logs are reconciled by $(\alpha, \text{mode}, \text{seed})$ and checked for equality before removal.

`paper/revision/check_boundary.py` checks the exact discrepancy, the sufficient sign certificate, the realizable conflict example, and the finite-step remainder on seeded linear encoders. These are algebraic checks of the new theory, separate from the archived neural-network experiments.

### A.1 Conditional Dynamical Models

**Definition 2** (Degenerate minimizer)**.** *A loss has a degenerate minimizer if a constant representation attains its minimum. For a differentiable interior minimum, stationarity is necessary but insufficient. Variance shrinkage has such minima; reconstruction with varying targets generally does not, be-*

*cause a constant code need not be stationary for a fixed decoder. Accordingly, we evaluate trained reconstruction heads empirically.*

**Theorem 2** (Conditional scalar collapse transition)**.** *Assume an independently justified scalar reduction*

$$\dot{\sigma} = -\alpha\kappa\sigma + b(\sigma), \qquad \kappa > 0, \tag{15}$$

*where $b(0) = 0$. Let $f(\sigma) = b(\sigma)/\sigma$ on $(0, \infty)$ tend to zero at both ends, increase strictly to a unique nondegenerate maximum $\lambda_\star = f(\sigma_\star) > 0$, and decrease strictly afterwards. Then $\alpha_c = \lambda_\star/\kappa$. For $0 < \alpha < \alpha_c$ there are two positive equilibria, the smaller unstable and the larger stable; initial values below the smaller one still collapse. At $\alpha_c$ the equilibria coalesce in a saddle-node; for $\alpha > \alpha_c$, every positive trajectory tends to zero.*

*Proof.* The vector field is $\sigma(f(\sigma) - \alpha\kappa)$. The crossings and their stability follow from monotonicity; the nonzero second derivative gives the nondegenerate fold. Above the maximum the field is strictly negative for every $\sigma > 0$, whose only possible limiting equilibrium is zero. □

**Lemma 1** (Departure from a primary optimum)**.** *At $G_p = 0$, any nonzero $G_a$ makes the joint gradient nonzero when $\alpha > 0$. This implies departure from the primary optimum, but neither collapse nor irreversibility.*

**Corollary 3** (Fixed-dose attenuation)**.** *In Theorem 2, a constant auxiliary multiplier $q \in (0, 1]$ replaces the effective dose by $q\alpha$ and the threshold by $\alpha_c/q$. Positive attenuation can therefore preserve the positive equilibrium at a fixed dose. Exact deletion removes the contraction term but is not necessary for fixed-dose protection.*

**Theorem 3** (First modal contraction versus full linear collapse)**.** *For the linear symmetric model $\dot{z} = (R - \alpha A)z$, assume $R \succ 0$ and $A \succeq 0$ in the same coordinates. Along an eigenvector of $R - \alpha A$ with eigenvalue $\nu_j$, the modal variance obeys*

$$\dot{\varsigma}_j^2 = 2\nu_j\varsigma_j^2. \tag{16}$$

*Let*

$$T = R^{-1/2}AR^{-1/2}, \qquad \rho_\star = \lambda_{\max}(T). \tag{17}$$

*Some mode contracts iff $\alpha\rho_\star > 1$. Every mode contracts exponentially iff $\alpha\lambda_{\min}(T) > 1$. Equality leaves a neutral mode.*

*Proof.* $R - \alpha A = R^{1/2}(I - \alpha T)R^{1/2}$ has the same inertia as $I - \alpha T$ by congruence. Since the generator is symmetric, its eigenvalue signs determine modal growth or decay. □

The scalar model and the symmetric linear model answer different questions. Neither follows from measuring the proxy $\hat{\rho}$ alone. For example, with $R = I$, $A = \mathrm{diag}(2, 1/2)$, and $\alpha = 1$, one mode contracts and another expands. This permits task performance on a surviving direction without preserving the full representation. The NYUv2 result is compatible with this mechanism but does not identify its dynamical eigenvectors.

## B Detailed Intervention Comparisons

The battery application uses a temporal encoder with a mixer and gated state-space block, feeding SOH, RUL, reconstruction, and distributional heads. The corrected protocol keeps train, validation, and test cells disjoint, selects checkpoints on validation SOH, and separates observed-lifetime MAE from censored shortfall. Across ten initializations, detachment minus joint training is $-3.47$ SOH percentage points (95% $t$ interval $[-4.20, -2.74]$) and $-34.8$ observed-RUL cycles ($[-57.8, -11.8]$). These intervals describe initialization variability on the fixed split.

On this MATR protocol (Severson et al., 2019), detachment improves SOH RMSE from $5.57$ to $2.10$ percentage points and observed-RUL MAE from $81.5$ to $46.7$ cycles (Table 5); selective filtering gives no further improvement. Joint training has relative covariance trace $3.55$, so the gain is a negative-transfer result without representation contraction. The rank arm's large trace and low participation rank also illustrate why amplitude and diversity need separate measurement. Checkpoint selection and censoring-aware evaluation changed together, so their separate contributions to the new ranking are unknown.

Table 4: Controlled optimizer comparisons, ten seeds per method. Cells give collapse rate and mean task metric. The optimizer grid uses $(m, \alpha) = (128, 0.1)$ for HAR, $(32, 3000)$ for battery, and $(64, 10000)$ for RadioML. Battery uses the controlled variance auxiliary. Detachment's collapse counts come from separate $m = 64$ controls at doses $100, 100, 3000$; its task metrics belong to those protocols (Tables 5–6). Detachment removes the encoder's dose dependence, but these widths differ from the optimizer grid.

| Combiner | HAR acc. ↑ | Battery log-SOH RMSE ↓ | RadioML acc. ↑ |
|---|---|---|---|
| JOINT (plain summation) | 1.00 (0.945) | 1.00 (0.611) | 1.00 (0.256) |
| *conflict resolution* | | | |
| PCGrad | 1.00 (0.585) | 1.00 (0.601) | 1.00 (0.091) |
| CAGrad ($c$=0.5) | 1.00 (0.952) | 1.00 (0.607) | 1.00 (0.091) |
| *Pareto* | | | |
| MGDA | 1.00 (0.200) | 1.00 (0.385) | 1.00 (0.091) |
| *gradient geometry* | | | |
| Aligned-MTL | 1.00 (0.865) | 1.00 (0.609) | 1.00 (0.091) |
| IMTL-G | 1.00 (0.419) | 1.00 (0.601) | 1.00 (0.091) |
| Nash-MTL | 1.00 (0.280) | 1.00 (0.552) | 1.00 (0.091) |
| *loss weighting* | | | |
| FAMO | 1.00 (0.182) | 1.00 (0.613) | 1.00 (0.091) |
| UW (uncert.) | 1.00 (0.940) | 1.00 (0.611) | 1.00 (0.091) |
| GradNorm | 1.00 (0.958) | 1.00 (0.611) | 1.00 (0.170) |
| IMTL-L | 1.00 (0.947) | 1.00 (0.611) | 1.00 (0.091) |
| Detachment (STF-0) | 0.0 | 0.0 | 0.0 |

Table 5: Corrected MATR test results: ten initialization seeds on a fixed cell-disjoint split, with validation-based checkpoint selection. $S = \mathrm{tr}\,\mathrm{Cov}(H)$; relative $S$ is the mean same-seed ratio to the SOH-only reference; PR is participation rank. Censored shortfall measures lower-bound violations in cycles. Table 12 reports uncertainty.

| Mode | SOH RMSE% ↓ | obs. RUL MAE ↓ | shortfall ↓ | $S$ | relative $S$ | PR |
|---|---|---|---|---|---|---|
| SOH-only | 0.61 | — | — | 1.01 | 1.00 | 1.35 |
| JOINT | 5.57 | 81.5 | 17.60 | 2.86 | 3.55 | 2.49 |
| STF-0 | **2.10** | **46.7** | **2.06** | 4.77 | 6.24 | 1.86 |
| STFCAL | 2.71 | 63.6 | 5.73 | 1.40 | 1.89 | 2.03 |
| STFCALRANK | 4.98 | 107.6 | 17.57 | 617.90 | 651.77 | 1.21 |

Figure 6 collects the dose sweeps, width crossings, NYUv2 gate boundary, intervention counts, and financial outcomes. Width changes both empirical crossing locations and probe calibration (Yang & Hu, 2021); Appendix D examines their scaling separately.

## C Financial Ranking Application

The motivating observation came from multi-task equity selection. A production model shared an encoder of A-share factors across pairwise ranking, distributional (MDN (Bishop, 1994)), recon-

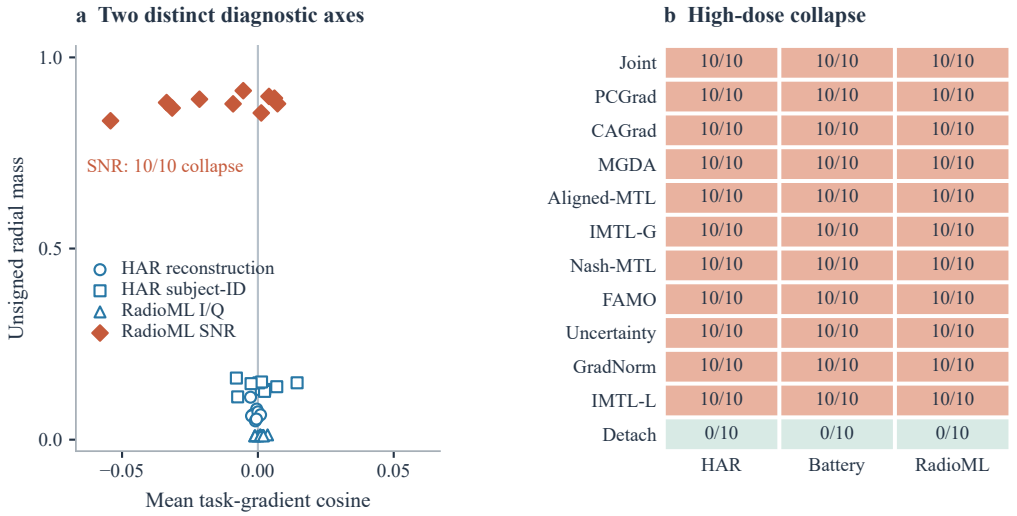



Figure 5: Task-gradient alignment leaves contraction unresolved. (a) Terminal probes of four real auxiliary heads, ten seeds each: mean per-example task-gradient cosine versus unsigned radial loading. SNR has high loading and $10/10$ collapsed endpoints; its signed flux is audited separately. (b) All ten optimizer baselines meet the scatter-collapse criterion in $10/10$ seeds at controlled doses $0.1$ (HAR), $3000$ (battery), and $10000$ (RadioML). Detachment uses separate ten-seed controls. Table 4 gives widths and task metrics.

Table 6: Real auxiliary heads at width $64$, ten seeds: mean primary accuracy and collapse rate in parentheses. The listed radial selective arms do not match detachment on high-dose SNR regression. The separate rank-3 arm also avoids collapse in all ten SNR seeds, but reaches only $0.211$ accuracy (Figure 8c).

| Auxiliary head | $\alpha$ | JOINT | STF-0 | STF-HARD | STF-SOFT | STFCAL |
|---|---|---|---|---|---|---|
| HAR input-reconstruction | 0.1 | .942 (0.0) | .943 (0.0) | .942 (0.0) | .942 (0.0) | .942 (0.0) |
| HAR subject-ID probe | 0.1 | .941 (0.0) | .943 (0.0) | .944 (0.0) | .943 (0.0) | .942 (0.0) |
| HAR subject-ID probe | 10 | .830 (0.0) | .940 (0.0) | .940 (0.3) | .939 (0.3) | .940 (0.0) |
| RadioML I/Q-reconstruction | 3000 | .343 (0.0) | .307 (0.0) | .329 (0.0) | .327 (0.0) | .315 (0.0) |
| RadioML SNR regression | 3000 | .205 (1.0) | **.307 (0.0)** | .208 (1.0) | .200 (0.9) | .199 (0.1) |

struction (AE), and view-contrastive (SCARF-style (Bahri et al., 2022)) heads. After deployment, predicted rankings became nearly constant under the unchanged training objective: cross-sectional prediction spreads fell by approximately $93\%$ relative to validation while individual training losses stayed low. The observation prompted us to measure feature and prediction variation alongside loss; identifying its cause required the controlled studies.

The separate financial experiment uses walk-forward training with real distributional, reconstruction, and contrastive heads. Its R5fin records set `fixed_aux=0`, so Table 15 evaluates real auxiliaries. Across ten folds and ten initialization seeds, mean out-of-sample (OOS) rank-IC is $0.060$ for joint training, $0.036$ for detachment, $0.063$ for the hard gate, and $0.056$ for the soft gate. Averaging seeds within each fold gives paired differences from joint of $-0.0237$, $+0.0029$, and $-0.0042$. Two-sided Wilcoxon $p$ values are $0.037$, $0.432$, and $0.695$; none survives Holm correction across the three comparisons. Temporal folds are dependent, so these comparisons are sensitivity analyses. Detachment costs accuracy in this sample, and no retained subspace shows a benefit. Script `finance_fold_audit.py` reproduces the comparisons from all 40 source files.

Table 7: Real HAR reconstruction ($m = R = 64$, ten seeds). Candidate ranks include closed-gate seeds; active ranks average only gate-open seeds. Accuracy and scatter average all ten seeds. PCA and auxiliary-gradient bases activate on identical seeds; these results establish neither superiority over joint training nor globally optimal deletion.

| $\alpha$ | basis | gate open | candidate rank | active rank | rel. scatter | accuracy |
|---|---|---|---|---|---|---|
| 12 | PCA | 4/10 | 57.0 | 62.00 | 0.842 | 0.9492 |
| 12 | Aux. grad. | 4/10 | 22.6 | 30.75 | 0.811 | 0.9497 |
| 30 | PCA | 8/10 | 62.8 | 63.62 | 0.754 | 0.9512 |
| 30 | Aux. grad. | 8/10 | 44.1 | 47.88 | 0.746 | 0.9531 |
| 100 | PCA | 10/10 | 64.0 | 64.00 | 0.630 | 0.9524 |
| 100 | Aux. grad. | 10/10 | 59.4 | 59.40 | 0.654 | 0.9520 |

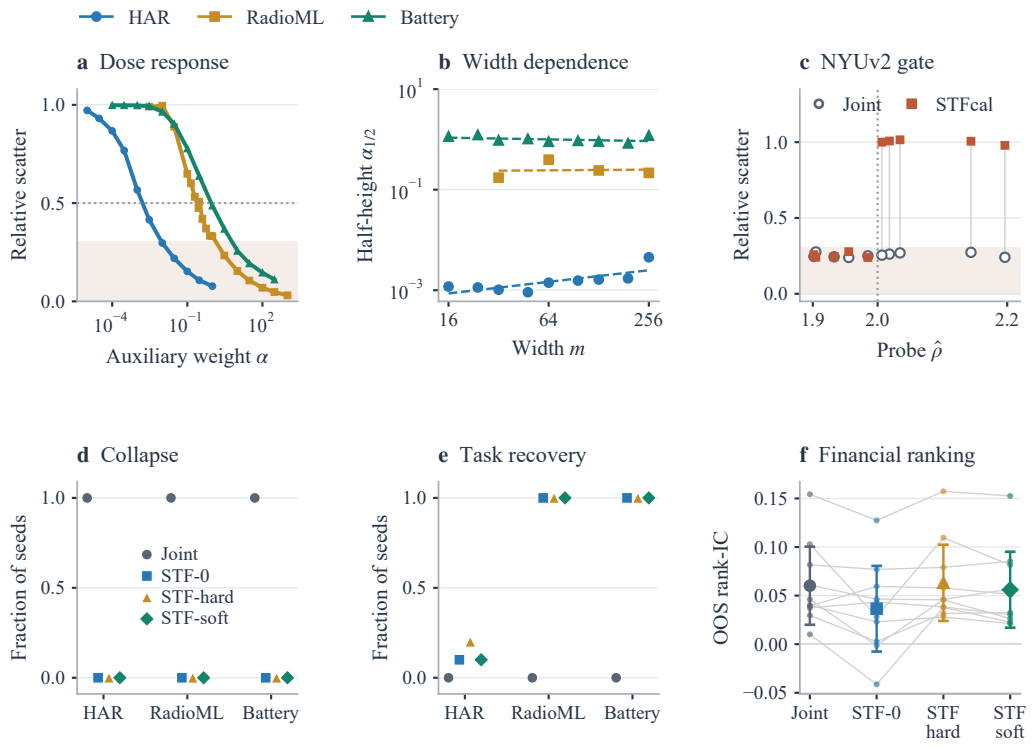



Figure 6: Additional diagnostics across domains. (a) Dose–scatter curves and empirical half-height crossings. (b) Width dependence. (c) Paired NYUv2 gate-boundary outcomes at $\alpha = 0.5$. (d,e) Collapse and task-recovery rates for the controlled interventions. (f) Financial rank-IC, paired by temporal fold. The corresponding tables specify protocols, seed counts, and the separate 0.3 collapse criterion.

In the separate synthetic testbed, initialization probes give $\rho = 0.74 \pm 0.19$ over 12 seeds and predicted critical weight $\alpha_c = \langle 1/\rho \rangle = 1.45 \pm 0.37$, consistent with the observed onset. Other domains use the projected-gradient diagnostic of Definition 1, but its numerical value depends on the task, encoder, and calibration time.

# D Proofs and Additional Analysis

## D.1 Statements Deferred from the Main Text

We formalize several statements discussed in the main text. Theorem numbering continues from the main text.

**Proposition 3** (Sufficient conditions for gradient vanishing)**.** *For a differentiable prediction map with Jacobian $J$ and output loss gradient $\ell$, the parameter gradient vanishes iff $J^\top \ell = 0$. Sufficient conditions are $\ell = 0$, or identical Jacobian rows together with $\mathbf{1}^\top \ell = 0$. Equal prediction values alone imply neither condition.*

*Proof.* Apply the chain rule. For a counterexample to the stronger assertion, $\hat{y}_i = \theta x_i$ is constant at $\theta = 0$ but has Jacobian rows $x_i$. The pairwise logistic loss $\log(1 + e^{-(\hat{y}_1 - \hat{y}_2)})$ has output gradient $(-1/2, 1/2)$ at a tie, so its parameter gradient is $(x_2 - x_1)/2$, generally nonzero. □

**Proposition 4** (Scope of positive attenuation)**.** *Under Theorem 2, a fixed positive multiplier $q$ leaves a finite threshold $\alpha_c/q$ but can prevent the transition at a specified dose. A dose-dependent positive multiplier can keep the effective dose subcritical for every $\alpha$: for $0 < \epsilon < 1$, take $q(\alpha) = (1 - \epsilon)/(1 + \alpha/\alpha_c)$.*

*Proof.* Substitute $q\alpha$ for $\alpha$. The proposed choice gives $q\alpha/\alpha_c = (1-\epsilon)\alpha/(\alpha+\alpha_c) < 1$. A positive stable equilibrium remains, although its lower basin boundary persists. Preserving gradient direction therefore does not imply that positive task weighting must fail. □

**Proposition 5** (Rank-dilution scaling)**.** *Assume the effective ratio scales as $\rho_\star \asymp (m_\Pi/m)^{\gamma_0}$, $\gamma_0 \geq 1$, and $m_\Pi \propto m^\beta$, $0 \leq \beta < 1$. Then $\alpha_c = 1/\rho_\star$ implies*

$$\gamma \;=\; \gamma_0\,(1-\beta), \qquad \gamma_0 \geq 1, \tag{18}$$

*The separable scaling is assumed; rank alone does not imply it. More generally, restoration $\asymp m^a$ and contraction $\asymp m^b m_\Pi^c$ yield $\gamma = a - b - c\beta$, with the displayed form requiring $c = a - b = \gamma_0$. If the toxic norm scales as $m_\Pi^p$, then $c = 2p$ only when contraction depends quadratically on that norm and has no additional rank dependence. The equality $\gamma_0 = 2p$ further requires $a - b = 2p$. Neither exponent is identified by the toxic eigenvalue spectrum alone. A fixed effective rank gives $\beta = 0$; a fixed per-head output dimension need not.*

**Corollary 4** (Composition of rank-one PSD operators)**.** *Let $T = \sum_k \rho_k u_k u_k^\top$, with unit $u_k$ and $\rho_k \geq 0$, in common restoration-normalized coordinates. If the directions are mutually orthogonal, $\lambda_{\max}(T) = \max_k \rho_k$; if aligned, it is $\sum_k \rho_k$. An additional orthogonal direction leaves the maximum unchanged only if its coefficient does not exceed the previous maximum.*

*Proof.* For orthogonal directions, the $u_k$ are eigenvectors. For aligned directions, the sum is a single rank-one operator. Neither case establishes the geometry of the production model's heads. □

For an explicit scalar example, take $b(\sigma) = \lambda\sigma^2/(1 + \sigma^2)$. Then

$$\sigma_\pm = \frac{\lambda \pm \sqrt{\lambda^2 - 4(\alpha\kappa)^2}}{2\alpha\kappa}, \qquad \alpha_c = \lambda/(2\kappa). \tag{19}$$

This verifies the fold and its basin dependence. Radial auxiliary behavior,

$$g_a = \kappa\delta h + o(\|\delta h\|), \tag{20}$$

requires isotropic curvature; stationarity at a constant representation is insufficient. For a general encoder, the Jacobian still determines its effect on variance through (4). In discrete time, linearizing a scalar stable equilibrium (Strogatz, 1994) with derivative $-\ell$ gives

$$e_{t+1} = (1 - \eta\ell)e_t, \qquad 0 < \eta\ell < 2, \tag{21}$$

where $e_t$ is a deviation from that equilibrium, not total representation scatter.

Power-law spectra do not determine the capacity exponent. For eigenvalues $\lambda_i = Ci^{-(1+s)}$ with $C$ independent of rank, $\|A_r\|_F^2 = C^2 \sum_{i=1}^r i^{-2(1+s)}$. Hence the Frobenius norm is bounded for $s >$

$-1/2$, grows as $\sqrt{\log r}$ for $s = -1/2$, and as $r^{-s-1/2}$ for $s < -1/2$. A decaying spectrum ($s > -1$) therefore has power exponent at most $1/2$; for the commonly intended $s > 0$ it has exponent zero. The trace instead involves $\sum_i i^{-(1+s)}$, and the operator norm is $C$ for a decreasing spectrum. Rank-dependent normalization $C(r, m)$ changes these conclusions. Thus neither a universal $\gamma_0(s)$ nor $\gamma_0 = 2p$ follows without specifying the norm, normalization, restoration and coupling to dynamics. Concretely, if $\kappa \asymp m^b \|A_r\|^q$, $\|A_r\| \asymp r^p$, and $r \asymp m^\beta$, then $\gamma = a - b - qp\beta$. The restricted form $\gamma_0(1 - \beta)$ requires $a - b = qp = \gamma_0$. Hence $\gamma_0 = 2p$ holds only under the additional quadratic coupling $q = 2$ and matching restoration exponent $a-b = 2p$; these are not implied by the measured width slope.

The original two-head generator has $W_k \in \mathbb{R}^{m\times r}$ with independent entries of variance $1/r$. Its combined span has rank $\min(m, 2r)$ almost surely. For the averaged quadratic loss the sample-normalized Hessian is $A = 2\sum_k W_k W_k^\top/(2r)$, with $\mathbb{E}A = 2I/r$. At $r = 32$, widths $32, 64, 128, 256$ therefore have ranks $32, 64, 64, 64$. Fixed per-head rank is not fixed combined rank over this range, and changing $r$ also changes curvature. The supplied orthonormal control instead constructs $A = Q\,\mathrm{diag}(\lambda)Q^\top$ at exactly specified rank and normalization.

The RS1 experiment fixes per-head rank $r \in \{4, 8, 16\}$ across widths $32, 48, 64, 96$, giving combined ranks $8, 16, 32$. At anchor $3 \times 10^{-4}$, descriptive half-height slopes are $-0.31$, $0.45$, and $3.02$; minimum relative scatter ranges are $0.040$–$0.068$, $0.013$–$0.018$, and $0.0056$–$0.0132$. Using the unpenalized anchor changes the slopes to $0.65$, $5.55$, and $3.88$. The fitted exponent is therefore sensitive to both rank and anchor. The analyzer retains all twelve curves and reports crossings and scatter floors separately.

The orthonormal control constructs the sample-normalized auxiliary Hessian $A = Q\,\mathrm{diag}(\lambda)Q^\top$ at exact rank $r$ and verifies its amplitude. Three normalizations hold different quantities fixed: $\lambda = 1$ fixes curvature per direction, so the Frobenius norm grows as $\sqrt{r}$; $\lambda = 1/r$ fixes the trace; and $\lambda = 1/\sqrt{r}$ fixes the Frobenius norm. The training grid tests the first two at $m \in \{32, 64, 128\}$, $r \in \{4, 16\}$, seven doses, and three seeds, holding the loss, optimizer, and data generator otherwise fixed. Three observations follow. First, at fixed width and per-direction curvature, the $r = 16$ crossing is lower than the $r = 4$ crossing by factors of $4.8$ at $m = 64$ and $228$ at $m = 32$. Second, the fitted width slope for $r = 4$ changes from $-0.29$ under $\lambda = 1$ to $+0.91$ under $\lambda = 1/r$. Third, minimum scatter is $3.5$–$11\%$ of the healthy reference for $r = 4$, versus $0.4$–$1.1\%$ for $r = 16$. Thus normalization affects the estimated exponent even when rank is controlled exactly. Each three-width fit has only one residual degree of freedom and cannot identify a precise exponent. The commands `controlled_rank_scan.py --self-test` and `run_controlled_rank_grid.py` verify the construction and reproduce the twelve configurations. The historical squared pairwise hinge has zero margin, so tied scores can attain zero primary loss. These controls concern that objective; they do not establish the same transition for informative ranking objectives. The separate `--primary-loss logistic` option removes this degeneracy, but no unfinished logistic sweep is counted as evidence.

The synthetic testbed uses random toxic projections of rank $\lfloor m/2 \rfloor$ and a low-rank primary task with 2–8 active directions across seeds (mean $5.3$). It sweeps 17 logarithmically spaced doses over $[3 \times 10^{-4}, 10]$ at six widths $m \in \{8, \ldots, 256\}$ under a pairwise-ranking loss. The toxic-rank schedule is proportional to width. The relative half-height rule gives $\gamma = 0.07$ across six widths and $\gamma = 0.01$ across the four widths used below. This nearly flat trend is consistent with the restricted law at $\beta = 1$, but does not independently establish that law.

Using a fixed absolute scatter threshold instead gives $\gamma = 1.53$ ($R^2 = 0.91$) on the same curves. At $\sigma_h = 1.5 \times 10^{-3}$, the crossing depends on the width-dependent healthy amplitude: the fixed-rank family has $\sigma_h(0) \propto m^{0.95}$. The absolute rule also gives nearly constant slopes $1.94, 2.02, 2.04, 1.96, 1.89$ across rank schedules $\beta = 0, \ldots, 1$, varying by only $0.15$ and largely masking schedule dependence. Using relative crossings, the fixed-per-head-rank family instead gives $\gamma_0 = 2.43$ ($R^2 = 0.90$, four widths). Its combined span can still grow as $\min(m, 2r)$, so this is not an exact fixed-effective-rank estimate.

Both synthetic families extend the dose grid to $\alpha = 0$ (21 points), using four widths $m \in \{32, 64, 128, 256\}$, five seeds, and 800 epochs. The fixed-rank family is the $\beta = 0$ case with constant per-head rank $r$. A corresponding ten-seed run yields crossings differing by

$+3.1\%, +1.8\%, -5.8\%, +19.7\%$ across these widths. This comparison quantifies sensitivity to seed count, including variation in the healthy anchor at larger widths.

Two checks qualify the crossing estimator. First, a half-height crossing can occur without near-total collapse. For fixed per-head rank $r \leq 4$, scatter plateaus at 2–23% of its healthy value; only $r \geq 8$ reaches a floor of at most $0.9\%$. These floors must be distinguished from the operational $0.3$ collapse threshold. Second, the synthetic fits use the first penalized dose, $\alpha = 3 \times 10^{-4}$, as the anchor. The unpenalized value is set by the ranking loss, is shared across toxic ranks, and lies 30–40% above the penalized trend. Varying the anchor over $\alpha \in [0, 3 \times 10^{-3}]$ preserves the decreasing ordering of $\gamma(\beta)$: $\gamma_0 \in [1.55, 3.23]$ at $\beta = 0$, versus $\gamma(1) \in [-0.06, +0.74]$. The ordering is more stable than the fitted coefficients.

The rank-schedule fits test a conditional scaling hypothesis. Given $\alpha_c \asymp (m/m_\Pi)^{\gamma_0}$ and an independently measured $m_\Pi \asymp m^\beta$, substitution gives $\gamma = \gamma_0(1 - \beta)$. Testing the premise requires measuring effective rank separately from the observed slope. The generator audit above specifies how per-head and combined ranks differ.

Crossing rules affect the fitted magnitudes. For HAR, an absolute threshold $T \in [0.20, 0.50]$ gives $\gamma \in [0.66, 0.82]$, compared with $0.66$ under the four-width relative rule. Battery requires a grid that reaches a healthy branch: earlier sweeps started at $\alpha = 300$, where relative scatter was already $0.07$–$0.11$, making thresholds left-censored. Extending to $\alpha \in [10^{-4}, 300]$ with five seeds gives a healthy branch at $\alpha \leq 10^{-3}$ (relative scatter $1.000 \pm 0.005$) and crossings $0.95, 0.89, 0.90, 1.18$ at $m = 32, \ldots, 256$. The fitted slope is $\gamma_{\text{bat}} = 0.10$ ($R^2 = 0.43$), varying over $[-0.10, +0.12]$ across crossing rules. These four widths are consistent with weak width dependence but do not resolve a precise exponent.

Treating each width's crossing as one observation gives four-width OLS 95% intervals $[-0.07, +4.93]$ for the $\beta = 0$ schedule, $[-0.16, +1.49]$ for HAR, $[-0.24, +0.43]$ for battery, and $[-1.16, +1.20]$ for RadioML. None excludes zero. Interior schedules yield positive slopes: $2.57$ (CI $[0.87, 4.27]$) at $\beta = 0.25$ and $1.86$ ($[0.81, 2.91]$) at $\beta = 0.50$; $\beta = 1$ remains consistent with zero. At the reported anchor, all four remaining schedule intervals contain $\gamma_0(1 - \beta)$, whereas the $\beta = 0.75$ interval excludes it at anchor $10^{-4}$. These results support schedule dependence without identifying an anchor-independent coefficient.

The denser SYN8 family supplies 26 completed $(m, \beta)$ cells, ten seeds each. Since $r(64) = 32$ for every schedule, we fit $\log \alpha_c = c + \gamma_0(1 - \beta)\log(m/64)$. At anchor $3 \times 10^{-4}$, the constrained fit gives $\gamma_0 = 2.50$ (nominal OLS 95% CI $[1.94, 3.06]$). Across anchors $\{10^{-4}, 3 \times 10^{-4}\}$ and inclusion/exclusion of $m = 128$, all four conditional intervals exclude zero (lowest bound $1.04$). In the unconstrained model $c + ax + b\beta x$, $x = \log(m/64)$, every interval for $a + b$ contains zero and $-b/a = 0.78$–$1.03$: the linear form is not rejected, not established. These are cell-level fit intervals, not independent-seed uncertainty: the schedules share seeds and the $m = 64$ curves are identical. Joint/anchor-row point estimates range from $1.35$ to $2.87$. At $m \geq 96$ the sampled curve never resolves the collapsed floor; half-height crossings interpolate the observed decline, not a verified collapse threshold. Thus SYN8 sharpens a conditional trend without supplying a precise physical exponent. `synth_beta_joint.py` reproduces the fits.

Nine-width HAR and battery sweeps use $m \in \{16, 24, 32, 48, 64, 96, 128, 192, 256\}$ with the same driver, seeds, epochs, and dose grids. Shared configurations agree to floating-point precision (maximum difference $2 \times 10^{-16}$). HAR gives $\gamma_{\text{HAR}} = 0.38$ (95% CI $[+0.10, +0.67]$, $R^2 = 0.60$, $n = 9$); excluding the two narrowest widths gives $0.60$. Battery gives $\gamma_{\text{bat}} = -0.06$ (CI $[-0.17, +0.06]$), consistent with weak width dependence. The denser grid narrows conditional fit intervals but does not eliminate crossing uncertainty. For example, HAR at $m = 32$, $\alpha = 10^{-3}$ has relative scatter $0.5026 \pm 0.0524$, placing the half-height crossing close to a sampled grid point and making interpolation sensitive to seed variation. We report both four- and nine-width estimates rather than treating either as an identified dynamical exponent.

We also examine probe scaling across encoders. Repeating calibration with a Transformer at $m \in \{32, 64, 128\}$ and ten seeds gives probe exponents $-0.93$ for HAR ($R^2 = 0.99$), $+0.39$ for battery ($R^2 = 0.87$), and $-0.17$ for RadioML ($R^2 = 0.79$). The corresponding MLP values are $-1.63$ ($R^2 = 0.96$), $+0.67$ ($R^2 = 0.86$), and $+0.28$ ($R^2 = 0.40$). Signs agree for HAR and battery; RadioML is consistent with zero for both encoders, and the 95% intervals overlap in all three

domains. This is evidence of broadly similar probe trends in the tested encoders, not architecture invariance. The script `probe_exponent.py` reproduces the analysis.

Probe exponents and scatter-crossing exponents estimate different quantities. Their MLP point estimates agree in sign for RadioML ($+0.28$ versus $+0.02$), but disagree for HAR ($-1.63$ versus $+0.38$) and battery ($+0.67$ versus $-0.06$). A half-height crossing locates a dose at which trained scatter reaches half its reference value; the probe estimates $1/\hat{\rho}$ at the warm-up anchor. Agreement within a factor of a few at one width (a factor of 7 for NYUv2 at $m = 64$) does not imply agreement in width scaling. Negative probe slopes do not satisfy the restricted nonnegative exponent assumed in (18), but that assumption is not a general constraint on empirical slopes. The width analysis here therefore reports crossing exponents separately, while Appendix G reports probe values as calibration statistics.

Several sweeps show bimodal per-seed scatter near the empirical crossing. Such outcomes are compatible with a saddle-node model, but can also arise from initialization, finite training time, and optimization. We report the heterogeneity without assigning it a unique dynamical cause.

The original GPU experiments use fixed integer seeds, bf16 autocast with TF32 matrix multiplication, and Adam (lr $= 10^{-3}$, batch 256) on an NVIDIA A100 80 GB. Exceptions are specified with their protocols. Operational collapse means scatter below $0.3$ times the same-seed single-task reference; the released recovery rate compares each seed with the mean joint-training metric. Code, seeds, hyperparameters, and per-fold results are included in the repository.

## E Generalizations and Finite-Size Theory

**Proposition 6** (Fixed attenuation in a specified model)**.** *For $T \succeq 0$ in restoration-normalized coordinates, uniform attenuation $T \mapsto (1-\lambda)T$, $0 \le \lambda < 1$, multiplies the first-modal-contraction threshold by $1/(1-\lambda)$. At fixed dose the smallest $\lambda$ meeting $\alpha(1-\lambda)\|T\| \le 1$ is $\max(0, 1 - 1/(\alpha\|T\|))$ when $\alpha\|T\| > 0$.*

*Proof.* All eigenvalues scale by $1 - \lambda$; solve the displayed inequality. This statement concerns a fixed PSD model, not arbitrary signed or nonsymmetric filtered dynamics. □

**Theorem 4** (Annihilation by orthogonal compression)**.** *Let $T \succeq 0$, $U = \mathrm{range}(T)$, and $Q = I - P_S$ be an orthogonal projector. Then $QTQ = 0$ iff $U \subseteq S$. Among such deletions the unique smallest deleted subspace is $U$, and $I - P_U$ preserves its orthogonal complement.*

*Proof.* $QTQ = (T^{1/2}Q)^\top (T^{1/2}Q) = 0$ iff $\mathrm{range}(Q) \subseteq \ker T$, equivalently $U \subseteq S$. A general one-sided backward operator $QT$ need not describe symmetric neural-network dynamics; a signed filter can even reverse contraction, so its operator norm cannot supply the missing implication. □

**Corollary 5** (Subspace estimation bound)**.** *If $\|P_U - P_{\hat{U}}\| \le \epsilon$ and $Q = I - P_{\hat{U}}$, then*

$$\|QT\| \le \epsilon\|T\|, \qquad \|QTQ\| \le \epsilon^2\|T\|. \tag{22}$$

*Proof.* Use $T = P_U T P_U$ and $\|QP_U\| \le \epsilon$ on one or both sides. Whitening must precede this argument if $T$ is restoration-normalized; an angle bound in other coordinates is insufficient. The bounds concern residual magnitude, not the existence of a residual. □

**Theorem 5** (Minimum rank for a PSD compression)**.** *Fix a PSD operator $T$ in specified Euclidean coordinates, with eigenvalues $\kappa_1 \ge \cdots \ge \kappa_d \ge 0$, and a threshold $\tau \ge 0$. Among orthogonal projectors $Q = I - P_S$ satisfying $\|QTQ\| \le \tau$, the minimum deletion rank is*

$$r^\star = \#\{j : \kappa_j > \tau\}.$$

*Deleting $S_c = \mathrm{span}\{u_j : \kappa_j > \tau\}$ attains it. Within the class of spectral deletions, $S_c$ also minimizes any inclusion-monotone deleted energy. Uniqueness of the energy optimum requires positive energy in every additionally deleted eigendirection.*

*Proof.* The spectral choice leaves only eigenvalues at most $\tau$. For any $\dim S < r^\star$, dimension counting gives a unit vector $v \in S^\perp \cap S_c$. Its Rayleigh quotient satisfies $v^\top QTQv = v^\top Tv > \tau$, violating feasibility. For spectral deletions feasibility requires containing $S_c$, giving the energy statement by inclusion. Rotated subspaces can also attain the same minimum rank; no global minimum-energy claim over rotations follows. □

**Corollary 6** (Saturation and its limits)**.** *For $\tau = 1/\alpha$, the optimal rank changes in steps as $\alpha$ crosses inverse positive eigenvalues and eventually equals $\dim U$. The limiting projector is $I - P_U$, which equals full detachment only on $U$, or globally when $U$ is the entire gradient space. Exhausting a finite probe depth does not prove saturation of $U$.*

The theorem concerns a specified operator, not a mean of gradient-norm ratios. For a linear reconstruction decoder $Wh$ with squared loss, $g_a = W^\top \delta$ (up to loss normalization) and the Hessian is $W^\top W$. The uncentered second moment relevant to normalized residual energy is

$$C_g = \mathbb{E}\left[\frac{g_a g_a^\top}{(\|g_p\| + \varepsilon)^2}\right] = W^\top \mathbb{E}\left[\frac{\delta\delta^\top}{(\|g_p\| + \varepsilon)^2}\right] W.$$

For any rank-$r$ projector $P$, the squared residual is $\mathrm{tr}((I - P)C_g)$, minimized by the top $r$ eigenvectors of $C_g$ (the Rayleigh–Ritz principle). Its square root upper-bounds the corresponding mean norm by Jensen's inequality. It does not generally minimize that mean itself. Hessian eigenvectors coincide with this basis under isotropic weighted residual covariance; representation PCA requires its leading subspace to coincide with the leading eigenspace of $C_g$. Commutation gives a common eigenbasis but does not ensure the same ordering. A nonlinear decoder additionally contributes residual-weighted second derivatives to its Hessian. A nearly flat PCA residual curve is therefore consistent with gradient energy outside leading representation components, without establishing orthogonality of the reconstruction residual to those components.

Table 9 gives the variance-shrinkage rank sweep. For real HAR reconstruction at $\alpha = 12$, auxiliary-gradient and PCA bases give mean candidate ranks $22.6$ and $57.0$. Both gates open in four of ten seeds, with active ranks $30.75$ and $62.0$ at similar aggregate accuracy (Table 7). This comparison isolates basis sensitivity. The artifact also includes the three-seed RMS-basis pilot and its negative accuracy result.

In the ten-seed battery cap32 experiment, `stfcalrank` reaches its cap in every seed and gives SOH RMSE $5.26$, versus $5.24$ for joint training. The larger cap therefore does not establish minimum-rank protection. A single-seed full-depth audit gives representation covariance traces $3.131$, $4.065$, $0.906$, and $0.821$ for joint training, detachment, the radial calibrated arm, and the rank arm, respectively, versus $1.147$ for the SOH-only reference. Corresponding normalized SOH prediction variances are $0.00826$, $0.01380$, $0.01253$, and $0.01019$, versus $0.01379$. These quantities separate variance from prediction spread and establish neither representation recovery nor constant predictions. Historical checkpoints were selected on the evaluation split and remain exploratory. The corrected cell-disjoint protocol fits preprocessing on training cells and evaluates test data after validation-based checkpoint selection.

Forward and backward-only filters differ even when their Jacobians agree. For fixed $U$, let $\widetilde{H}_f = H - C_B HUU^\top$ and $\widetilde{H}_b = \mathrm{sg}(H) + D - C_B DUU^\top$, where $D = H - \mathrm{sg}(H)$ and sg stops gradients. Both induce the backward map $G \mapsto G - C_B GUU^\top$, but only $\widetilde{H}_b$ preserves the auxiliary input. Consequently, the loss gradients are evaluated at different forward values and need not produce the same update. Even at full rank, both retain a batch-mean gradient channel, unlike full detachment.

On HAR reconstruction at $\alpha = 100$, $m = R = 64$, paired forward and backward-only filters share warm-up, seed, basis, and selected rank. All ten gates open. Forward filtering gives primary accuracy $0.9507 \pm 0.0033$, reconstruction MSE $0.05112 \pm 0.01083$, and relative scatter $0.681 \pm 0.039$; backward-only filtering gives $0.9478 \pm 0.0058$, $0.02053 \pm 0.00115$, and $0.597 \pm 0.061$. Auxiliary errors use unfiltered features. The paired accuracy difference (backward minus forward) is $-0.00285$ (95% $t$ interval $[-0.00815, 0.00245]$); reconstruction improves in every seed while scatter decreases. Holm-adjusted Wilcoxon $p$ values across these outcomes are $0.242$, $0.00586$, and $0.0195$. Thus the forward choice changes the auxiliary-fit/scatter tradeoff, with no resolved primary-accuracy advantage. Joint and detached accuracy in the same harness are $0.9461$ and $0.9431$. A three-seed $\alpha = 12$ pilot, including two closed-gate pairs, is also retained. Scripts `filter_semantics_audit.py`

and `summarize_filter_semantics.py` provide the checks and paired records. Probe algebra and eigendecomposition run in fp32, with invariance to enclosing bf16 autocast checked separately.

A complementary experimental intervention uses $v = \nabla_\theta V$, proposed update $u_0 = G_p + \alpha G_a$, and slack $h = V - V_{\min}$. For $k > 0$ and $v \neq 0$, the closest update satisfying $v^\top u \leq kh$ is

$$u^\star = u_0 - \frac{(v^\top u_0 - kh)_+}{\|v\|^2} v.$$

The KKT conditions prove this Euclidean halfspace projection. Under a well-defined continuous flow, $\dot{h} \geq -kh$ preserves feasibility on this *fixed batch*. At $v = 0$ feasibility requires $kh \geq 0$. Discrete updates require checking the actual post-step variance: the supplied SGD implementation backtracks and restores parameters on failure. The argument covers neither unseen data nor Adam without modeling its applied update. The projection is the established constrained-gradient construction (Allibhoy & Cortés, 2024); its empirical benefit is untested here.

Table 8 compares deletion with positive attenuation on UCI-HAR ($m = 64$, sixty epochs). The calibrated multiplier uses the unprojected mean gradient ratio, so its $c = 1$ is a calibration setting rather than an identified dynamical threshold. The $c = 0.1$ setting is competitive with the tested fixed multipliers. This exploratory grid has three seeds per configuration and includes non-finite runs. At $\alpha = 300$, seed 0 diverges for every arm, including detachment, whose auxiliary heads still train during detached warm-up. We record numerical divergence separately from representation collapse.

Table 8: Matched attenuation on UCI-HAR ($m = 64$, sixty epochs). Accuracy and calibrated multiplier $q$ are averaged over completed seeds; `div` indicates non-finite evaluation. There are three seeds per configuration, except at $\alpha = 300$, where seed 0 diverges for every arm and is excluded. Bold identifies the best fixed setting and the best calibrated setting when it exceeds the best fixed accuracy.

| | accuracy | | | multiplier $q$ | | |
|---|---|---|---|---|---|---|
| arm | 30 | 100 | 300 | 30 | 100 | 300 |
| plain summation | .9293 | `div` | `div` | 1 | 1 | 1 |
| detachment (STF-0) | .9276 | .9276 | `div` | 0 | 0 | 0 |
| fixed $10^{-4}$ | .9275 | .9276 | .9272 | — | — | — |
| fixed $3{\cdot}10^{-3}$ | .9278 | **.9304** | .9286 | — | — | — |
| fixed $10^{-2}$ | **.9301** | .9299 | **.9293** | — | — | — |
| fixed $3{\cdot}10^{-2}$ | **.9301** | .9294 | .9265 | — | — | — |
| fixed $10^{-1}$ | .9295 | .9291 | .9260 | — | — | — |
| fixed $3{\cdot}10^{-1}$ | .9242 | .8842 | `div` | — | — | — |
| calibrated $c = 0.1$ | .9299 | **.9306** | .9289 | .076 | .016 | .004 |
| calibrated $c = 0.3$ | .9261 | .9285 | .9281 | .228 | .049 | .012 |
| calibrated $c = 1.0$ | .9257 | .9266 | .9276 | .759 | .163 | .039 |

**Proposition 7** (Sampling fluctuation of a frozen statistic)**.** *For independent identically distributed scalar diagnostics $X_i$ with finite variance $\sigma_X^2$, $\hat{\rho} = B^{-1}\sum_i X_i$ has variance $\sigma_X^2/B$ and obeys a central limit theorem. Under a normal approximation the probability that the* gate $\alpha\hat{\rho} > 1$ *opens is* $\Phi(\sqrt{B}(\rho - 1/\alpha)/\sigma_X)$. *This is not the probability of training collapse. Batch-dependent centering or fitted bases require additional analysis or an independent frozen calibration sample.*

*Proof.* Use variance additivity and the central limit theorem. For a ratio $A/P$, the delta-method variance includes $-2\mu_A \operatorname{Cov}(A, P)/\mu_P^3$ as well as the two marginal variance terms. Mean per-example ratios and ratios of means are distinct estimands. □

**Proposition 8** (Conditional calibration bound)**.** *Assume independent $X_i \in [0, M]$ with common mean $\rho$ and population variance at most $v_0$. Bernstein's inequality gives*

$$\mathbb{P}(|\hat{\rho} - \rho| \geq \varepsilon) \leq 2\exp\left[-\frac{B\varepsilon^2}{2v_0 + 2M\varepsilon/3}\right]. \tag{23}$$

Table 9: Selected rank versus auxiliary dose on UCI-HAR ($m = 64$, five seeds, probe depth $R = 64$). Candidate ranks are given for representation-PCA and auxiliary-gradient bases. A closed gate removes no directions; full detachment blocks all 64. Relative scatter is $\mathcal{S}_h/\mathcal{S}_{single}$ for `stfcalrank`. The PSD optimum in Theorem 5 additionally requires identifying the operator and basis.

| $\alpha$ | gate | $r^\star$ (principal) | $r^\star$ (aux. basis) | rel. scatter (rank arm) | rel. scatter (JOINT) |
|---|---|---|---|---|---|
| 0.003 | 0 | – | – | 0.404 | 0.402 |
| 0.01 | 0 | – | – | 0.287 | 0.280 |
| 0.03 | 1 | 1.6 | 1.6 | 1.007 | 0.200 |
| 0.1 | 1 | 3.8 | 3.8 | 1.026 | 0.138 |
| 0.3 | 1 | 4.2 | 4.2 | 1.018 | 0.101 |
| 1.0 | 1 | 11.6 | 11.6 | 0.978 | 0.070 |
| 3.0 | 1 | 25.8 | 25.8 | 0.987 | 0.051 |
| 10 | 1 | 39.8 | 40.0 | 0.986 | 0.035 |
| 30 | 1 | 47.4 | 47.8 | 0.968 | 0.025 |
| 100 | 1 | 61.6 | 61.0 | 0.950 | 0.016 |

Table 10: Historical battery results with evaluation-selected checkpoints, retained as exploratory comparisons. The legacy RUL score combines errors against observed lifetimes and censored lower bounds. Reference rows use four seeds; other rows use ten. SOH and legacy RUL are separate outcomes; single-task rows provide references.

| | MATR | | NASA | |
|---|---|---|---|---|
| Mode | SOH RMSE (%) | legacy RUL score | SOH RMSE (%) | legacy RUL score |
| single-task reference | 0.51 | 694.6 | 9.17 | 18.5 |
| JOINT (aux heads) | $5.24 \pm 0.54$ | $209.0 \pm 59.8$ | $11.42 \pm 0.85$ | $17.5 \pm 1.1$ |
| JOINT (toxifier) | $1.23 \pm 0.44$ | $306.6 \pm 48.2$ | $9.66 \pm 0.44$ | $17.6 \pm 2.4$ |
| STF-0 | $1.68 \pm 0.41$ | $574.7 \pm 80.7$ | $9.35 \pm 0.55$ | $18.0 \pm 2.0$ |
| STF-0W (tri-weight) | $1.54 \pm 0.44$ | $541.2 \pm 91.3$ | $9.40 \pm 0.62$ | $18.6 \pm 1.4$ |
| STFCAL ($\hat{\rho}$=17.4/0.40) | $1.77 \pm 0.47$ | $419.8 \pm 33.6$ | $11.40 \pm 0.78$ | $18.3 \pm 1.7$ |

All MATR rows: censoring-aware RUL supervision, all 185 cells retained; bf16 autocast training, fp32 evaluation.

*Thus an error probability at most $\delta$ follows if*

$$B \geq (2v_0 + 2M\varepsilon/3)\varepsilon^{-2}\log(2/\delta). \tag{24}$$

*A nonzero primary-gradient denominator alone does not provide a uniform bound $M$. Population bounds cannot be replaced by observed sample moments without a separate valid empirical-concentration result.*

*Proof.* Apply the bounded-variable Bernstein inequality to the centered independent sum and solve for $B$. □

**Corollary 7** (Certifying a frozen gate or nested rank rule)**.** *Under Proposition 8, a gate with population margin $d = |\rho - 1/\alpha| > 0$ has error probability at most $2\exp[-Bd^2/(2v_0 + 2Md/3)]$. For $R+1$ fixed residual statistics with uniformly positive margins, a union bound controls all their threshold decisions and hence the selected nested rank. This does not identify the statistics with Hessian eigenvalues, certify an adaptively fitted basis without conditioning, or certify a subsequent training trajectory.*

*Proof.* A wrong threshold decision requires an estimation error at least its margin; apply the bound and then the union bound. Strict margins exclude equality cases. □

**Proposition 9** (Local linear noise floor)**.** *For deviations $e_{t+1} = ae_t + \eta\xi_t$ from a fixed equilibrium, assume $|a| < 1$ and independent mean-zero innovations of variance $D$. The stationary variance is $\eta^2 D/(1-a^2)$.*

Table 11: Frozen-anchor sampling experiment on HAR ($m = 64$, four detached warm-up epochs, five seeds). Each seed uses 2000 batches of size $B$, sampled with replacement. Standard deviation and CV describe variation across batches at a fixed anchor. "Meas." is the observed frequency of $|\hat{\rho} - \rho| > \eta\rho$; "bound" is the Bernstein plug-in envelope from (23), using estimated probe moments ($\kappa^2 = 2\rho^2 c_0^2 + \frac{2}{3} R_{\max}\varepsilon$). Values above one are uninformative probability bounds. The envelopes exceed observed frequencies but are not population confidence guarantees. The fitted CV slope is $-0.500$ ($R^2 = 0.96$), and the mean slope is $+0.006$. The norm of $\sum_i g_a^{(i)}$ is at most $4 \times 10^{-8}$ of $\sum_i \|g_a^{(i)}\|$. Raising nominal confidence from 90% to 99.9% multiplies the sample-size expression by $\ln(2/10^{-3})/\ln(2/0.1) = 2.54$.

| | | | | $\eta = 0.2$ | | $\eta = 0.1$ | |
|---|---|---|---|---|---|---|---|
| $B$ | $\hat{\rho}$ | $\mathrm{sd}(\hat{\rho})$ | $\mathrm{CV}(\hat{\rho})$ | meas. | bound | meas. | bound |
| 16 | 74.30 | 69.52 | 0.947 | 0.820 | 1.94 | 0.911 | 1.98 |
| 32 | 76.69 | 50.40 | 0.664 | 0.740 | 1.90 | 0.866 | 1.97 |
| 64 | 77.03 | 36.00 | 0.471 | 0.657 | 1.83 | 0.836 | 1.94 |
| 128 | 77.00 | 25.34 | 0.332 | 0.526 | 1.71 | 0.756 | 1.90 |
| 256 | 77.74 | 18.24 | 0.237 | 0.378 | 1.49 | 0.657 | 1.83 |
| 512 | 77.71 | 12.92 | 0.168 | 0.221 | 1.15 | 0.538 | 1.69 |
| 1024 | 77.97 | 9.20 | 0.119 | 0.094 | 0.70 | 0.384 | 1.45 |
| 2048 | 78.00 | 6.34 | 0.082 | 0.022 | 0.29 | 0.218 | 1.07 |
| 4096 | 77.90 | 4.54 | 0.059 | 0.003 | 0.06 | 0.095 | 0.61 |

*Proof.* The second-moment recursion is $\mathbb{E}e_{t+1}^2 = a^2\mathbb{E}e_t^2 + \eta^2 D$; sum the convergent geometric series. This is variance of deviations, not total representation scatter. Divergence of a local linear approximation as $|a| \to 1$ does not prove a universal ordering of half-height and binary-collapse thresholds in a nonlinear experiment. □

At a frozen HAR anchor, 2000 resampled batches per size over five seeds give slopes $-0.500$ for probe CV, $-0.494$ for standard deviation, and $+0.006$ for the mean. Table 11 compares observed fluctuations with plug-in envelopes based on estimated variance and range. Shared batch centering and estimated moments limit these envelopes to diagnostic use. The records also distinguish mean per-example ratios from ratios of batch-mean gradients: the latter can vanish for centered variance losses.

### E.1 Prediction Variance Normalization and Target $z$-Scoring

For $\hat{y} = w^\top h + b$, shrinking centered features by $1/c$ and expanding $w$ by $c$ preserves predictions. Shrinking both changes the regression loss, and constant predictions generally cannot fit varying targets. This scale ambiguity motivates reporting prediction variance and across-example feature covariance alongside task error, and testing prediction normalization and target scaling as controls.

The historical regression recipe standardizes targets and divides predictions by a detached batch standard deviation. This changes optimization and can reduce sensitivity to scale, but detached normalization does not implement the derivative of a truly scale-invariant loss. Softmax also does not normalize logit scale: changing that scale changes probabilities. These operations therefore require empirical ablations, with scaling statistics fit only on training data.

## F Additional Tables and Protocol

The corrected battery protocol uses cell-disjoint train/validation/test partitions: 130/27/28 MATR cells and 24/4/5 NASA cells. MATR test data contain 1069 observed and 1797 censored cycles; all 344 NASA test cycles are observed. Variation is over initializations on these fixed splits. Checkpoints minimize validation SOH, prioritizing that task. Participation rank is $(\mathrm{tr}\,\Sigma)^2/\,\mathrm{tr}(\Sigma^2)$; prediction variance uses squared SOH percentage points. Trace measures amplitude and participation rank measures spectral concentration; both depend on the encoder–head parameterization. Script `current_evidence_audit.py` checks split disjointness, seed pairing, and source hashes and

Table 12: Corrected battery test results. Means ± population standard deviations over ten initialization seeds on one fixed cell split. RUL is observed-only MAE; $S$ is across-example covariance trace, PR is participation rank, and $V_{\text{pred}}$ is SOH prediction variance in squared percentage points. The SOH-only head does not train RUL.

| Data | Mode | SOH RMSE (%) | RUL MAE | $S$ | PR | $V_{\text{pred}}$ |
|---|---|---|---|---|---|---|
| MATR | SOH-only | $0.61 \pm 0.19$ | — | $1.01 \pm 0.50$ | $1.35 \pm 0.14$ | $141.67 \pm 5.92$ |
| MATR | JOINT | $5.57 \pm 0.48$ | $81.54 \pm 26.84$ | $2.86 \pm 0.57$ | $2.49 \pm 0.65$ | $66.10 \pm 8.59$ |
| MATR | STF-0 | $2.10 \pm 0.62$ | $46.73 \pm 11.32$ | $4.77 \pm 1.33$ | $1.86 \pm 0.22$ | $109.50 \pm 13.01$ |
| MATR | STFCAL | $2.71 \pm 0.54$ | $63.59 \pm 19.43$ | $1.40 \pm 0.44$ | $2.03 \pm 0.20$ | $99.26 \pm 9.69$ |
| MATR | STFCALRANK | $4.98 \pm 1.09$ | $107.58 \pm 31.10$ | $617.90 \pm 489.76$ | $1.21 \pm 0.25$ | $70.11 \pm 20.06$ |
| NASA | SOH-only | $12.66 \pm 0.99$ | — | $3.39 \pm 1.57$ | $2.59 \pm 0.74$ | $183.51 \pm 44.46$ |
| NASA | JOINT | $14.43 \pm 0.64$ | $20.05 \pm 1.59$ | $5.41 \pm 0.85$ | $3.25 \pm 0.49$ | $127.49 \pm 24.32$ |
| NASA | STF-0 | $12.74 \pm 1.20$ | $19.43 \pm 2.66$ | $4.38 \pm 1.28$ | $3.16 \pm 0.58$ | $135.70 \pm 37.07$ |
| NASA | STFCAL | $14.47 \pm 1.22$ | $19.20 \pm 2.40$ | $4.55 \pm 1.42$ | $2.75 \pm 0.62$ | $109.72 \pm 24.59$ |
| NASA | STFCALRANK | $14.47 \pm 1.22$ | $19.20 \pm 2.40$ | $4.55 \pm 1.42$ | $2.75 \pm 0.62$ | $109.72 \pm 24.59$ |

regenerates the table. The remaining tables give datasets, historical censoring and head comparisons, auxiliary-weight sweeps, and the training protocol.

**Algorithm 2** STF training protocol (per domain, per configuration)

1: **Input:** domain data $\mathcal{D}$, encoder width $m$, toxicity schedule $\{\alpha\}$, seeds
2: **Phase 1 – health anchor:** train the single-task encoder ($\alpha = 0$), record healthy scatter $\mathcal{S}_{single} = \mathcal{S}_h(\theta^*)$ and healthy spectral ratio $\rho_0$.
3: **Phase 2 – calibration:** sweep $\alpha$ with mode JOINT; record $\rho$, relative scatter $\mathcal{S}_h/\mathcal{S}_{single}$, collapse flag; locate $\alpha_c$ by the threshold crossing.
4: **Phase 3 – repair:** at a selected dose, train each arm on $L_p(H) + \alpha L_a(\widetilde{H})$, using its specified forward map $\widetilde{H}$ (or detached $H$ for STF-0); backpropagate this loss and monitor the gradient diagnostic and relative scatter.
5: **Report:** collapse rate, recovery rate, $\rho$, $\mathcal{S}_h/\mathcal{S}_{single}$.

On NASA, the corrected protocol gives SOH RMSE $14.43$ for joint and $12.74$ for detachment, against the SOH-only reference $12.66$. The paired difference is $-1.69$ percentage points (95% $t$ interval $[-2.54, -0.83]$, nine of ten seeds improve). Observed-RUL MAE changes from $20.05$ to $19.43$ cycles, but the paired interval $[-3.37, 2.13]$ includes zero (exact two-sided Wilcoxon $p = 0.770$). The calibrated gate stays closed in all ten seeds: its two arms coincide with each other ($14.47$ SOH, $19.20$ RUL), but not with joint because their encoder warm-up is detached. NASA has only five test cells; initialization intervals cannot establish population generalization.

In the historical ten-seed MATR experiment, the radial calibrated arm improves SOH over joint training ($1.77 \pm 0.47$ versus $5.24 \pm 0.54$), while detachment reaches $1.68 \pm 0.41$. The cap32 rank arm remains near joint ($5.26$), so the radial result does not validate minimum-rank protection; Corollary 6 locates the saturation limit at full-width deletion. Censored RUL requires separate observed-error and lower-bound-shortfall metrics; a lower legacy MAE need not indicate better lifetime prediction. NASA's closed gate and its discrepancy from detachment further limit the diagnostic. Because these historical checkpoints were evaluation-selected, the corrected validation protocol provides the primary battery evidence.

A four-seed MATR ablation removes each weighting factor in turn. Omitting the rate factor gives fast-charge SOH RMSE $0.95$; omitting the temperature factor gives extreme-temperature SOH RMSE $2.35$. These results support the intended association between factors and operating regimes. A sweep over $\omega \in \{0.5, 1, 2\}$ evaluates sensitivity around the chosen value $\omega = 1$.

Table 24 reports the composition experiment. On HAR, accuracy changes from $0.943$ with filtering alone to $0.933$ with PCGrad and $0.940$ with CAGrad. Relative scatter is at most $0.15$ without filtering, compared with $0.97$–$1.14$ after filtering. These measurements support compatibility for the tested filter-then-combiner order.

**Algorithm 3** Released calibrated forward filters: radial STFCAL and capped STFCALRANK

1: **Input:** data $\mathcal{D}$, encoder width $m$, operating weight $\alpha$, probe depth $R$, warm-up length $W$
2: **Warm-up:** train $\theta$ with the auxiliary branch detached ($\nabla_\theta L_a \equiv 0$) for $W$ epochs; record the healthy anchor $\mathcal{S}^\star$ of (10)
3: **Probe:** obtain per-example gradients on an unfiltered frozen batch, radial projectors $\Pi_{c,i}$, and principal basis $U$. With $\varepsilon = 10^{-8}$, compute $\hat\rho = B^{-1}\sum_i \|\Pi_{c,i} g_a^{(i)}\|/(\|g_p^{(i)}\| + \varepsilon)$ and $\hat\rho_{\mathrm{res}}[r] = B^{-1}\sum_i \|(I - U_r U_r^\top) g_a^{(i)}\|/(\|g_p^{(i)}\| + \varepsilon)$ for $r = 0, \dots, R$.
4: **Calibrate:** $g \leftarrow \mathbf{1}[\alpha\hat\rho \ge 1]$; $\mathcal{R} \leftarrow \{r \le R : \alpha\hat\rho_{\mathrm{res}}[r] < 1\}$; choose $r \leftarrow \max(1, \min \mathcal{R})$ if nonempty, else $r \leftarrow R$. Record whether a feasible rank exists.
5: **Latch:** freeze $g, r, U$. If $g = 0$, pass $H$ to the auxiliary head. Otherwise, STFCALRANK passes $H - C_B H U_r U_r^\top$; STFCAL passes $h_i - \langle h_i - \bar h, u_i\rangle u_i$, $u_i = (h_i - \bar h)/\max(\|h_i - \bar h\|, \varepsilon)$. Differentiate the resulting auxiliary loss through this map.
6: **Report:** gate, $r$, $\hat\rho$, $\mathcal{S}_h/\mathcal{S}^\star$

Table 13: The sixteen controlled toxifier cells behind Figure 7a. Every training auxiliary is a fixed projected-variance toxifier; the probe uses an isotropic variance surrogate. $\alpha_{\mathrm{probe}} = 1/\bar\rho$ is the inverse mean warm-up probe, while $\alpha_{1/2}$ interpolates the mean dose–scatter curve at relative scatter 0.5. Their ratio is a calibration-transfer diagnostic. Each probe value is a ten-seed mean of the archived warm-up records.

| Domain | Head | Width $m$ | Probe $\bar\rho$ | $\alpha_{\mathrm{probe}}$ | $\alpha_{1/2}$ | Ratio |
|---|---|---|---|---|---|---|
| HAR | toxifier | 32 | 30.2 | $3.32 \times 10^{-2}$ | $1.02 \times 10^{-3}$ | 32.5 |
| HAR | toxifier | 64 | 110.2 | $9.07 \times 10^{-3}$ | $1.41 \times 10^{-3}$ | 6.4 |
| HAR | toxifier | 128 | 552.0 | $1.81 \times 10^{-3}$ | $1.63 \times 10^{-3}$ | 1.1 |
| HAR | toxifier | 256 | 1504.9 | $6.65 \times 10^{-4}$ | $4.52 \times 10^{-3}$ | 0.15 |
| Battery | toxifier | 32 | 1.10 | $9.07 \times 10^{-1}$ | $9.51 \times 10^{-1}$ | 0.95 |
| Battery | toxifier | 64 | 0.71 | 1.42 | $8.95 \times 10^{-1}$ | 1.58 |
| Battery | toxifier | 128 | 0.85 | 1.18 | $9.03 \times 10^{-1}$ | 1.31 |
| Battery | toxifier | 256 | 0.28 | 3.56 | 1.18 | 3.01 |
| RadioML | toxifier | 32 | 0.14 | 7.13 | $1.73 \times 10^{-1}$ | 41.1 |
| RadioML | toxifier | 64 | 0.08 | 11.9 | $3.95 \times 10^{-1}$ | 30.3 |
| RadioML | toxifier | 128 | 0.05 | 19.8 | $2.41 \times 10^{-1}$ | 82.2 |
| RadioML | toxifier | 256 | 0.04 | 24.9 | $2.16 \times 10^{-1}$ | 115.6 |
| NYUv2 | toxifier | 32 | 2.26 | $4.43 \times 10^{-1}$ | $1.14 \times 10^{-1}$ | 3.9 |
| NYUv2 | toxifier | 64 | 2.01 | $4.98 \times 10^{-1}$ | $6.89 \times 10^{-2}$ | 7.2 |
| NYUv2 | toxifier | 128 | 1.64 | $6.11 \times 10^{-1}$ | $3.56 \times 10^{-2}$ | 17.1 |
| NYUv2 | toxifier | 256 | 1.29 | $7.78 \times 10^{-1}$ | $2.39 \times 10^{-2}$ | 32.6 |

The synthetic testbed uses 17 log-spaced doses, six widths, and random projections of rank $\lfloor m/2 \rfloor$. Its near-flat relative-crossing slope ($\gamma = 0.07$) contrasts with the absolute-threshold estimate (1.53), as analyzed above. In RadioML, a five-seed sweep over $\alpha \in [3\times10^{-3}, 1]$, refined near the transition, gives crossings $\{0.17, 0.39, 0.24, 0.22\}$ at widths $\{32, 64, 128, 256\}$. The eightfold width range changes the crossing by at most a factor of 2.3, much less than the factor of eight implied by $\gamma = 1$. At $m = 64$, per-seed crossings span 0.29–0.54. Together with the near-zero fit $R^2$, this variation limits inference about the width slope.

The repository supplies the scripts and recorded results listed in its reproduction manifest; data-dependent reruns additionally require the stated datasets and preprocessing. New audits are identified separately from historical experiments. The drivers `cross_domain_mtl.py`, `battery_bmtl_v3.py`, and `nyu_bmtl.py` implement the STF family for the unified, battery,

Table 14: Four scientific domains and the financial discovery application. Sample counts describe the listed experimental datasets; individual protocols specify their partitions. Controlled scientific experiments use variance-shrinkage auxiliaries. The financial application uses real heads, and the battery and NYUv2 application studies also evaluate practical reconstruction heads.

| Domain | Signal | Primary task | Samples | Metric |
|---|---|---|---|---|
| HAR (UCI-HAR) | motion (561-d) | 6-class activity | 7,352 / 2,947 | accuracy |
| RadioML (2016.10A) | I/Q (256-d) | 11-class modulation | 110,000 / 110,000 | accuracy |
| Battery (NASA+MATR) | charge/discharge | SOH + RUL | 18,696 / 3,210 | RMSE / MAE |
| Finance (A-share) | 1,680-d factors | 5-d return rank | 800 stocks $\times$ 2,059 d | rank-IC |
| NYUv2 | RGB-D (144$\times$192) | 13-class seg + depth + normals | 795 / 654 scenes | pxacc / mIoU / RMSE / angle |

Table 15: Financial application: mean OOS rank-IC $\pm$ population s.d. over 100 fold-seed cells (ten temporal folds, ten initialization seeds). Cells sharing a fold are dependent. After averaging seeds within folds and correcting three paired comparisons, no method differs significantly from joint training. These records use real auxiliary heads without the fixed variance toxifier.

| Mode | OOS rank-IC | vs. joint |
|---|---|---|
| JOINT | $+0.060 \pm 0.055$ | — |
| STF-0 (`.detach()`) | $+0.036 \pm 0.056$ | $-39\%$ |
| STF-HARD | $+0.063 \pm 0.052$ | $+5\%$ |
| STF-SOFT | $+0.056 \pm 0.053$ | $-7\%$ |

and dense-perception experiments. Algorithm 3 specifies the calibrated forward maps; each driver records application-specific details. PCGrad, CAGrad, and FAMO share these experiment pipelines, including the composition study. The remaining methods (MGDA, Aligned-MTL, Nash-MTL, IMTL-G, IMTL-L, GradNorm, and uncertainty weighting) use `moo_combiners.py`, which ports kernels from the authors' reference implementations. Comparisons use the recorded optimizer and schedule with published method defaults; batch-size exceptions are stated below.

JSON job manifests specify each experimental configuration by domain, width, mode, and seed, together with its command, output path, and log path. A deterministic runner skips existing outputs. The manifests cover collapse scans, width sweeps, filter composition, real auxiliary heads, battery weight sweeps, calibrated validation, NYUv2 experiments, and the three-domain multi-task baseline comparison. Experiments use fixed integer seeds, with counts reported for each study. Where available, JSON records retain individual seed outcomes; aggregate-only historical records are identified in the artifact audit.

The original unified runs use an NVIDIA A100 80 GB, Adam ($\mathrm{lr} = 10^{-3}$), bf16 autocast, and TF32 matrix multiplication. Calibration runs in fp32 outside autocast on detached features. Batch size is 256 for the unified protocol, real-head comparison, regularizer sweep, and MATR application; 128 for NASA conflict baselines versus 256 for its plain-summation runs; 32 for NYUv2; and 512 for financial walk-forward training. The NASA batch-size difference qualifies those historical comparisons. A unified-protocol configuration takes a few GPU-minutes; the battery application uses 150 epochs per configuration. Corrected battery runs use Adam at $2 \times 10^{-3}$, the parameter-constraint pilot uses SGD, and exact-rank controls use AdamW on CPU. Individual scripts and JSON configurations record these optimizer differences. Calibration records contain each seed's $\hat{\rho}$, residual statistics $\hat{\rho}_{\mathrm{res}}[0..R]$, gate, and selected rank. The script `calib_cert_b.py` repeats probes on a frozen healthy anchor at different batch sizes to produce Table 11.

Table 19 measures per-epoch training cost on UCI-HAR at four widths. We report the median of per-repetition median epoch times over ten repetitions, exclude epoch 0, and synchronize the device within each timed epoch. Runs begin after three consecutive idle-device readings, and method order rotates between repetitions to reduce timing drift. Contention materially affects the result: a non-idle run measured the reference at $0.180\,\mathrm{s}$ rather than $0.099\,\mathrm{s}$ and FAMO overhead at $0.0\%$ rather than approximately $31\%$.

On the idle device, hard and soft filtering cost at most $21.8\%$ on HAR. At $m = 256$, their overheads are $9.5\%$ and $12.2\%$, respectively. For the tested radial implementation, projection costs $O(Bd)$ compared with the encoder's $O(Bd^2)$; this accounting does not include arbitrary full-rank eigende-

Table 16: Historical MATR censoring ablation: SOH RMSE (%) and legacy RUL MAE (cycles), with leave-cell-out evaluation, $d = 128$, bf16 training, and fp32 evaluation. Legacy rows use four seeds; the censoring-aware row uses ten. Legacy supervision treats censored bounds as exact targets; the EOL-only variant additionally excludes $96/185$ cells, leaving 14 test cells. Censoring-aware supervision applies a one-sided hinge to the $59.2\%$ censored samples and retains all cells. Evaluation-selected checkpoints and legacy RUL scoring limit interpretation.

| | SOH RMSE (%) | | RUL MAE (cyc) | | |
|---|---|---|---|---|---|
| Mode | JOINT | STF-0 | JOINT | STF-0 | cells used |
| legacy (RUL=0 labels) | $5.56 \pm 0.17$ | $2.05 \pm 0.38$ | $94.1 \pm 5.7$ | $97.0 \pm 15.3$ | 185 |
| legacy + `--eol-only` | $1.53 \pm 0.13$ | $0.53 \pm 0.03$ | $77.5 \pm 2.3$ | $76.6 \pm 4.1$ | 89 |
| censoring-aware | $5.24 \pm 0.54$ | $1.68 \pm 0.41$ | $209.0 \pm 59.8$ | $574.7 \pm 80.7$ | 185 |

The legacy rows reproduce the `nocens` supervision, every other setting held fixed.

Table 17: Historical battery-head comparison at $\alpha = 10$ (five seeds): SOH RMSE (%), legacy all-sample RUL MAE (cycles), and "scatter" $\mathbb{E}_b[\mathrm{Var}_j(h_j^{(b)})]$ (sample correction $d - 1$, no centering across examples), not (10). PCGrad, FAMO, and CAGrad yield higher SOH error than detachment in both datasets. Scatter alone does not establish operational collapse; Table 4 reports separate controlled-toxifier experiments. These evaluation-selected results are exploratory.

| | MATR | | | NASA | | |
|---|---|---|---|---|---|---|
| Combiner | SOH RMSE (%) | RUL MAE | scatter | SOH RMSE (%) | RUL MAE | scatter |
| single-task reference | 0.51 | 694.6 | – | 9.17 | 18.5 | – |
| JOINT (sum, plain) | $6.43 \pm 0.19$ | 208.8 | 0.126 | $13.33 \pm 1.07$ | 18.7 | 0.214 |
| PCGrad | $4.02 \pm 0.98$ | 172.9 | 0.094 | $11.89 \pm 0.73$ | 18.6 | 0.164 |
| FAMO | $10.10 \pm 2.17$ | 295.3 | 0.435 | $16.02 \pm 0.50$ | 32.4 | 0.474 |
| CAGrad | $6.33 \pm 0.45$ | 198.0 | 0.166 | $13.04 \pm 0.75$ | 17.7 | 0.198 |
| STF-0 (detach) | $1.83 \pm 0.51$ | 623.4 | 0.198 | $8.81 \pm 0.72$ | 17.1 | 0.225 |

compositions. Detachment and calibrated radial filtering reduce measured time by up to $65.0\%$ and $47.8\%$. PCGrad, CAGrad, and FAMO cost 21.1–$41.6\%$, 38.5–$75.2\%$, and 19.6–$31.1\%$, respectively. Across five domains and four widths, spectral-variant overhead is 6.3–$26.0\%$, versus 5.8–$98.2\%$ for the comparison methods. Hard and soft filtering are cheaper than PCGrad and CAGrad in every measured configuration; FAMO is an exception to the broader ordering on CelebA and NYUv2 at $m = 256$. These are empirical timings for the specified implementations.

## G NYUv2: Dense Perception Beyond Time Series

NYUv2 (Silberman et al., 2012; Ladický et al., 2014) contains 795 training and 654 test scenes with semantic segmentation (13 classes), depth, and surface normals. Images are resized to $144 \times 192$; segmentation pixels labeled $-1$ and zero-norm normal targets are masked. The convolutional encoder uses four stride-2 blocks with batch normalization and ReLU ($3 \to 32 \to 64 \to 128 \to 256$ channels), followed by a $d = 128$ representation and linear task heads. Losses are cross-entropy, $L_1$ depth error, and cosine normal error. We report final-epoch pixel accuracy, mIoU, depth RMSE/MAE, and mean angular error over ten seeds, without test-based checkpoint selection. The *proj* auxiliary is a fixed random variance-shrinkage projection to $d/4$, evaluated at $\alpha = 1$. The *ae* auxiliary is a trainable three-layer input-reconstruction head, evaluated at $\alpha = 250$, or 3.4 times its probe-based threshold. A separate unified-protocol experiment uses an MLP and a $9 \times 12$ coarse depth grid (108 outputs), with the optimizer, probe, and collapse criterion of Section 4.1.

After four detached warm-up epochs, the unified MLP probe gives $\hat{\rho} = 2.26 \pm 0.16$, $2.01 \pm 0.10$, $1.64 \pm 0.09$, and $1.29 \pm 0.05$ at widths $32, 64, 128, 256$. Per-seed ranges are 2.06–2.65, 1.90–2.20, 1.51–1.77, and 1.22–1.38. The inverse-mean calibration values are 0.44, 0.50, 0.61, and 0.78, yielding a probe-based width slope of 0.27 ($R^2 = 0.98$, 95% CI $[+0.15, +0.40]$).

Table 18: Historical MATR auxiliary-weight sweep (seed 0): SOH RMSE (%) and legacy RUL MAE (cycles), with censoring-aware supervision and bf16 training/fp32 evaluation. Joint training uses reconstruction/distributional weight $\lambda_{\text{aux}}$; detachment makes encoder learning independent of this weight. SOH error generally increases with dose, although the $d = 32$ curve is not strictly monotone. Single-task rows are width-matched references; legacy RUL scores and evaluation-selected checkpoints are not independent lifetime-error estimates.

| | | SOH RMSE (%)↓ | | | RUL MAE (cyc)↓ | | |
|---|---|---|---|---|---|---|---|
| Mode | $\lambda_{\text{aux}}$ | $d$=32 | $d$=128 | $d$=256 | $d$=32 | $d$=128 | $d$=256 |
| JOINT | $10^{-2}$ | 2.72 | 1.36 | 1.52 | 478.0 | 386.3 | 377.5 |
| | $10^{-1}$ | 2.69 | 2.23 | 2.87 | 230.4 | 399.7 | 411.0 |
| | 1 | 5.96 | 4.69 | 5.53 | 285.6 | 198.0 | 176.4 |
| | 10 | 6.52 | 6.58 | 6.01 | 178.5 | 197.0 | 182.0 |
| STF-0 (dose-free) | — | 2.36 | 1.18 | 1.51 | 607.2 | 627.4 | 540.2 |
| single-task reference | — | 0.60 | 0.33 | 0.57 | 951.8 | 761.7 | 325.8 |

Table 19: UCI-HAR per-epoch overhead relative to plain summation, batch 256. Entries are medians over ten repetitions of median epoch time, excluding epoch 0 and synchronizing each epoch. Method order rotates after three idle-device readings. A paired-ratio estimator differs by at most five percentage points. The first row gives reference times; negative overhead indicates speedup for the measured implementation.

| Method | $m$=32 | $m$=64 | $m$=128 | $m$=256 |
|---|---|---|---|---|
| plain summation (s/epoch) | 0.0813 | 0.0986 | 0.1283 | 0.1454 |
| *overhead relative to plain summation* | | | | |
| STF-0 (detach) | −35.8% | −48.1% | −59.5% | −65.0% |
| STF-HARD | +21.8% | +18.1% | +14.3% | +9.5% |
| STF-SOFT | +21.8% | +19.8% | +8.9% | +12.2% |
| STFCAL | −0.7% | −20.1% | −37.7% | −47.8% |
| PCGrad | +41.6% | +31.4% | +26.6% | +21.1% |
| CAGrad | +75.2% | +60.2% | +46.0% | +38.5% |
| FAMO | +31.0% | +31.1% | +23.9% | +19.6% |

This is not the scatter-crossing exponent. Half-height crossings on the same widths yield $\gamma = -0.77$; varying the crossing level from 0.3 to 0.7 gives $[-0.94, -0.73]$ with $R^2 \geq 0.99$, whereas HAR remains positive ($+0.34$ to $+0.46$). At $m = 64$, the crossing and calibration values are 0.069 and 0.50, a factor of 7 apart. Extending the grid four decades lower restores relative scatter to 0.99–1.01 at $\alpha = 10^{-6}$, compared with 0.64–0.83 at the previous lowest dose 0.01, without materially changing the crossing. The opposite width trends therefore cannot be explained solely by the original grid's lower endpoint.

We report the probe because it determines the implemented gate, while retaining the crossing estimate as a separate diagnostic. The probe slope varies over a range of 0.17 across calibration times on NYUv2, versus 2.55 on HAR (Appendix H.3); this relative stability does not make it a capacity exponent. For the convolutional encoder, the *proj* probe gives $\hat{\rho} = 3.27 \pm 0.54$ (range 2.45–3.92), predicting approximately 0.31, while the *ae* probe gives $0.0136 \pm 0.0015$, predicting approximately 73.

Table 20 shows joint collapse in $10/10$ seeds for every tested $\alpha \geq 0.5$. Calibrated filtering achieves $0/10$ only for $\alpha \geq 1$, when all gates open and relative scatter is approximately 1.00; the recorded candidate rank is 3. At $\alpha = 0.5$, the gate opens in the five seeds with $\hat{\rho} > 2$, and all five avoid collapse (candidate rank 1). This is an empirical threshold comparison, subject to Corollary 7's scope restrictions. At lower doses, closed-gate outcomes are close to joint training.

The task metric can conceal shrinkage: joint depth RMSE falls overall from 1.335 to 1.092 (approximately 18%) while relative scatter falls from 0.62 to 0.04. Thus the high-dose joint model has lower depth error than the filtered models despite meeting the representation-collapse criterion. At every

Table 20: NYUv2 under the unified protocol (MLP, $m = 64$, ten seeds): depth RMSE in meters on a $9 \times 12$ grid and scatter relative to the same-seed single-task anchor. Calibration gives mean $\hat{\rho} = 2.01$ and inverse-mean threshold $0.50$. Recorded candidate ranks are 1 at $\alpha \leq 0.5$ and 3 at $\alpha \geq 1$; the radial STFCAL arm does not use this rank. Full detachment yields RMSE $1.353$ and relative scatter $1.000$ at every dose.

| $\alpha$ | 0.03 | 0.1 | 0.3 | 0.5 | 1 | 3 | 10 | 30 | 100 |
|---|---|---|---|---|---|---|---|---|---|
| JOINT RMSE | 1.335 | 1.336 | 1.316 | 1.307 | 1.305 | 1.280 | 1.221 | 1.100 | 1.092 |
| JOINT rel. scatter | 0.62 | 0.45 | 0.30 | 0.26 | 0.22 | 0.16 | 0.10 | 0.04 | 0.04 |
| JOINT collapsed | 0/10 | 0/10 | 5/10 | 10/10 | 10/10 | 10/10 | 10/10 | 10/10 | 10/10 |
| STFCAL RMSE | 1.339 | 1.327 | 1.314 | 1.328 | 1.354 | 1.353 | 1.353 | 1.354 | 1.354 |
| STFCAL rel. scatter | 0.63 | 0.45 | 0.31 | 0.63 | 1.01 | 1.01 | 1.01 | 1.00 | 1.01 |
| STFCAL collapsed | 0/10 | 0/10 | 4/10 | 5/10 | 0/10 | 0/10 | 0/10 | 0/10 | 0/10 |
| calibrated gate open | 0/10 | 0/10 | 0/10 | 5/10 | 10/10 | 10/10 | 10/10 | 10/10 | 10/10 |

Table 21: NYUv2 three-task comparison, shared convolutional encoder ($d = 128$), split, and ten seeds; final-epoch evaluation. Blocks give single-task and auxiliary-free references, projection auxiliary (*proj*, $\alpha = 1$), and reconstruction (*ae*, $\alpha = 250$). Depth error is in meters and normal error in radians. Recorded "scatter" is within-example $\mathbb{E}_b[\mathrm{Var}_j(h_j^{(b)})]$, with correction $d - 1$; it measures a different quantity from across-example scatter in (10). Absolute task scores depend on this small-backbone protocol.

| Configuration | seg pxacc | seg mIoU | depth RMSE | normal angle | scatter |
|---|---|---|---|---|---|
| single-seg | 0.359 | 0.083 | 2.780 | 1.574 | 2.22 |
| single-depth | 0.076 | 0.032 | 1.159 | 1.573 | 5.08 |
| single-normal | 0.074 | 0.031 | 3.346 | 0.593 | 12.43 |
| joint3 (no aux) | 0.356 | 0.100 | 1.130 | 0.586 | 7.30 |
| + proj aux ($\alpha$=1) | 0.365 | 0.090 | 1.085 | 0.580 | 1.64 |
| STF-0 | 0.356 | 0.100 | 1.131 | 0.585 | 7.30 |
| STFCAL | 0.357 | 0.100 | 1.134 | 0.586 | 7.37 |
| STFCALRANK | 0.357 | 0.095 | 1.115 | 0.583 | 2.65 |
| + ae aux ($\alpha$=250) | 0.350 | 0.092 | 1.168 | 0.580 | 3.99 |
| STF-0 | 0.357 | 0.099 | 1.133 | 0.586 | 7.19 |
| STFCAL | 0.338 | 0.091 | 1.171 | 0.594 | 3.84 |

tested $\alpha \geq 1$, paired Wilcoxon tests on relative scatter give $p \approx 2 \times 10^{-3}$, the smallest two-sided value attainable with ten nonzero, same-sign paired differences.

Table 21 compares dense tasks under a shared small-backbone protocol. Its recorded statistic is mean within-example feature variance, which differs from the across-example quantity in (10). With the projection auxiliary, the amplitude ratio is $0.22$: segmentation mIoU decreases while depth RMSE improves, and radial filtering returns amplitude near the auxiliary-free reference. With reconstruction at $\alpha = 250$, filtered amplitude is $3.84$, compared with $3.99$ for joint training and $7.19$ for detachment. These task and amplitude comparisons motivate an across-example covariance audit before assigning a collapse mechanism.

## H Supplementary Experimental Results

Ten-seed calibrated experiments support Section 5. At high doses, joint versus STFCAL relative scatter is $0.016$ versus $0.990$ for HAR ($\alpha = 100$), $0.147$ versus $1.001$ for battery ($\alpha = 100$), and

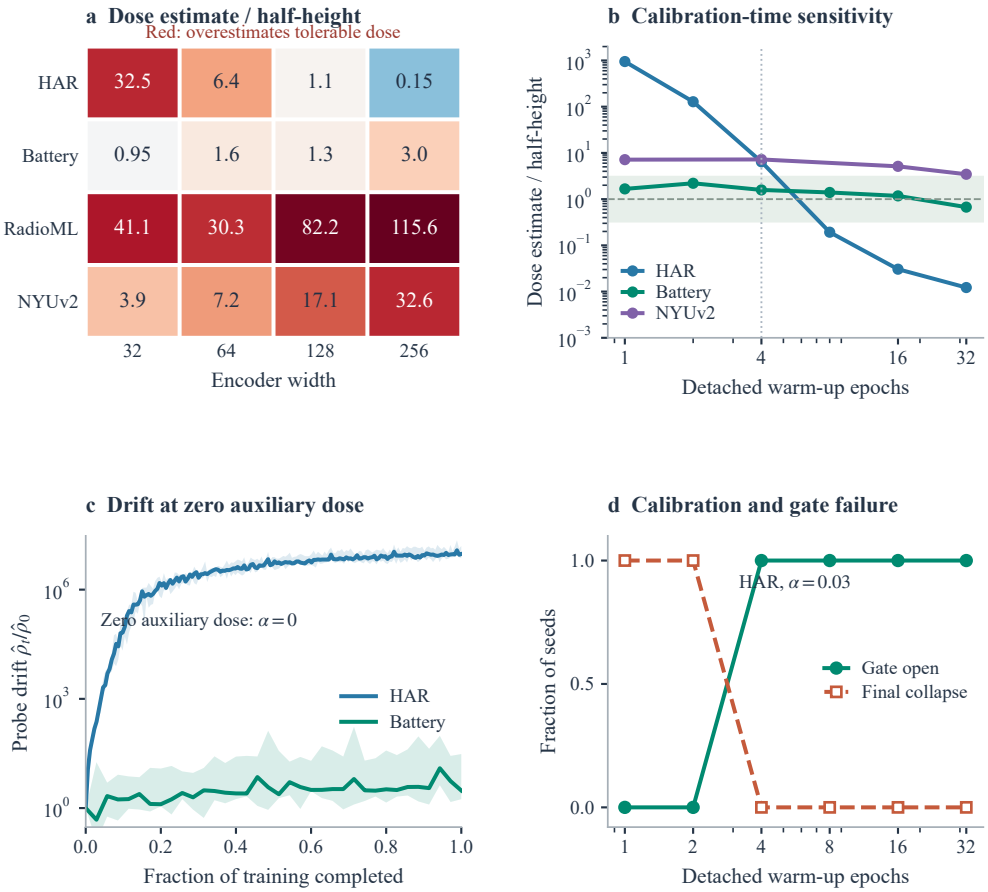



Figure 7: A frozen radial probe does not transfer uniformly. (a) $\alpha_{\text{probe}}/\alpha_{1/2}$ for 16 controlled domain–width cells. Red indicates an optimistic dose estimate; blue, a conservative one. (b) Calibration-time sensitivity at $m = 64$; shading marks a factor-3 reference band. Both panels use ten-seed mean probes. (c) Zero-dose probe drift: median and full range over five seeds, normalized to the initial value. (d) HAR at $\alpha = 0.03$: early calibration leaves all ten gates closed and all runs collapse; $W \geq 4$ opens every gate and prevents collapse. Values and crossing definitions are in Table 13 and Appendices H.2–H.3.

$0.014$ versus $0.982$ for RadioML ($\alpha = 3000$). In each case, collapse rates are $10/10$ versus $0/10$. At low doses with closed gates, outcomes are close: HAR at $\alpha = 0.001$ gives $0.525$ versus $0.528$, and battery at $\alpha = 0.01$ gives $0.963$ versus $0.965$.

With Transformer encoders at $\alpha = 100$, HAR scatter changes from $0.030$ to $1.041$ and battery scatter from $0.036$ to $1.000$, again reducing collapse from $10/10$ to $0/10$. Battery SOH RMSE improves from $0.447$ to $0.267$. On CelebA at $\alpha = 100$ (five seeds), scatter increases from $0.003$ to $0.934$ and collapse falls from $5/5$ to $0/5$.

The real-head experiments use $m = 64$ and ten seeds (Table 6). Neither reconstruction head collapses at its tested dose; filtering has little effect or modestly reduces primary accuracy. The training-subject classifier presents a different trade-off: joint training achieves identity accuracy $0.646$ against chance $1/21$, while filtering reduces it to $0.055$ and improves primary accuracy. This is a training-head measurement, not a held-out privacy attack. For high-dose SNR regression, joint training collapses in every seed; hard and soft filtering do not restore accuracy, and detachment performs best among the tested interventions. Probe magnitudes differ substantially across these heads, but the false negatives discussed in Section 4.4 preclude a universal decision rule. Figure 9 shows the

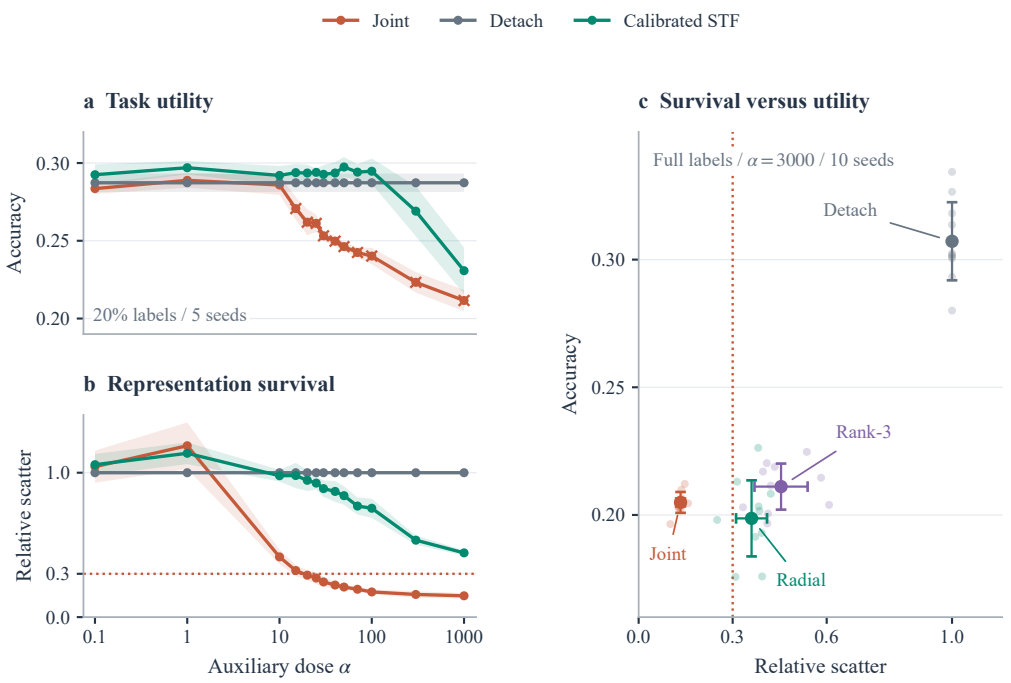



Figure 8: Filtering can retain an auxiliary gain, but scatter protection alone is insufficient. RadioML SNR regression, width 64. (a,b) With 20% primary labels, STF exceeds detachment at intermediate doses; the gain disappears at extreme doses despite scatter above 0.3. Five unique seeds per dose; bands show seed SD and crosses mark at least one collapsed joint run. (c) In the separate fully labeled dose-3000 protocol, rank-3 filtering avoids collapse in all ten seeds but underperforms detachment. Small points are seeds; large points and bars show means and SD. The dotted line marks the 0.3 criterion.

Table 22: Intervention families and their evaluation requirements. Performance comparisons under the stated implementations and protocols appear in Tables 4 and 25.

| Family | Intervention | Relevant limitation |
|---|---|---|
| Task weighting (Chen et al., 2018; Liu et al., 2023) | scalar auxiliary strength | positive attenuation can help at a fixed dose |
| Gradient geometry (Yu et al., 2020; Liu et al., 2021a; Navon et al., 2022) | task-gradient combination | alignment does not determine variance change |
| Variance regularization (Bardes et al., 2022) | representation penalty | penalty weight and primary-task trade-off |
| Full detachment | remove auxiliary encoder gradient | also removes useful auxiliary learning |
| Selective filtering | remove selected directions | basis identification and subsequent drift |
| Parameter variance constraint | constrain the actual parameter update | fixed guard batch; finite-step feasibility must be checked |

battery repair outcome, Table 23 the corresponding collapse and recovery rates, and Table 22 the intervention families.

The results identify several distinct regimes: task error can improve as scatter shrinks, real auxiliaries can remain benign over the tested grid, and calibration can miss contraction. Frozen-basis error and sampling variation provide additional diagnostic checks. Numerical instability and loss of plasticity (Dohare et al., 2024) remain outside the fixed-operator model.

Table 23: Controlled high-dose collapse and task-recovery rates for three domains. Recovery is the fraction of seeds whose primary metric improves over the mean joint-training metric. HAR can retain high accuracy despite substantial shrinkage, so scatter recovery does not imply better accuracy. Battery uses 20 seeds per mode and RadioML ten; the historical HAR file provides aggregate rates. The joint recovery rate is set to zero by convention. All plotted rates are reproduced directly from the released records.

| Domain | Metric | JOINT coll. / rec. | STF-0 coll. / rec. | STF-HARD coll. / rec. | STF-SOFT coll. / rec. |
|---|---|---|---|---|---|
| HAR | accuracy (flat) | 1.0 / 0.0 | 0.0 / 0.1 | 0.0 / 0.2 | 0.0 / 0.1 |
| RadioML | accuracy | 1.0 / 0.0 | 0.0 / 1.0 | 0.0 / 1.0 | 0.0 / 1.0 |
| Battery | log-SOH RMSE | 1.0 / 0.0 | 0.0 / 1.0 | 0.0 / 1.0 | 0.0 / 1.0 |

The battery encoder maps a per-cycle window $x \in \mathbb{R}^{T\times D}$ of normalized voltage, current, temperature, and resistance to $h \in \mathbb{R}^d$. It combines a TSMIXER feature/temporal-mixing block (Ekambaram et al., 2023) with a gated state-space block (Gu et al., 2022; Gu & Dao, 2024). SOH ($L_2$), RUL ($L_1$), reconstruction, and distributional heads share this representation. The real auxiliary heads are evaluated empirically and are not assumed to satisfy Definition 2.

RUL targets require careful handling of right censoring. If a cell has not reached end of life when recording stops, its remaining lifetime is known only to exceed a lower bound. Treating that bound as an exact target can label a healthy cell with zero remaining life. In MATR, $59.2\%$ of samples are censored and $96/185$ cells do not reach EOL within their records. Initialization zeros in summary arrays cause a further $48/185$ cells to receive an EOL index of zero under a naive threshold search. Restricting the search to each trajectory's physically valid suffix recovers these cells. We retain censored cells and use $|\hat{r} - r|$ for observed lifetimes and $\max(0, r_{\mathrm{lb}} - \hat{r})$ for censored lower bounds. This one-sided loss is motivated by censored regression (Tobin, 1958), but is not a Tobit likelihood. Excluding cells without observed EOL would leave only 14 test cells in the historical split (Section 5).

To emphasize fast-charge, extreme-temperature, and rapid-degradation windows, BatteryMTL uses three multiplicative weighting factors:

$$\begin{aligned} w_i &= w_i^{(\mathrm{rate})} \cdot w_i^{(\mathrm{temp})} \cdot w_i^{(\mathrm{degrad})}, \\ w_i^{(\mathrm{rate})} &= 1 + \omega\, \mathbf{1}[\text{C-rate}_i > r_q], \\ w_i^{(\mathrm{temp})} &= 1 + \omega\, \mathbf{1}[T_i < t_{lo} \vee T_i > t_{hi}], \\ w_i^{(\mathrm{degrad})} &= 1 + \omega\, \tfrac{|\Delta\mathrm{SOH}_i|}{\max_j |\Delta\mathrm{SOH}_j|}, \end{aligned} \tag{25}$$

Equation (25) amplifies windows in which several factors are elevated, while also up-weighting windows with a single elevated factor. Thresholds are estimated from the training fold: $r_q$ is the 70th C-rate percentile, and $t_{lo}, t_{hi}$ are the 10th and 90th temperature percentiles. These relative thresholds distinguish regimes within MATR, where $99.8\%$ of cycles exceed $1.5$C and temperatures lie within 31–36°C. In contrast, fixed cuts at $1.5$C and $0/40$°C would mark nearly all cycles for rate and none for temperature. Section 5 reports regime-specific ablations; SOH-only and RUL-only models provide task-specific references, not guaranteed performance ceilings.

### H.1 Composition and Regime Weighting

The following measurements expand the composition and weighting results in Sections 5 and 5.

At the tested high doses, applying PCGrad or CAGrad to the unfiltered battery branch leaves collapse rate at $1.0$ and relative scatter at $0.029/0.030$ (Table 24). Hard filtering alone reduces collapse to $0.0$, restores relative scatter to $1.00$, and improves log-SOH RMSE from $0.611$ to $0.294$. Adding PCGrad or CAGrad afterward gives RMSE $0.283/0.291$ and scatter up to $1.18$. HAR and RadioML show the same qualitative preservation of scatter under composition. These results establish compatibility

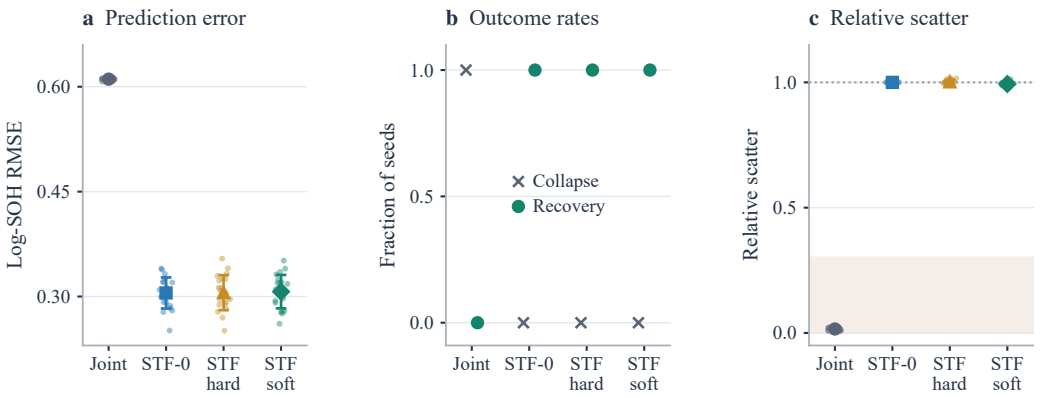



Figure 9: Controlled battery contraction can be prevented by filtering. NASA PCoE unified protocol, $m = 128$, $\alpha = 10^4$, 20 seeds per mode. (a) Mean log-SOH RMSE falls from 0.611 to about 0.305. (b) Collapse and task-recovery rates. (c) Scatter relative to the same-seed single-task reference; the dotted line marks 1.0 and shading the region below 0.3. Small points are seeds; large markers and whiskers show means and population SD. Log-SOH error and percentage-point SOH error in the separate application have different units.

Table 24: Filter-then-combiner experiments at high auxiliary doses (five seeds, three domains). Unfiltered branches retain collapse rates of 1.0 and relative scatter at most 0.15. The hard filter, alone or followed by PCGrad/CAGrad, avoids the collapse criterion in every seed. Primary metrics change modestly after composition; the reverse order was not tested. Bold identifies the filter-only reference within each block.

| | | unfiltered (JOINT) | | | filtered (STF-HARD) | | |
|---|---|---|---|---|---|---|---|
| | combiner | sum | PCGrad | CAGrad | sum | PCGrad | CAGrad |
| Battery | log-SOH RMSE↓ | 0.611 | 0.592 | 0.603 | **0.294** | 0.283 | 0.291 |
| | collapse rate | 1.0 | 1.0 | 1.0 | 0.0 | 0.0 | 0.0 |
| | relative scatter | 0.02 | 0.03 | 0.03 | **1.00** | 1.05 | 1.18 |
| HAR | accuracy↑ | 0.941 | 0.667 | 0.952 | 0.943 | 0.933 | 0.940 |
| | collapse rate | 1.0 | 1.0 | 1.0 | 0.0 | 0.0 | 0.0 |
| | relative scatter | 0.15 | 0.00 | 0.03 | **0.97** | 1.14 | 1.03 |
| RadioML | accuracy↑ | 0.293 | 0.091 | 0.109 | **0.301** | 0.296 | 0.295 |
| | collapse rate | 1.0 | 1.0 | 1.0 | 0.0 | 0.0 | 0.0 |
| | relative scatter | 0.02 | 0.00 | 0.00 | **0.97** | 0.99 | 1.04 |

for the tested order; they neither show that sequencing is necessary nor establish why the unfiltered combiners fail.

BatteryMTL's multiplicative regime weights use training-data percentiles for C-rate and temperature, together with degradation rate. The aggregate SOH results change little, while factor ablations associate improvements in fast-charge and extreme-temperature windows with the rate and temperature terms, respectively (errors 0.95 and 2.35 when the corresponding terms are removed). Table 10 reports the historical aggregate results.

Table 25: Anti-collapse regularizers at the doses and seeds of Table 4 (ten seeds per configuration). Penalties of weight $\lambda$ are added to joint training. Entries report collapse rate, primary metric in parentheses, and relative scatter (1.0 is the single-task reference). Joint training, detachment, and hard filtering are evaluated in the same runs. Avoiding the scatter threshold need not improve the primary task.

| | HAR | | Battery | | RadioML | |
|---|---|---|---|---|---|---|
| | coll. / acc.↑ | rel. scat. | coll. / RMSE↓ | rel. scat. | coll. / acc.↑ | rel. scat. |
| JOINT | 1.00 (0.945) | 0.15 | 1.00 (0.610) | 0.022 | 1.00 (0.256) | 0.008 |
| STF-0 | 0.00 (0.945) | 1.00 | 0.00 (0.297) | 1.00 | 0.00 (0.341) | 1.00 |
| STF-HARD | 0.00 (0.946) | 0.99 | 0.00 (0.299) | 1.00 | 0.00 (0.339) | 0.97 |
| *VICReg (variance + covariance)* | | | | | | |
| $\lambda$=1 | 1.00 (0.950) | 0.22 | 1.00 (0.610) | 0.023 | 1.00 (0.249) | 0.007 |
| $\lambda$=3 | 0.20 (0.946) | 0.31 | 1.00 (0.610) | 0.023 | 1.00 (0.253) | 0.007 |
| $\lambda$=10 | 0.00 (0.944) | 0.36 | 1.00 (0.609) | 0.024 | 1.00 (0.256) | 0.008 |
| $\lambda$=30 | 0.00 (0.926) | 0.36 | 1.00 (0.604) | 0.033 | 1.00 (0.260) | 0.008 |
| $\lambda$=100 | 0.00 (0.902) | 0.36 | 1.00 (0.451) | 0.14 | 1.00 (0.301) | 0.011 |
| $\lambda$=300 | 0.00 (0.873) | 0.36 | 0.00 (0.339) | 0.59 | 1.00 (0.362) | 0.035 |
| $\lambda$=$10^3$ | 0.00 (0.843) | 0.36 | 0.00 (0.390) | 1.95 | 1.00 (0.320) | 0.15 |
| $\lambda$=$3 \times 10^3$ | 0.00 (0.831) | 0.36 | 0.00 (0.423) | 3.70 | 0.10 (0.207) | 0.45 |
| $\lambda$=$10^4$ | 0.00 (0.821) | 0.36 | 0.00 (0.423) | 4.54 | 0.00 (0.215) | 1.29 |
| *Barlow Twins (redundancy reduction)* | | | | | | |
| $\lambda$=1 | 1.00 (0.935) | 0.11 | 1.00 (0.607) | 0.11 | 1.00 (0.367) | 0.016 |
| $\lambda$=3 | 1.00 (0.765) | 0.022 | 1.00 (0.607) | 0.13 | 1.00 (0.330) | 0.024 |
| $\lambda$=10 | 1.00 (0.435) | 0.006 | 1.00 (0.606) | 0.15 | 1.00 (0.121) | 0.074 |
| $\lambda$=30 | 1.00 (0.352) | 0.006 | 1.00 (0.606) | 0.16 | 1.00 (0.105) | 0.096 |
| $\lambda$=100 | 1.00 (0.338) | 0.006 | 1.00 (0.605) | 0.16 | 1.00 (0.126) | 0.072 |
| $\lambda$=300 | 1.00 (0.347) | 0.006 | 1.00 (0.606) | 0.16 | 1.00 (0.141) | 0.043 |
| $\lambda$=$10^3$ | 1.00 (0.343) | 0.006 | 1.00 (0.606) | 0.16 | 1.00 (0.140) | 0.042 |
| $\lambda$=$3 \times 10^3$ | 1.00 (0.365) | 0.006 | 1.00 (0.607) | 0.16 | 1.00 (0.148) | 0.053 |
| $\lambda$=$10^4$ | 1.00 (0.339) | 0.006 | 1.00 (0.607) | 0.16 | 1.00 (0.153) | 0.044 |

### H.2 PROBE DYNAMICS DURING TRAINING

The frozen calibration in Section 5 is motivated by the behavior of a continuously recomputed probe. We repeat unfiltered joint training with `--rho-trace`, using five seeds per configuration over 1730 HAR steps and 350 battery steps. Monitoring does not alter the measured trajectories: all 28 shared BSCALE configurations match exactly in task metric, collapse rate, relative scatter, and $\hat{\rho}$ (maximum difference zero).

Three observations matter. First, HAR probe values rise sharply during training: from $0.017$ to $1.9 \times 10^5$ at $m = 32$, $0.010$ to $1.0 \times 10^5$ at $m = 64$, and $0.006$ to $3.5 \times 10^4$ at $m = 256$. Second, similar terminal values occur when the toxifier is inactive ($\alpha = 0$) and at the lowest positive dose ($10^{-4}$), so a large probe value alone does not establish auxiliary-induced contraction. At $m = 32$, increasing the dose instead lowers the terminal probe, to $5.3 \times 10^2$ at $10^{-2}$ and $1.1 \times 10^2$ at $3 \times 10^{-2}$. Third, in ten HAR configurations whose final scatter lies between $0.50$ and $1.04$ of the reference, the gate $\alpha\hat{\rho} \geq 1$ opens in every seed between steps 110 and 590. A gate using this running statistic therefore activates on runs that do not meet the collapse criterion at the recorded endpoint.

Battery behaves differently: probe values remain $O(1)$, ranging from $0.29$–$0.49$ initially to $0.3$–$2.9$ at the final step. Among 35 seeds whose final scatter lies above the half-height level, the gate opens only once, at the last step. These domain-dependent trajectories motivate a specified calibration time; they do not establish that a fixed probe is universally preferable to every adaptive rule. The script `repro/scripts/rho_trace_analyze.py` reproduces both analyses.

### H.3 Sensitivity to Calibration Time

The default calibration measures $\hat{\rho}$ after four detached warm-up epochs and fixes the gate and rank. We vary only this measurement time: $W \in \{1, 2, 4, 8, 16, 32\}$ at five widths 32–512 for HAR and battery, and $W \in \{1, 4, 16, 32\}$ at four widths for NYUv2, with ten seeds per cell. The $W = 4$ configuration exactly reproduces all 100 shared HAR and battery probe values. NYUv2 gives $2.258/2.008/1.637/1.285$, matching the rounded values in Appendix G.

Probe values depend on the training state. Across the warm-up sweep, HAR at $m = 32$ increases from $0.63$ to $6.9 \times 10^4$, and at $m = 512$ from $51.9$ to $1.6 \times 10^4$. Battery remains $O(1)$: $0.96$ to $1.92$ at $m = 32$ and $0.058$ to $0.67$ at $m = 512$. NYUv2 at $m = 32$ changes from $1.94$ to $4.19$, a factor of 2.2. Accordingly, width slopes fitted to $1/\hat{\rho}$ vary from $-2.00$ to $+0.55$ for HAR, $+0.37$ to $+0.97$ for battery, and $+0.11$ to $+0.27$ for NYUv2. These are calibration-dependent probe slopes, not interchangeable estimates of capacity scaling. Differences in primary-gradient magnitude, which appears in the denominator, can contribute to this sensitivity.

Measurement time also changes gate activation. For HAR at $m = 64$, the inverse-mean probe value is $1.34$ at $W = 1$, $0.181$ at $W = 2$, $9.07 \times 10^{-3}$ at $W = 4$, and $1.7 \times 10^{-5}$ at $W = 32$, compared with the measured transition interval $[0.001, 0.01]$. In the sixty-epoch, ten-seed experiment at $\alpha = 0.03$, gates remain closed for $W = 1$ and $W = 2$: both settings collapse in $10/10$ seeds, with accuracy $0.947$ and relative scatter $0.19/0.20$, close to joint training $(0.949, 0.19)$. For $W \geq 4$, all gates open, no seed collapses, and scatter returns to $0.99$–$1.00$. At $\alpha = 100$, all six warm-up settings activate the gate and avoid collapse.

A short warm-up therefore fails at the marginal dose in this experiment. Detached warm-up also changes the training trajectory, so closed-gate runs need not equal joint training from initialization. Checking whether $\alpha\hat{\rho} \geq 1$ verifies activation at the chosen operating dose; it does not independently validate protection. The script `repro/scripts/warmup_sensitivity.py` reproduces these results.

### H.4 Scale-Invariant Audit of the Scatter Criterion

The operational criterion in (10) is a coordinate-dependent, scale-sensitive proxy. To measure what a contraction preserves, we run the released HAR protocol at three doses and audit the frozen representation with covariance geometry and linear probes on raw, per-dimension-standardized, whitened, 10%-label, and new-label-space features. Thirty cells use $m = 64$, 60 epochs, Adam ($10^{-3}$, batch 256), and seeds 0–9; the driver is `repro/scripts/representation_audit.py`, and the records are in `results/REPAUDIT/`.

Table 26: Scale-invariant audit of the controlled HAR toxifier. Relative scatter uses the same-seed healthy reference. Covariance trace is absolute; participation rank (PR), effective (entropy) rank, and standardized probes are scale-invariant. At $\alpha = 100$ the scatter criterion records $10/10$ collapses, while the standardized probe and the new-label-space probe are unchanged and PR rises.

| $\alpha$ | rel. scatter | cov. trace | PR | eff. rank | standardized probe | primary acc. |
|---|---|---|---|---|---|---|
| 0 (healthy) | 1.000 | 578.7 | 2.00 | 2.57 | 0.947 | 0.941 |
| 0.1 | 0.130 | 8.51 | 4.97 | 5.52 | 0.958 | 0.956 |
| 100 | 0.016 | 0.125 | 4.90 | 5.02 | 0.955 | 0.952 |

The trace falls by $4{,}600\times$ and every seed meets the scatter criterion, yet per-dimension standardisation preserves activity-probe accuracy. A static-versus-dynamic activity probe stays at $0.999$. A one-hidden-layer MLP probe (64 units, 120 epochs) similarly reaches $0.945$ at $\alpha = 0$ and $0.954$ at $\alpha = 100$ over five seeds. These measurements show a large change in feature amplitude with retained access to the measured task information.

### H.5 Small-Dose Operator Audit

The main-text comparison holds each operator at the domain's supercritical dose. The complementary question is whether the default scalarizations contract the representation when the auxiliary weight is far below the calibrated threshold. Table 27 reports

five operators at three doses on HAR ($m = 64$, 60 epochs, three seeds); the driver is `repro/scripts/lowalpha_operator_audit.py`, and the records are in `results/LOWALPHA/`.

Table 27: Accuracy (scatter-collapse count over three seeds) at small auxiliary doses, HAR, $m = 64$. Plain summation is healthy at $\alpha = 0.001$, while the default scalarizations contract in every seed. This is operator conditioning rather than toxifier dose.

| $\alpha$ | joint | MGDA | IMTL-G | Aligned | Nash | FAMO |
|---|---|---|---|---|---|---|
| 0.001 | 0.943 (0/3) | 0.551 (3/3) | 0.242 (3/3) | 0.927 (3/3) | 0.425 (3/3) | 0.176 (3/3) |
| 0.01 | 0.950 (2/3) | 0.409 (3/3) | 0.301 (3/3) | 0.884 (3/3) | 0.461 (3/3) | 0.176 (3/3) |
| 0.1 | 0.953 (3/3) | 0.182 (3/3) | 0.183 (3/3) | 0.832 (3/3) | 0.565 (3/3) | 0.181 (3/3) |

At $\alpha = 0.001$, the tested operators contract features even though the auxiliary gradient is tiny and plain summation remains healthy. The low-dose behavior therefore depends on the default scalarization as well as the auxiliary dose. MGDA still collapses after lowering the learning rate to $10^{-4}$ (Table 28), which rules out the tested step-size reduction as a remedy. The fixed-dose comparisons establish the outcomes of these implementations; identifying their general performance requires separate optimization studies.

Table 28 compares the default learning rate with a tenfold smaller one at $\alpha = 0.001$. The loss-weighting families (UW, GradNorm, IMTL-L) become healthy at this low dose, as does plain summation; the gradient-geometry operators (PCGrad, CAGrad, FAMO, IMTL-G, Nash, Aligned, MGDA) collapse in every seed at both learning rates, and several of their accuracies worsen at the lower rate. The low-dose contraction is therefore operator-specific, not a single step-size artifact. At the high doses of Table 4, all ten operators collapse.

Table 28: Small-dose operator audit at $\alpha = 0.001$, HAR, $m = 64$, three seeds. Accuracy (scatter-collapse count) at the default $10^{-3}$ and a tenfold smaller $10^{-4}$ learning rate. The weighting families become healthy at low dose; the gradient-geometry operators collapse at both rates.

| Method | $lr = 10^{-3}$ | $lr = 10^{-4}$ |
|---|---|---|
| joint (sum) | 0.943 (0/3) | 0.934 (0/3) |
| PCGrad | 0.660 (3/3) | 0.656 (3/3) |
| CAGrad | 0.954 (3/3) | 0.953 (3/3) |
| FAMO | 0.176 (3/3) | 0.176 (3/3) |
| UW | 0.941 (0/3) | 0.935 (0/3) |
| GradNorm | 0.954 (0/3) | 0.934 (0/3) |
| IMTL-L | 0.944 (0/3) | 0.935 (0/3) |
| IMTL-G | 0.242 (3/3) | 0.187 (3/3) |
| Nash-MTL | 0.425 (3/3) | 0.338 (3/3) |
| Aligned-MTL | 0.927 (3/3) | 0.790 (3/3) |
| MGDA | 0.551 (3/3) | 0.186 (3/3) |

### H.6 Readout Transfer Under Contraction

The next experiment measures the compatibility of a readout fitted to healthy features. For five HAR seeds, we train healthy ($\alpha = 0$) and contracted ($\alpha = 100$) encoders and split training examples into readout-fitting and calibration subsets. Table 29 evaluates four maps on held-out test data: the healthy reference; the unchanged healthy head and statistics on contracted features; the same head with input statistics refitted without new labels; and a new head fitted to the contracted features.

Proposition 1 isolates the scale mechanism. A unique bias-only winner gives a constant predicted class at sufficiently small positive scale; ties need a specified convention. Fixed input centering contributes to the effective offset even with a bias-free head. Figure 10 summarizes finite $c_i^{\mathrm{bias}}$ by their within-seed median and then the median across seeds. This statistic describes bias exit, distinct from the first prediction flip when shrinking from $c = 1$. The standard-deviation floor is inactive over the plotted range; exact invariance for every $c > 0$ assumes unfloored statistics.

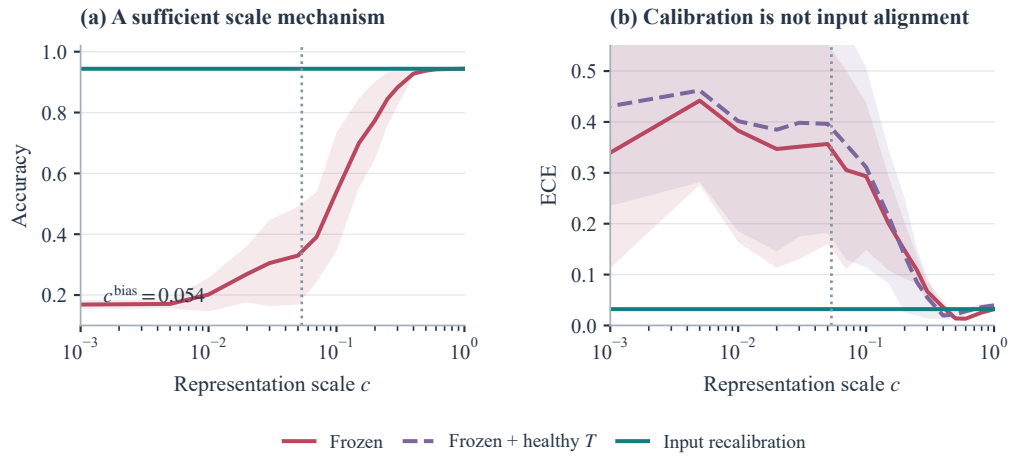



Figure 10: Pure scale change can invalidate a fixed affine readout. Healthy HAR features are multiplied by $c$ before readout. (a) Frozen accuracy degrades; refitting input statistics cancels the imposed scale change. (b) ECE before and after applying the healthy temperature map. Five seeds, $m = 64$; bands show seed SD. The dashed line summarizes finite bias-exit thresholds: median of seed medians $c^{\text{bias}} = 0.054$, seed range 0.020–0.207. Bias exit and the first flip when shrinking from $c = 1$ are distinct boundaries.

Table 29: Historical transfer between separately trained healthy and contracted HAR encoders (five seeds, means). "Frozen" applies the healthy head and input statistics unchanged; the reported ECE is the raw softmax ECE, and applying the healthy temperature map raises frozen ECE to 0.512. Input recalibration reuses the weights but refits the feature statistics; retraining fits a new head. The frozen readout degrades with large NLL/ECE, while cheap input recalibration recovers accuracy but not the NLL of a retrained head. The last two rows apply the same healthy readout to the STF-0 and STF-cal encoders.

| Readout | accuracy | NLL | ECE |
|---|---|---|---|
| healthy reference | 0.945 | 0.187 | 0.031 |
| frozen (healthy statistics) | 0.241 | 2.516 | 0.430 |
| input recalibration | 0.952 | 0.396 | 0.043 |
| retrained on contracted features | 0.955 | 0.190 | 0.034 |
| STF-0 encoder (same readout) | 0.946 | 0.188 | 0.031 |
| STF-cal encoder (same readout) | 0.943 | 0.201 | 0.033 |

The historical readout-transfer study trains healthy and contracted encoders separately from one initialization, so it measures compatibility across fitted models rather than a continuation. Its calibration subset is withheld from readout fitting but participates in supervised encoder training. Historical nominal-90% conformal coverage (0.802 healthy, 0 frozen, 0.806 recalibrated, 0.834 retrained) is therefore descriptive, not a split-conformal guarantee; its quantile also omits the finite-sample order-statistic correction. The paired continuation study below excludes calibration examples from encoder and head fitting and uses the corrected quantile. The official HAR subject split still changes the test population, so we report empirical coverage without assuming calibration–test exchangeability.

The unchanged readout models a fixed deployment interface behind a changing encoder. Its NLL and ECE rise sharply; input-statistic recalibration restores accuracy and most of ECE, while a new head also restores NLL. Applying the same healthy readout to STF-0 and STF-cal encoders gives 0.946 and 0.943 accuracy (`stf_readout_audit.py`). The gap also appears at $m = 32$ (frozen 0.236, recalibrated 0.887, refitted 0.909), $m = 128$ (0.166, 0.955, 0.954), and with a Transformer (0.162, 0.950, 0.952). These tested widths and encoders exhibit the same separation between decodabil-

Table 30: Historical real-head transfer between separately trained encoders. Frozen reuses the healthy head and statistics; recalibration refits statistics; retrained fits a new head; detached reuses the healthy readout on the STF-0 encoder. Relative $S$ is the same-seed scatter ratio; five seeds for SNR and subject-ID, three for I/Q.

| Head | $\alpha$ | rel. $S$ | healthy | frozen | recal. | retrained | detached |
|---|---|---|---|---|---|---|---|
| RadioML SNR | 3000 | 0.137 | 0.315 | 0.099 | 0.094 | 0.229 | 0.315 |
| RadioML I/Q | 3000 | 1.053 | 0.312 | 0.195 | 0.195 | 0.351 | 0.312 |
| HAR subject-ID | 10 | 0.932 | 0.944 | 0.384 | 0.466 | 0.901 | 0.944 |

ity and compatibility. Driver: `repro/scripts/readout_transfer_audit.py`; records: `results/READOUT/`.

### H.7 Paired Checkpoint Continuation

The paired experiment updates the encoder behind an already fitted consumer. We use the official HAR subject split and a prepared RadioML split with the same three-layer, width-64 MLP; the RadioML source-split audit appears below. A class-stratified $20\%$ training reservation is made before any encoder update for calibration. The remaining $80\%$ fits the encoder, primary head, and replacement probes. Reference readouts use per-feature standardization and the historical regularized LBFGS fit. Temperature fitting follows the fixed reference predictor; raw conformal scores use the corrected $\lceil (n+1)0.9 \rceil$ order statistic. Because HAR separates subjects between training and test, we report empirical coverage without assuming calibration–test exchangeability.

Each seed first trains a healthy encoder for 60 HAR or 25 RadioML epochs. A real SNR head is trained on detached features during this warm-up. We clone all model weights and Adam moments, retain the fitted consumer readout, and continue each branch for 30 HAR or 15 RadioML epochs. The learning rate is $10^{-3}$, batch size 256, and every operation uses CPU fp32 except the double-precision linear readout fit and evaluation. Future minibatch permutations are generated once per seed and reused for joint training, detachment, STF-cal, and primary-only continuation. Initial parameter, optimizer-state, and minibatch-order hashes verify the pairing; all detached encoder and primary-head endpoints match primary-only continuation exactly. Evaluation never selects a checkpoint. The isotropic variance objective matches the historical readout audit; a fixed projected-variance objective tests the benchmark operator separately. The first five HAR variance seeds are followed by a predeclared replication on seeds 5–9, with the recipe unchanged and all outcomes retained.

Table 31 gives every sampled dose for both variance objectives. At isotropic $\alpha = 3$, frozen accuracy is $0.453$, current-head accuracy $0.948$, and recalibrated accuracy $0.947$. Continuation thus causes a large compatibility loss while task decodability remains high. The fitted scalar residual shows additional feature drift, which matters at $\alpha = 100$: recalibration reaches $0.829$ versus $0.935$ after refitting. The imposed affine sweep and the trained continuations provide complementary tests of the scale component.

The SNR continuation provides a protocol boundary. At $\alpha = 3, 10, 30$, frozen accuracy is $0.293$, $0.269$, and $0.243$, against $0.304$ detached. At 30, recalibration reaches $0.253$, refitting $0.285$, and relative scatter $0.870$. The severe collapse seen in historical from-initialization training is absent at these nominal doses after healthy warm-up. Calibration split, auxiliary state, warm-up, and total horizon also differ; the head-state factorial below tests one of these factors.

At the first minibatch of specified continuation epochs, we evaluate $V$ and its gradient $v$ on a fixed 256-example fitting batch. Loss gradients use the actual update minibatch. We record flow $r - \alpha c$, the applied increment $s_\theta$, $v^\top s_\theta$, and the exact post-step change in $V$. A primary-only Adam counterfactual starts from identical pre-step weights and moments; its difference from the joint step measures the auxiliary's conditional effect at that optimizer state. Across SNR joint branches, flow and actual-step signs disagree on $33/75$ sampled updates. Table 32 includes negative-$c$ checkpoints with a contracting auxiliary counterfactual, demonstrating why flow attribution and applied-step attribution need separate records.

Table 31: Paired checkpoint continuation with held-out calibration. Means ± population seed standard deviations. Each branch starts from the same healthy model and Adam state; the future minibatch order is shared. Trace is relative to the initial healthy checkpoint. HAR uses 60 healthy plus 30 continuation epochs; RadioML uses 25 plus 15. The real auxiliary head is trained on detached features during warm-up.

| Case / dose | arm | frozen | input recal. | retrained | own | rel. trace |
|---|---|---|---|---|---|---|
| *HAR variance, 10 seeds* | | | | | | |
| 0 | primary only | $0.943 \pm 0.007$ | $0.943 \pm 0.006$ | $0.945 \pm 0.005$ | $0.944 \pm 0.005$ | 1.0897 |
| 0.1 | joint | $0.789 \pm 0.161$ | $0.951 \pm 0.005$ | $0.953 \pm 0.002$ | $0.951 \pm 0.007$ | 0.0213 |
| 0.1 | STF-cal | $0.943 \pm 0.007$ | $0.943 \pm 0.006$ | $0.945 \pm 0.005$ | $0.944 \pm 0.005$ | 1.0897 |
| 3 | joint | $0.453 \pm 0.135$ | $0.947 \pm 0.005$ | $0.952 \pm 0.004$ | $0.948 \pm 0.006$ | 0.0037 |
| 3 | STF-cal | $0.943 \pm 0.007$ | $0.943 \pm 0.005$ | $0.945 \pm 0.005$ | $0.941 \pm 0.009$ | 1.0868 |
| 100 | joint | $0.221 \pm 0.067$ | $0.829 \pm 0.062$ | $0.935 \pm 0.016$ | $0.796 \pm 0.060$ | 0.0003 |
| 100 | STF-cal | $0.941 \pm 0.009$ | $0.942 \pm 0.005$ | $0.944 \pm 0.005$ | $0.941 \pm 0.006$ | 1.0918 |
| *HAR projected, 5 seeds* | | | | | | |
| 0 | primary only | $0.941 \pm 0.007$ | $0.940 \pm 0.004$ | $0.944 \pm 0.004$ | $0.943 \pm 0.004$ | 1.0636 |
| 0.1 | joint | $0.844 \pm 0.128$ | $0.932 \pm 0.022$ | $0.950 \pm 0.005$ | $0.945 \pm 0.007$ | 0.3460 |
| 0.1 | STF-cal | $0.941 \pm 0.007$ | $0.940 \pm 0.004$ | $0.944 \pm 0.004$ | $0.943 \pm 0.004$ | 1.0636 |
| 3 | joint | $0.652 \pm 0.125$ | $0.782 \pm 0.077$ | $0.946 \pm 0.003$ | $0.936 \pm 0.011$ | 0.3489 |
| 3 | STF-cal | $0.939 \pm 0.011$ | $0.940 \pm 0.004$ | $0.945 \pm 0.004$ | $0.943 \pm 0.004$ | 1.0737 |
| 100 | joint | $0.214 \pm 0.056$ | $0.398 \pm 0.033$ | $0.851 \pm 0.083$ | $0.724 \pm 0.110$ | 0.0640 |
| 100 | STF-cal | $0.939 \pm 0.011$ | $0.940 \pm 0.003$ | $0.944 \pm 0.005$ | $0.943 \pm 0.004$ | 1.0717 |
| *RadioML SNR, 5 seeds* | | | | | | |
| 0 | primary only | $0.304 \pm 0.011$ | $0.306 \pm 0.015$ | $0.320 \pm 0.010$ | $0.319 \pm 0.009$ | 1.1045 |
| 3 | joint | $0.293 \pm 0.011$ | $0.300 \pm 0.014$ | $0.314 \pm 0.012$ | $0.312 \pm 0.011$ | 0.8280 |
| 3 | STF-cal | $0.296 \pm 0.012$ | $0.304 \pm 0.015$ | $0.318 \pm 0.012$ | $0.317 \pm 0.012$ | 0.9778 |
| 10 | joint | $0.269 \pm 0.017$ | $0.281 \pm 0.016$ | $0.302 \pm 0.011$ | $0.298 \pm 0.013$ | 0.7274 |
| 10 | STF-cal | $0.292 \pm 0.015$ | $0.301 \pm 0.015$ | $0.317 \pm 0.011$ | $0.315 \pm 0.011$ | 0.8520 |
| 30 | joint | $0.243 \pm 0.020$ | $0.253 \pm 0.016$ | $0.285 \pm 0.012$ | $0.277 \pm 0.014$ | 0.6394 |
| 30 | STF-cal | $0.286 \pm 0.014$ | $0.297 \pm 0.013$ | $0.312 \pm 0.011$ | $0.309 \pm 0.010$ | 0.7111 |

Table 32: Actual-step audit on a fixed 256-example fitting batch, using the first minibatch of predeclared continuation epochs. Disagree counts opposite signs for the gradient-flow derivative and actual Adam variance change. Aux. contracts compares the joint step with a primary-only Adam step from the same optimizer state. The final column counts negative flow auxiliary flux with a contracting auxiliary counterfactual. Checkpoints within a trajectory are dependent.

| Case | $\alpha$ | updates | disagree | aux. contracts | $c < 0$, aux. contracts |
|---|---|---|---|---|---|
| HAR variance | 0.1 | 80 | 28 | 80 | 0 |
| HAR variance | 3 | 80 | 22 | 79 | 2 |
| HAR variance | 100 | 80 | 23 | 76 | 3 |
| HAR projected | 0.1 | 40 | 14 | 31 | 0 |
| HAR projected | 3 | 40 | 11 | 25 | 1 |
| HAR projected | 100 | 40 | 8 | 29 | 2 |
| RadioML SNR | 3 | 25 | 10 | 12 | 1 |
| RadioML SNR | 10 | 25 | 13 | 10 | 0 |
| RadioML SNR | 30 | 25 | 10 | 10 | 1 |

Each new run uses one CPU thread, with at most four concurrent workers and explicit memory and time limits. The remote directories contain numerical code, inputs, checkpoints, and logs; analysis, writing, figures, and compilation run locally. Driver `paired_readout_continuation.py` writes the complete per-seed records to `repro/results/CPUCONT/`. Script `summarize_cpu_evidence.py` checks the paired controls and regenerates the numerical macros, tables, and figure. Seed intervals describe variation conditional on these datasets.

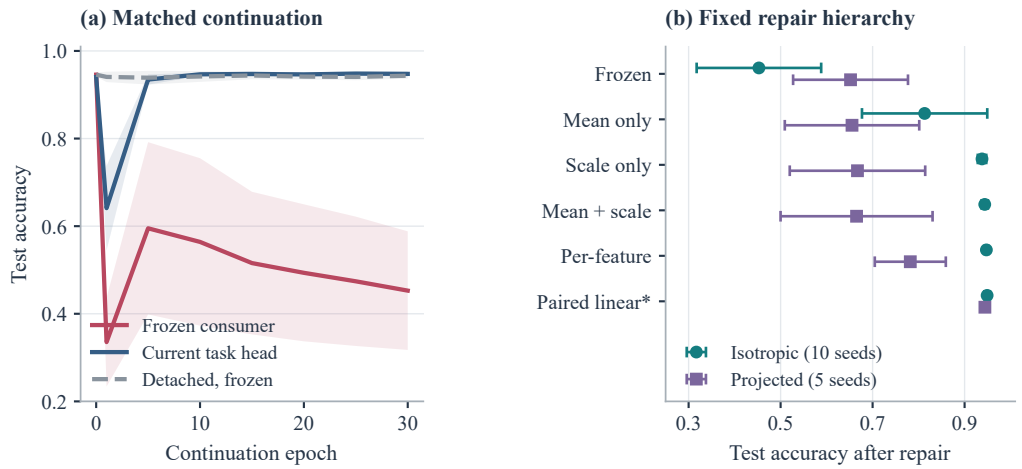



Figure 11: Input repair reveals the scale component of a learned compatibility failure. (a) Frozen-consumer and current-head trajectories at $\alpha = 3$, sharing healthy weights, Adam state, and future minibatches. (b) Fixed repairs fitted to training features without new labels. Global scale correction largely repairs isotropic contraction; projected contraction needs richer alignment. *Paired linear alignment requires old/new features of the same anchors. Shading and bars show population seed SD: 10 isotropic and 5 projected seeds.

### H.8 A Fixed Unlabeled Repair Hierarchy

To distinguish scale from richer drift, we apply a fixed repair hierarchy to every saved endpoint from the 10 isotropic HAR, 5 projected HAR, and 5 SNR continuation seeds. Deterministic replay reconstructs each healthy readout and reproduces its accuracy, NLL, ECE, and all archived frozen and recalibrated endpoints within $10^{-7}$. The reconstructed weights and statistics are saved; the encoders remain fixed throughout the audit.

Let $R = H_0 - \mathbf{1}\mu_0^\top$ and $X = H_1 - \mathbf{1}\mu_1^\top$ denote the old and new fitting features after centering. The global scale estimate is $c_{\mathrm{RMS}} = \|X\|_F/\|R\|_F$. Mean-only correction feeds $h - \mu_1 + \mu_0$ to the original preprocessing; scale-only feeds $h/c_{\mathrm{RMS}}$; their combination feeds $(h - \mu_1)/c_{\mathrm{RMS}} + \mu_0$. Per-coordinate recalibration replaces each input mean and standard deviation. These estimators use marginal fitting statistics, with no new labels or test-statistic fitting.

The richer repairs are fixed post-hoc controls. Paired diagonal regression predicts each old coordinate from its new counterpart; scaled orthogonal Procrustes fits a rotation and global scale. Covariance alignment maps centered features by $C_1^{-1/2}C_0^{1/2}$, following CORAL (Sun et al., 2016), with eigenvalues floored at $10^{-6}$ of the largest. Paired ridge maps new standardized anchors to old standardized anchors with $\lambda = 10^{-3}$. These constants are fixed independently of test outcomes. Covariance and marginal-statistic repairs need only marginal features; paired regressions and Procrustes need old/new features for the same anchors. Jointly trained BC-Aligner (Hu et al., 2022b) has a different fitting procedure.

At isotropic dose 3, scale-only correction reaches $0.938$, against $0.813$ for mean correction, $0.944$ for their combination, and $0.947$ for per-coordinate statistics. A scale-sensitive component survives substantial feature drift. At projected dose 3, scale-only reaches $0.667$, per-coordinate statistics $0.782$, and paired ridge $0.944$, close to the refitted probe. At isotropic dose 100, the corresponding scores are $0.603$, $0.829$, and $0.915$. The effective repair therefore depends on the drift. Covariance matching also leaves covariance-preserving rotations unidentified. Numerical controls recover known shifts, scales, positive diagonal changes, and scaled rotations, and include a covariance-identical sign flip.

Table 33: Fixed repair hierarchy on archived paired checkpoints. Means across 10 HAR isotropic, 5 HAR projected, and 5 RadioML SNR seeds. All maps use fitting features only, without labels or test-statistic fitting. PD, Proc., and ridge require paired old/new anchors; covariance alignment uses only marginal second moments. Ridge is fixed at $10^{-3}$ in standardized coordinates; covariance eigenvalue flooring is $10^{-6}$ of the largest eigenvalue. Full seed dispersions, NLL, ECE, prediction agreement, and fitting/held-out residuals accompany the records.

| Case | $\alpha$ | frozen | mean | scale | mean +scale | diag. | PD | Proc. | cov. | ridge |
|---|---|---|---|---|---|---|---|---|---|---|
| HAR isotropic | 0 | 0.943 | 0.941 | 0.943 | 0.941 | 0.943 | 0.943 | 0.941 | 0.945 | 0.946 |
| HAR isotropic | 0.1 | 0.789 | 0.941 | 0.946 | 0.950 | 0.951 | 0.952 | 0.949 | 0.930 | 0.950 |
| HAR isotropic | 3 | 0.453 | 0.813 | 0.938 | 0.944 | 0.947 | 0.949 | 0.948 | 0.912 | 0.949 |
| HAR isotropic | 100 | 0.221 | 0.280 | 0.603 | 0.782 | 0.829 | 0.841 | 0.855 | 0.774 | 0.915 |
| HAR projected | 0 | 0.941 | 0.938 | 0.941 | 0.938 | 0.940 | 0.940 | 0.940 | 0.944 | 0.944 |
| HAR projected | 0.1 | 0.844 | 0.896 | 0.850 | 0.898 | 0.932 | 0.924 | 0.931 | 0.553 | 0.950 |
| HAR projected | 3 | 0.652 | 0.655 | 0.667 | 0.665 | 0.782 | 0.711 | 0.862 | 0.493 | 0.944 |
| HAR projected | 100 | 0.214 | 0.362 | 0.333 | 0.354 | 0.398 | 0.354 | 0.431 | 0.383 | 0.769 |
| RadioML SNR | 0 | 0.304 | 0.306 | 0.303 | 0.306 | 0.306 | 0.302 | 0.310 | 0.312 | 0.310 |
| RadioML SNR | 3 | 0.293 | 0.296 | 0.294 | 0.299 | 0.300 | 0.291 | 0.302 | 0.302 | 0.303 |
| RadioML SNR | 10 | 0.269 | 0.275 | 0.271 | 0.283 | 0.281 | 0.267 | 0.280 | 0.285 | 0.287 |
| RadioML SNR | 30 | 0.243 | 0.246 | 0.247 | 0.254 | 0.253 | 0.233 | 0.248 | 0.257 | 0.266 |

All nine arms, all doses, and the primary-only, detached, and STF-cal endpoints are retained. Records include accuracy, NLL, ECE, agreement with healthy predictions, fitting/calibration/test alignment residuals, and map hashes. Fitting residuals and evaluation residuals are reported separately. Drivers: `checkpoint_repair_audit.py` and `readout_alignment.py`; records: `repro/results/CPUCONT/mechanism/d*.json`.

### H.9 Real Reconstruction Under Matched Continuation

We extend the unchanged paired-continuation driver to a real reconstruction decoder, $64 \to 256 \to d_{\text{input}}$ with a ReLU hidden layer and mean squared error. All five seeds and the fixed grids are retained: HAR doses $1, 10, 100$, with 60 healthy and 30 continuation epochs; RadioML I/Q doses $30, 300, 3000$, with 25 healthy and 15 continuation epochs. During warm-up the decoder learns on detached features. Every continuation branch starts from identical encoder, primary head, decoder, and Adam states and reuses the same future minibatch order. All detached encoder and primary-head endpoints again match primary-only continuation bitwise.

The trained decoder is evaluated on fitting, calibration, and test data at the fixed endpoint. The reported ratio averages seed-wise test-MSE ratios to the dose-matched detached decoder. STF-cal evaluation uses its forward filter and frozen gate on ordered batches of 256; raw-input decoder error is also archived. Reconstruction progress, primary-task benefit, and frozen-consumer preservation are scored separately.

At HAR dose 100, reconstruction improves while frozen accuracy is $0.826$ and current-head accuracy $0.941$; the primary-only current head reaches $0.943$. The paired frozen difference is $-0.115$, with descriptive seed-level $95\%$ interval $[-0.216, -0.014]$. Relative scatter is $0.449$ against primary-only continuation, with no seed below $0.3$. Per-coordinate repair reaches $0.863$ against $0.941$ after refitting. Reconstruction thus improves while the consumer breaks, and input repair recovers most of it; primary-task benefit is measured in the low-label study. RadioML I/Q is a near-null control: at 3000, frozen accuracy is $0.301$ versus $0.304$ detached and trace is $1.061$ of healthy. All grid cells and paired seed intervals are retained.

The prepared RadioML cache is checked read-only against the original dictionary using the data loader's per-modulation 50/50 split, NumPy `RandomState(0)`, and original dictionary iteration order. Every input, modulation label, and SNR target tensor must match exactly; the record stores source/cache and index hashes. This is a seeded benchmark partition, not a claim of a unique official RadioML split. Calibration is then reserved within its training portion before encoder updates.

Table 34: Paired continuation with real reconstruction auxiliaries (five seeds per domain). All branches share the healthy encoder, primary head, Adam state, and future batches. MSE ratio compares held-out reconstruction error with the dose-matched detached decoder; values below one indicate improved reconstruction. Trace is relative to the healthy encoder. All entries are seed means.

| Case | $\alpha$ | arm | frozen | recal. | refit | own | rel. trace | MSE ratio |
|---|---|---|---|---|---|---|---|---|
| HAR recon. | 0 | primary | 0.941 | 0.940 | 0.944 | 0.943 | 1.064 | – |
| HAR recon. | 1 | joint | 0.941 | 0.940 | 0.945 | 0.946 | 1.073 | 0.886 |
| HAR recon. | 1 | detach | 0.941 | 0.940 | 0.944 | 0.943 | 1.064 | 1.000 |
| HAR recon. | 1 | STF-cal | 0.942 | 0.942 | 0.944 | 0.945 | 1.027 | 3.779 |
| HAR recon. | 10 | joint | 0.930 | 0.932 | 0.943 | 0.943 | 0.658 | 0.784 |
| HAR recon. | 10 | detach | 0.941 | 0.940 | 0.944 | 0.943 | 1.064 | 1.000 |
| HAR recon. | 10 | STF-cal | 0.939 | 0.941 | 0.944 | 0.945 | 0.756 | 3.799 |
| HAR recon. | 100 | joint | 0.826 | 0.863 | 0.941 | 0.941 | 0.208 | 0.765 |
| HAR recon. | 100 | detach | 0.941 | 0.940 | 0.944 | 0.943 | 1.064 | 1.000 |
| HAR recon. | 100 | STF-cal | 0.914 | 0.932 | 0.941 | 0.937 | 0.288 | 3.778 |
| RadioML I/Q | 0 | primary | 0.304 | 0.306 | 0.320 | 0.319 | 1.105 | – |
| RadioML I/Q | 30 | joint | 0.303 | 0.305 | 0.319 | 0.319 | 1.106 | 0.995 |
| RadioML I/Q | 30 | detach | 0.304 | 0.306 | 0.320 | 0.319 | 1.105 | 1.000 |
| RadioML I/Q | 30 | STF-cal | 0.303 | 0.305 | 0.319 | 0.319 | 1.106 | 0.995 |
| RadioML I/Q | 300 | joint | 0.303 | 0.305 | 0.319 | 0.318 | 1.098 | 0.987 |
| RadioML I/Q | 300 | detach | 0.304 | 0.306 | 0.320 | 0.319 | 1.105 | 1.000 |
| RadioML I/Q | 300 | STF-cal | 0.303 | 0.305 | 0.319 | 0.318 | 1.098 | 0.987 |
| RadioML I/Q | 3000 | joint | 0.301 | 0.305 | 0.319 | 0.318 | 1.061 | 1.007 |
| RadioML I/Q | 3000 | detach | 0.304 | 0.306 | 0.320 | 0.319 | 1.105 | 1.000 |
| RadioML I/Q | 3000 | STF-cal | 0.301 | 0.305 | 0.319 | 0.318 | 1.061 | 1.007 |

Drivers: `reconstruction_task_audit.py` and `radioml_split_audit.py`; full records: `repro/results/CPUCONT/{mechanism,task_validation}/`.

### H.10 Waveform CNN Continuation

The paired real-head studies above use small MLPs. We therefore predeclare a waveform CNN on the identical prepared RadioML I/Q inputs, with five seeds and reconstruction doses $30, 300, 3000$. The flattened vector is reshaped to $2 \times 128$. The encoder uses a padded kernel-7 convolution ($2 \to 8$ channels), ReLU and max-pooling by two; a padded kernel-5 convolution ($8 \to 16$), ReLU and pooling by two; and a linear map from the resulting $16 \times 32$ features to width $64$. There is no stochastic layer or normalization. The linear primary head and $64 \to 256 \to 256$ reconstruction decoder are unchanged.

We reuse the unchanged paired training engine: 25 healthy epochs with a detached auxiliary, 15 continuation epochs, batch 256, Adam at $10^{-3}$, independent calibration reservation, complete model/optimizer cloning, matched future orders, and actual-step counterfactual audits. This study tests joint training, detachment, and primary-only continuation, for seven branches per seed; it does not test STF-cal on the CNN. The source records contain both the unchanged engine hash and the wrapper/architecture hash. Healthy heads are replayed deterministically and checked against their archived accuracy, NLL, and ECE before the consumer audit.

The healthy CNN readout reaches $0.424$, compared with the MLP's $0.297$ on the same split. At dose $3000$, frozen accuracy is $0.415$ versus $0.418$ detached; recalibration gives $0.425$, refitting $0.442$, and the current head $0.437$. Relative trace is $1.032$ and the reconstruction MSE ratio to detachment is $0.917$. The full dose grid and a pure-scale sweep for each healthy CNN readout are recorded. Consumer damage is small for this encoder under the tested recipe, which bounds how far the MLP failure transfers. Driver: `cnn_reconstruction_continuation.py`; records: `repro/results/CPUCONT/frontier/c*.json`.

Table 35: Waveform CNN reconstruction continuation on the same prepared RadioML inputs and split as the MLP study. Five seeds; 25 healthy plus 15 continuation epochs; the original heads, optimizer, cloned-state and batch controls are retained. Entries are means, with population seed SD for frozen and own accuracy; all dispersions are archived. MSE ratio compares reconstruction error with the matched detached decoder. All metrics here use the full original test corpus, separately from the monitor/remainder audit.

| $\alpha$ | arm | frozen | recal. | refit | own | rel. trace | MSE ratio |
|---|---|---|---|---|---|---|---|
| 0 | primary | $0.418 \pm 0.018$ | 0.428 | 0.442 | $0.436 \pm 0.015$ | 1.058 | – |
| 30 | joint | $0.417 \pm 0.018$ | 0.427 | 0.443 | $0.437 \pm 0.015$ | 1.060 | 0.918 |
| 30 | detach | $0.418 \pm 0.018$ | 0.428 | 0.442 | $0.436 \pm 0.015$ | 1.058 | 1.000 |
| 300 | joint | $0.418 \pm 0.018$ | 0.428 | 0.443 | $0.437 \pm 0.015$ | 1.053 | 0.851 |
| 300 | detach | $0.418 \pm 0.018$ | 0.428 | 0.442 | $0.436 \pm 0.015$ | 1.058 | 1.000 |
| 3000 | joint | $0.415 \pm 0.023$ | 0.425 | 0.442 | $0.437 \pm 0.015$ | 1.032 | 0.917 |
| 3000 | detach | $0.418 \pm 0.018$ | 0.428 | 0.442 | $0.436 \pm 0.015$ | 1.058 | 1.000 |

Table 36: Conditional SNR head-state factorial at dose 30 (five seeds). Shared encoder/primary parameters and their Adam states, the frozen consumer, and future minibatches are identical at branching. Auxiliary weights and auxiliary Adam state form the four cells. Warm/retained replays the original model/Adam endpoint bitwise. Fresh probes share a fixed initialization seed across cells. Means with population seed SD for frozen accuracy; full paired intervals and early actual-step audits are archived.

| Weights | aux. Adam | frozen | recal. | refit | own | rel. trace | SNR MSE |
|---|---|---|---|---|---|---|---|
| warm | retained | $0.243 \pm 0.020$ | 0.253 | 0.284 | 0.277 | 0.639 | 0.460 |
| warm | reset | $0.241 \pm 0.020$ | 0.254 | 0.283 | 0.277 | 0.643 | 0.460 |
| initial | retained | $0.244 \pm 0.025$ | 0.245 | 0.281 | 0.275 | 0.726 | 0.459 |
| initial | reset | $0.226 \pm 0.033$ | 0.227 | 0.283 | 0.277 | 0.888 | 0.460 |

### H.11 Conditional Auxiliary Head-State Factorial

To test whether auxiliary-head state explains the difference between historical SNR training and warm continuation, we branch from five archived healthy references. Encoder and primary weights, their Adam state, the frozen consumer, and future minibatches stay identical. At dose 30, warm versus initial pre-warm auxiliary weights are crossed with retained versus reset auxiliary Adam state, followed by 15 continuation epochs. The initial/retained cell isolates the interaction. The design precedes outcome computation; seed 0 is the complete implementation check and remains in the five-seed analysis. The dose is fixed throughout.

Warm/retained reproduces the original model and Adam endpoint bitwise in all five seeds. Resetting auxiliary moments changes frozen accuracy from $0.243$ to $0.241$. Initial/reset reaches $0.226$ with trace $0.888$, versus $0.639$ for warm/retained. More retained variance therefore need not give higher consumer accuracy. No factorial cell reproduces the historical severe scatter collapse: auxiliary state alone leaves the protocol difference unexplained at this reference and horizon.

Dense early audits and selected later audits record 620 dependent step observations, not 620 independent trials. First-step primary-only counterfactual increments match across all four cells; changing only auxiliary moments also leaves the first encoder increment unchanged, as expected from the parameterwise optimizer. Shared-state and batch hashes, CPU checkpoint validation, bitwise endpoint replays, and seed-paired intervals are archived in `head_state_verification.json` and `head_state_summary.json`. The 20 endpoint states include five exact replays and 15 new conditional trajectories; they are outside the fixed consumer-certificate bank below. Numerical driver: `auxiliary_state_factorial.py`.

### H.12 Readout-Nullspace Protection with a Useful Real Auxiliary

The old consumer applies $z_0(x) = W_0 D_0^{-1}(h_0(x) - \mu_0) + b_0$, with fixed positive diagonal $D_0 = \mathrm{diag}(s_0)$. Let $A = W_0 D_0^{-1}$ and $P_\perp = I - A^\dagger A$, the orthogonal projector onto $\ker A$. We use

Table 37: All predeclared primary-accuracy contrasts in percentage points, ten independent confirmation seeds. Paired two-sided 95% descriptive t intervals are conditional on the fixed dataset/split/recipe. GD controls the guarded architecture itself; GJ–GD isolates its auxiliary benefit. Noninferiority was declared at $-0.5$ point, not selected from the outcomes.

| Contrast | mean (pp) | paired 95% interval | required lower limit |
|---|---|---|---|
| J–D | 0.97 | [0.63, 1.31] | 0 |
| GJ–GD | 0.89 | [0.60, 1.18] | 0 |
| GJ–D | 1.25 | [0.92, 1.58] | 0 |
| GJ–J | 0.28 | [0.08, 0.47] | −0.50 |

established projection and backward-compatibility ideas (Farajtabar et al., 2020; Shen et al., 2020) as a constructive control for measuring retained benefit and compatibility.

**Proposition 10** (Compatibility and remaining feature freedom)**.** *For any updated encoder $h_\theta$ and retained deterministic reference encoder $h_0$, define*

$$\widetilde{h}_\theta(x) = h_0(x) + P_\perp\big(h_\theta(x) - h_0(x)\big). \tag{26}$$

*The frozen affine consumer has exactly the reference logits on every input in real arithmetic. It leaves $m - \operatorname{rank}(A)$ feature directions free. It is also the unique Euclidean-nearest feature vector to $h_\theta(x)$ among all vectors with the reference logits. Conversely, every feature displacement preserving these logits belongs to $\ker A$; if $A$ has full column rank, no nonzero compatible displacement remains.*

*Proof.* Since $AP_\perp = 0$, the logit difference is $AP_\perp(h_\theta - h_0) = 0$. Rank–nullity gives the free dimension and the converse. The construction is the orthogonal projection onto the affine set $h_0(x) + \ker A$, proving the minimum-distance statement. The fixed normalizer's mean and the head's bias cancel in the difference. □

The row-vector implementation is $\widetilde{H} = H_0 + (H_\theta - H_0)P_\perp$. A float64 SVD of the normalized head $A$ fixes the projector before continuation. Because the reference branch remains present, the logit identity holds for later encoder parameters and arbitrary inputs in real arithmetic. It also preserves fixed post-processing and its resulting NLL and ECE. Corpus-level float64 checks measure the numerical residual. This functional identity is stronger than the checkpoint-local flow statement and covers a different object from the sampled disagreement bound.

The committed plan separates development from confirmation. Development seeds 200–202 test HAR reconstruction with $10\%$ of fitting primary labels at doses $1, 10, 100$, and RadioML SNR with $20\%$ at $0.1, 1, 10$. A training reservation made before any model or readout fit supplies validation only. Candidates share healthy states, Adam moments, and future minibatches. Healthy readouts use labeled fitting examples; all fitting inputs and auxiliary targets remain available. HAR reconstructs the $561$-dimensional input, while SNR metadata are standardized on fitting data. The rule maximizes the minimum mean validation gain among J–D, GJ–GD, and GJ–D. The frozen choice record retains all six candidates, including null and adverse results; development computes no test metrics.

The selected recipe is RadioML SNR dose $1$, width $64$, Adam $10^{-3}$, batch $256$, $25$ healthy plus $25$ continuation epochs. Ten new seeds 300–309 receive this unchanged recipe and all four branches: ordinary detachment/joint training (D/J), and guarded detachment/joint training (GD/GJ). Classification logits and the projector use float64 in every arm; encoder/auxiliary parameters and Adam states remain fp32. The guard and losses are fixed before continuation. Training-anchor prediction drift is logged at fixed epochs with a $0.05$ disagreement threshold; it is a diagnostic trace rather than an adaptive controller or warning-lead-time claim. Endpoints are fixed; neither checkpoint nor seed selection uses test outcomes.

Confirmation requires five predeclared conditions: positive paired lower $95\%$ limits for J–D, GJ–GD, and GJ–D; GJ–J above $-0.5$ percentage point; and logit error below $10^{-8}$ with zero prediction changes. All five pass. Intervals are paired seed-level $t$ intervals conditional on this dataset, split, and recipe. Current-head accuracy is $0.288$ for GJ, $0.285$ for J, $0.279$ for GD, and $0.275$ for D. The guarded comparison GJ–GD isolates the auxiliary's contribution, with SNR MSE $0.766$ of matched guarded detachment.

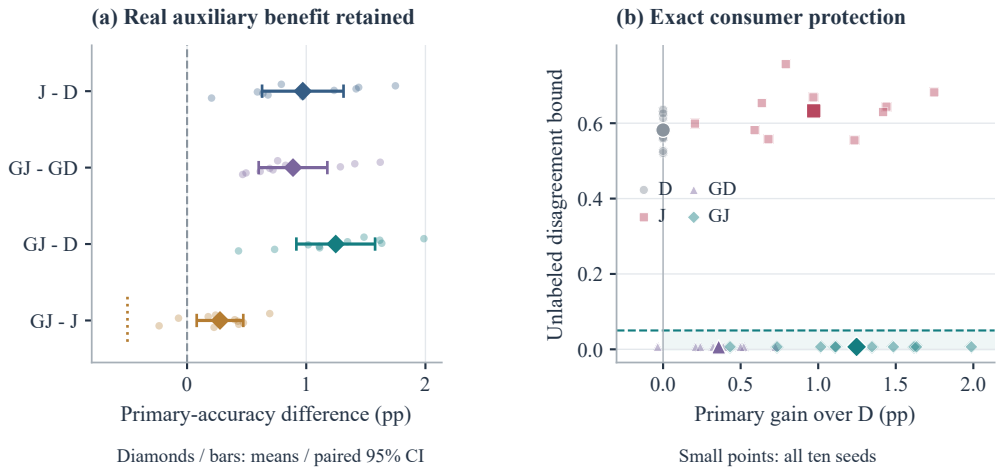



Figure 12: Independent confirmation retains auxiliary benefit and old outputs. (a) All predeclared primary-accuracy contrasts across ten new seeds, with paired 95% intervals. Zero marks the benefit threshold; $-0.5$ applies to the final noninferiority contrast. (b) All forty consumers: primary gain over D versus the unlabeled disagreement bound. The dashed line is the five-point relative-error budget; shading marks positive gain with a certified consumer. Ordinary consumers remain uncertified despite nearly unchanged aggregate frozen accuracy. Bounds cover the fixed corpus remainder.

The separate confirmation bank contains forty fixed consumers, total tail probability $0.05$, an independent $1{,}024$-input monitor per seed, and a $0.05$ relative classification-error budget. GD and GJ each certify in $10/10$ seeds with a bound near $0.0065$; D and J each certify in $0/10$. Ordinary consumers change many predictions while aggregate frozen accuracy remains nearly unchanged. J's frozen ECE rises from $0.048$ to $0.155$; guarded predictions, NLL, and ECE match the reference, with maximum logit error $4.6 \times 10^{-14}$. This bank is distinct from the earlier $3{,}015$-candidate family.

All ten normalized heads have rank $11$, leaving $53$ free directions. Stacking normalized matrices protects several affine heads on the same reference features, at the cost of a smaller nullspace; full column rank leaves no compatible displacement. Nonlinear consumers need their own constraint. The reference encoder remains at inference. On one Xeon Platinum 8570 CPU thread, batch $1{,}024$, the mean ratio of per-seed median latencies is $2.37\times$, using nine repetitions after three warm-ups. Reference fitting features are cached during training. Thus the confirmed low-label result includes the cost of retaining and evaluating the reference.

A direct dual-routing baseline also retains both encoders, serving the old consumer from $h_0$ and the current head from J. It inherits J's primary metric and the reference legacy metrics. The guarded control instead emits one shared fixed-width vector for both heads and adds a projection. Its distinctive constraint is a common representation; it has no measured efficiency advantage over dual routing. The nearest-vector property concerns Euclidean feature distance.

The full study verifies $104$ development/confirmation model–Adam states, shared initial states, primary-label masks, future-order hashes, and all consumer bounds. An independent process reconstructs all forty saved consumers and replays their accuracy/NLL/ECE and bounds exactly. Numerical drivers are `benefit_compatibility_cpu.py` and `verify_benefit_consumers_cpu.py`; local analysis is `summarize_benefit_compatibility.py`. The committed plan, frozen validation choice, full records, verification and all negative development cases accompany the artifact.

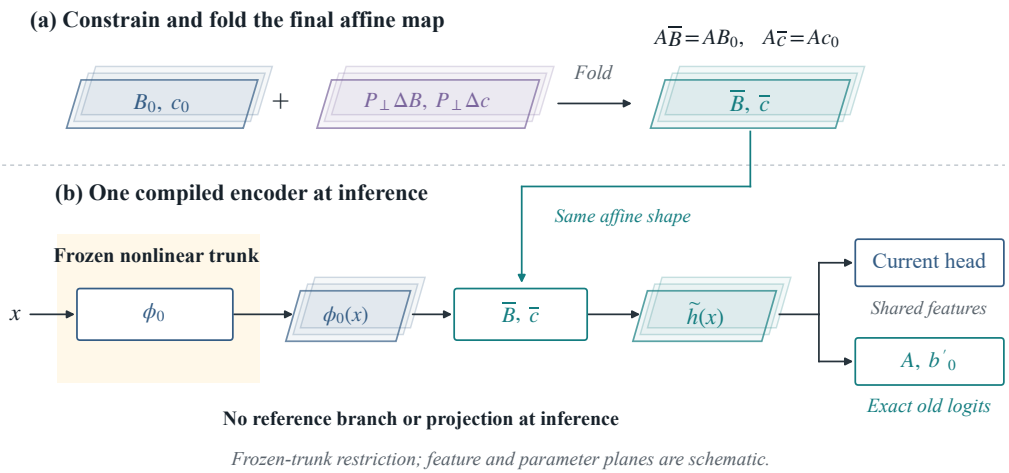



Figure 13: A frozen trunk allows the compatibility constraint to fold into one encoder. With $\Delta B = B - B_0$ and $\Delta c = c - c_0$, constrained weights and biases retain their native shapes (Equation 27); $b'_0 = b_0 - A\mu_0$ is the consumer's effective bias. Inference evaluates one encoder while preserving the reference logits. Freezing the nonlinear trunk restricts feature learning. Planes are schematic.

### H.13 Folding Compatibility into a Single Encoder

The reference branch makes compatibility expensive. Freezing the nonlinear trunk lets us fold the constraint into its final affine map, retaining ordinary inference shapes but restricting feature learning. A new plan, committed after the earlier confirmation and before execution, tests this tradeoff as a separate study.

**Corollary 8** (Foldable affine compatibility)**.** *Let $h_0(x) = B_0\phi_0(x) + c_0$, with a retained deterministic trunk $\phi_0$. For arbitrary trainable $B, c$, set*

$$\overline{B} = B_0 + P_\perp(B - B_0), \qquad \overline{c} = c_0 + P_\perp(c - c_0). \tag{27}$$

*Then $\overline{h}(x) = \overline{B}\phi_0(x) + \overline{c}$ has the reference affine-consumer logits for every input in real arithmetic. Deploying $\overline{B}, \overline{c}$ needs one trunk and one final affine map of the original shapes, with no reference branch or projection. If the trunk has $d$ outputs, this parameterization leaves $(m - \operatorname{rank} A)(d + 1)$ affine degrees of freedom. Changing the trunk is outside this guarantee.*

*Proof.* Both $A(\overline{B} - B_0)$ and $A(\overline{c} - c_0)$ vanish. Folding is a substitution of one weight and bias, not a second inference operation. Each of the $d + 1$ weight/bias columns can vary in $\ker A$. □

The construction uses constrained affine adaptation, related to nullspace updates and mergeable frozen-backbone methods (Farajtabar et al., 2020; Pham et al., 2026; Hu et al., 2022a). With a linear current head it remains a linear predictor of fixed trunk features. Folding preserves the constrained function exactly; whether that function retains auxiliary benefit is the empirical question.

RadioML uses SNR regression, dose 1 and 20% of fitting primary labels, inherited from the prior validation choice. HAR uses input reconstruction, dose 10 and 10% labels, the positive max-min candidate in the previously archived three-seed HAR validation grid. No new grid is searched. Each task uses ten new seeds 400–409, the same prepared test split, a stratified 20% training reservation before any model/head fit, width 64, Adam $10^{-3}$ and batch 256. HAR has 60 healthy plus 30 continuation epochs; RadioML has $25 + 25$. The last encoder affine map is natively float64 in every arm, as are classification logits; trunk and auxiliary parameters are fp32. The healthy consumer is fitted to that same mixed-precision interface. This explicit precision change prevents pooling with the earlier encoder-fp32 experiment.

Six branches clone each healthy model and full Adam state, with identical fitting-label masks and future minibatches: full-network D/J; frozen-trunk, unconstrained-last-layer LD/LJ; and frozen-trunk,

Table 38: Folded single-encoder continuation, ten new seeds per task. D/J update the full encoder; LD/LJ freeze the nonlinear trunk; CD/CJ additionally constrain the last affine map to preserve the normalized healthy consumer. All use the same native float64 final affine and float32 trunk. Means with population seed SD; the 120-consumer bank uses a five-point relative-error budget.

| Task | arm | own accuracy | frozen accuracy | frozen NLL | ECE | certified |
|---|---|---|---|---|---|---|
| SNR | D | $0.267 \pm 0.008$ | $0.243 \pm 0.032$ | 2.262 | 0.169 | 0/10 |
| | J | $0.278 \pm 0.009$ | $0.248 \pm 0.026$ | 2.198 | 0.166 | 0/10 |
| | LD | $0.242 \pm 0.015$ | $0.221 \pm 0.025$ | 2.261 | 0.139 | 0/10 |
| | LJ | $0.242 \pm 0.015$ | $0.219 \pm 0.029$ | 2.255 | 0.132 | 0/10 |
| | CD | $0.242 \pm 0.015$ | $0.244 \pm 0.015$ | 2.061 | 0.039 | 10/10 |
| | CJ | $0.242 \pm 0.015$ | $0.244 \pm 0.015$ | 2.061 | 0.039 | 10/10 |
| HAR recon. | D | $0.918 \pm 0.008$ | $0.913 \pm 0.012$ | 0.482 | 0.062 | 6/10 |
| | J | $0.919 \pm 0.005$ | $0.898 \pm 0.019$ | 0.540 | 0.069 | 2/10 |
| | LD | $0.918 \pm 0.006$ | $0.913 \pm 0.011$ | 0.440 | 0.059 | 6/10 |
| | LJ | $0.918 \pm 0.006$ | $0.906 \pm 0.015$ | 0.450 | 0.062 | 5/10 |
| | CD | $0.918 \pm 0.006$ | $0.918 \pm 0.005$ | 0.355 | 0.052 | 10/10 |
| | CJ | $0.918 \pm 0.007$ | $0.918 \pm 0.005$ | 0.355 | 0.052 | 10/10 |

Table 39: All declared folded-control accuracy contrasts in percentage points, mean with paired descriptive 95% interval. Conditional on each fixed dataset/split/recipe. The first four rows have required lower limits $0, 0, 0, -0.5$ respectively; all other rows diagnose freezing and constraint costs regardless of sign.

| Contrast | RadioML SNR | HAR reconstruction |
|---|---|---|
| J–D | $+1.11$ [0.90, 1.32] | $+0.04$ [$-0.30$, 0.37] |
| CJ–CD | $+0.01$ [$-0.02$, 0.04] | $+0.01$ [$-0.08$, 0.09] |
| CJ–D | $-2.42$ [$-3.20$, $-1.64$] | $-0.03$ [$-0.43$, 0.36] |
| CJ–J | $-3.53$ [$-4.23$, $-2.83$] | $-0.07$ [$-0.47$, 0.33] |
| LD–D | $-2.45$ [$-3.21$, $-1.69$] | $-0.02$ [$-0.43$, 0.39] |
| LJ–J | $-3.53$ [$-4.21$, $-2.85$] | $-0.08$ [$-0.47$, 0.30] |
| CJ–LJ | $-0.003$ [$-0.039$, 0.033] | $+0.010$ [$-0.078$, 0.099] |
| CD–LD | $+0.018$ [$-0.025$, 0.061] | $-0.020$ [$-0.093$, 0.052] |
| LJ–LD | $+0.028$ [0.001, 0.055] | $-0.024$ [$-0.068$, 0.021] |

constrained-last-layer CD/CJ. Heads continue learning in every branch. The constraint is fixed before continuation; the fixed training-anchor diagnostic is not an adaptive policy. Validation/test inputs never train the encoder or healthy consumer. Frozen trunk parameters *and their Adam states* are checked unchanged. Every branch uses the declared endpoint and every seed is retained.

The five criteria require positive lower paired $95\%$ limits for J–D, CJ–CD, and CJ–D; CJ–J above $-0.5$ percentage point; and preserved old logits and predictions. Neither task passes all five. On SNR, unrestricted auxiliary gain is $1.11$ points, $[0.90, 1.32]$, while compiled gain is $0.01$, $[-0.02, 0.04]$. CJ–J is $-3.53$, $[-4.23, -2.83]$, below the noninferiority margin. The unconstrained frozen-trunk comparison LJ–J is $-3.529$ points, $[-4.213, -2.846]$; adding the constraint, CJ–LJ, changes accuracy by $-0.003$, $[-0.039, 0.033]$. Freezing accounts for the dominant measured cost in this recipe. LJ–LD retains a small $0.028$-point gain, $[0.0006, 0.0554]$, far below unrestricted training.

HAR gives J–D of $0.04$ points, $[-0.30, 0.37]$, and CJ–CD of $0.01$, $[-0.08, 0.09]$. Both leave auxiliary benefit unresolved. CJ–J meets the half-point margin. Table 39 gives every planned contrast, including adverse and near-null outcomes. All intervals describe paired variation over ten seeds on each fixed dataset, split, and recipe.

All $40$ CD/CJ consumers retain old predictions, NLL, and ECE, with maximum logit error $3.6 \times 10^{-14}$. Folded and training-time features are bitwise identical on the scored corpus. A separate fixed family of $120$ consumers uses total $\delta = 0.05$, $1{,}024$ uniformly sampled monitor inputs per seed, and a five-point relative-error budget. All $40$ compiled consumers certify on the corpus remainder; monitor information stays outside training.

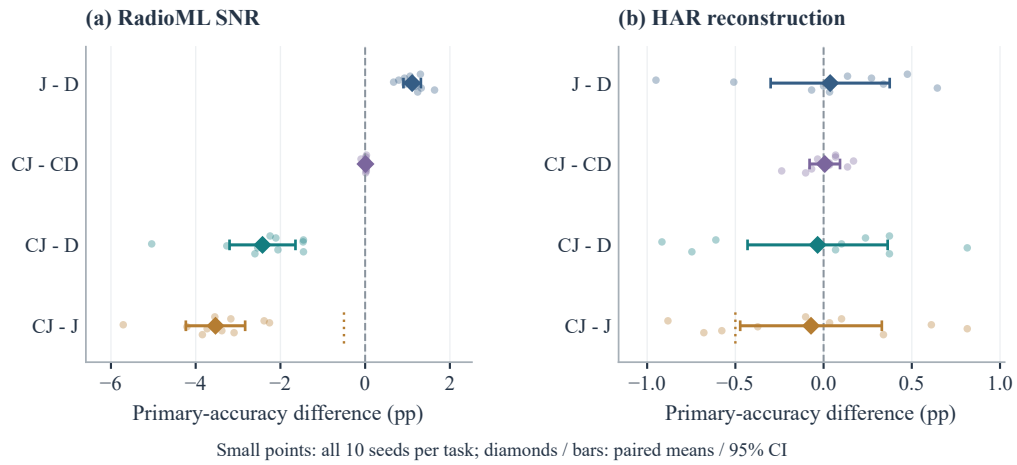



Figure 14: Folding retains compatibility but loses the SNR auxiliary gain. All four declared primary-accuracy contrasts, ten new seeds per task. Diamonds and bars show means and paired descriptive 95% intervals. D/J update the full encoder; CD/CJ freeze the trunk and constrain its last affine map. The $-0.5$ marker applies only to CJ–J. SNR retains an unrestricted gain; neither task passes the full five-criterion target. Panel scales differ.

Deployed encoders retain exactly three affine evaluations and the baseline parameter shapes, with no reference or projector parameters. At batch 1, mean ratios of per-seed median CPU latency are $1.00\times$ for SNR and $1.00\times$ for HAR. At batch 1,024, they are $1.01\times$ and $1.00\times$. Measurements cover the encoder and current head, using one Xeon Platinum 8570 thread, five warm-ups and 21 alternating timed repetitions per path; the baseline uses the same mixed precision. The corresponding two-encoder paths average $2.22\times$ and $2.16\times$ at batch 1,024. These deployment measurements establish the cost reduction, not useful adaptation.

Local checks cover all 180 healthy, endpoint model–Adam, and compiled states. A separate tensor-based process reconstructs all 120 consumers and exactly replays accuracy, NLL, ECE, and sampling bounds. Structural checks cover the normalized-head constraint, Adam steps, folding, reference and projector removal, nonzero gradients, and failure under trunk drift. The artifact includes the committed plan, drivers, complete outcomes, and verification. This control removes reference computation but loses the confirmed SNR benefit, motivating a trainable nonlinear channel.

### H.14 A Compatible Nonlinear Channel: Development Gate and Limits

A narrow side network allows nonlinear learning while the base remains fixed. We combine side-tuning (Zhang et al., 2020) with established compatibility projections (Farajtabar et al., 2020; Shen et al., 2020):

$$\widetilde{h}(x) = f_0(x) + P_\perp g_\psi(x), \qquad A\widetilde{h}(x) = Af_0(x). \tag{28}$$

The healthy base $f_0$ stays fixed, and a two-ReLU side MLP $g_\psi$ learns its nonlinear prefix. Both heads receive one shared 64-vector. Projection preserves affine-consumer logits; the following experiment measures whether the added flexibility also retains useful learning.

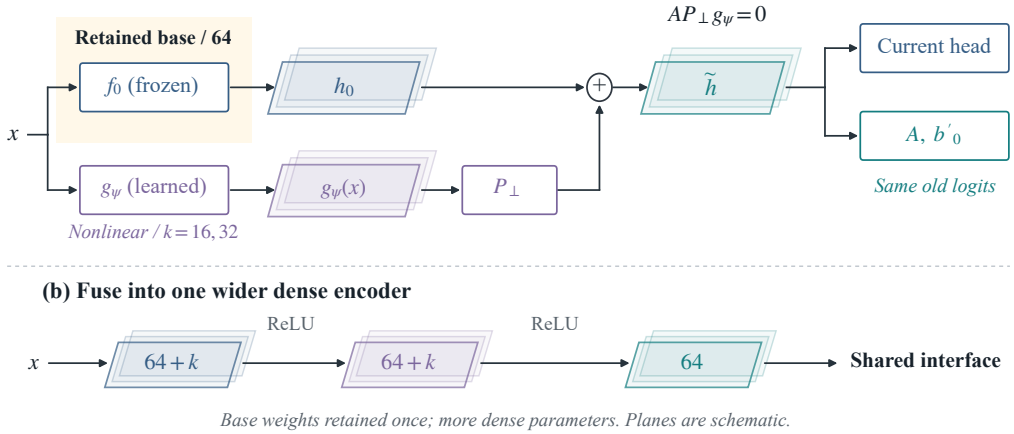



Figure 15: A compatible nonlinear channel adds trainable capacity. (a) The base stays fixed while the side learns, with $AP_\perp g_\psi = 0$. (b) Equal-depth paths fuse into a dense width-$64 + k$ MLP: the middle matrix is block diagonal and the final side block is projected. The base is embedded once, all dense zero blocks count toward storage and computation, and the output stays 64-dimensional. Planes are schematic; parameter ratios and the validation screen are reported separately.

For base weights $(U_1, U_2, U_3)$ and side weights $(V_1, V_2, V_3)$, the fused matrices are

$$\overline{U}_1 = \begin{bmatrix} U_1 \\ V_1 \end{bmatrix}, \quad \overline{U}_2 = \begin{bmatrix} U_2 & 0 \\ 0 & V_2 \end{bmatrix}, \quad \overline{U}_3 = [U_3 \quad P_\perp V_3]\,. \tag{29}$$

Stacking the first two biases and using final bias $b_3 + P_\perp c_3$ gives the same function with three affine layers and two ReLUs. The fused model embeds the base once and absorbs the separate projection. Its wider dense matrices still count all zero blocks: for side widths $16/32$, encoder parameter ratios are $1.301/1.623$ on SNR and $1.279/1.569$ on HAR, at native float64. Full-data runtime evaluation was contingent on passing the development screen. The side's final weight and bias start at zero; prefix weights copy the first $k$ healthy units and their $k \times k$ second-layer block.

The new plan was committed after the earlier studies and before numerical execution. Development uses seeds 500–502, the original width-64 base, Adam $10^{-3}$, batch 256, and the same stratified $20\%$ training reservation before model/head fitting. SNR uses $20\%$ of fitting primary labels, dose 1, and $25 + 25$ healthy/continuation epochs. HAR uses a new $1\%$ fitting-label reconstruction regime, doses $1, 10$, and $60 + 30$ epochs. Both test side widths $16, 32$. All feature-encoder arithmetic is float64, with fp32 head parameters and float64 classification logits. These changes prevent pooling with earlier confirmation results.

Each candidate compares full-network D/J, unconstrained side UD/UJ, and constrained side TD/TJ, with detached/joint auxiliary gradients. Healthy base/head parameters and their Adam states are retained; new side Adam states start empty. The common parameter ordering, paired side initialization, fitting-label masks and future minibatches are checked. The declared score is the minimum of three mean validation gains (J–D, TJ–TD, TJ–D) and the mean TJ–J contrast plus $0.5$ point; ties prefer smaller width, then dose. A task enters ten-new-seed test confirmation only if its best score is strictly positive. No development test outcomes are computed.

The highest-scoring SNR candidate uses $k = 32$, dose 1. Its validation TJ–TD is $0.32$ points, with descriptive paired $95\%$ interval $[-0.17, 0.80]$, but TJ–D is $-0.69$, $[-1.93, 0.55]$, giving a negative score. The highest-scoring HAR candidate uses $k = 16$, dose 1: TJ–TD is $0.70$, $[-1.76, 3.17]$, but J–D is exactly tied in aggregate correct counts, giving score zero. Neither qualifies for confirmation, and the artifact retains all six candidates.

The strict-positive gate is evaluated from integer correct counts and exact rational means. Before confirmation, an arithmetic check corrected a floating mean near $3.7 \times 10^{-17}$ for the tied HAR

Table 40: Complete nonlinear-channel development grid: three seeds per task, validation only. D/J are full-network branches; TD/TJ are constrained side branches. All entries are percentage-point means. The score is $\min\{\overline{\mathrm{J}-\mathrm{D}}, \overline{\mathrm{TJ}-\mathrm{TD}}, \overline{\mathrm{TJ}-\mathrm{D}}, \overline{\mathrm{TJ}-\mathrm{J}}+0.5\}$. Confirmation requires strictly positive score, evaluated from integer correct counts. No candidate qualifies. HAR uses a new 1% fitting-label regime, SNR 20%.

| Task | $k$ | $\alpha$ | J–D | TJ–TD | TJ–D | TJ–J | score | confirm |
|---|---|---|---|---|---|---|---|---|
| SNR | 16 | 1 | $+0.34$ | $+0.53$ | $-1.73$ | $-2.07$ | $-1.73$ | no |
| SNR | 32 | 1 | $+0.34$ | $+0.32$ | $-0.69$ | $-1.02$ | $-0.69$ | no |
| HAR | 16 | 1 | $+0.00$ | $+0.70$ | $+0.34$ | $+0.34$ | 0.00 | no |
| HAR | 32 | 1 | $+0.00$ | $+0.39$ | $+0.25$ | $+0.25$ | 0.00 | no |
| HAR | 16 | 10 | $+1.50$ | $+0.32$ | $-0.05$ | $-1.54$ | $-1.04$ | no |
| HAR | 32 | 10 | $+1.50$ | $-0.29$ | $-0.43$ | $-1.93$ | $-1.43$ | no |

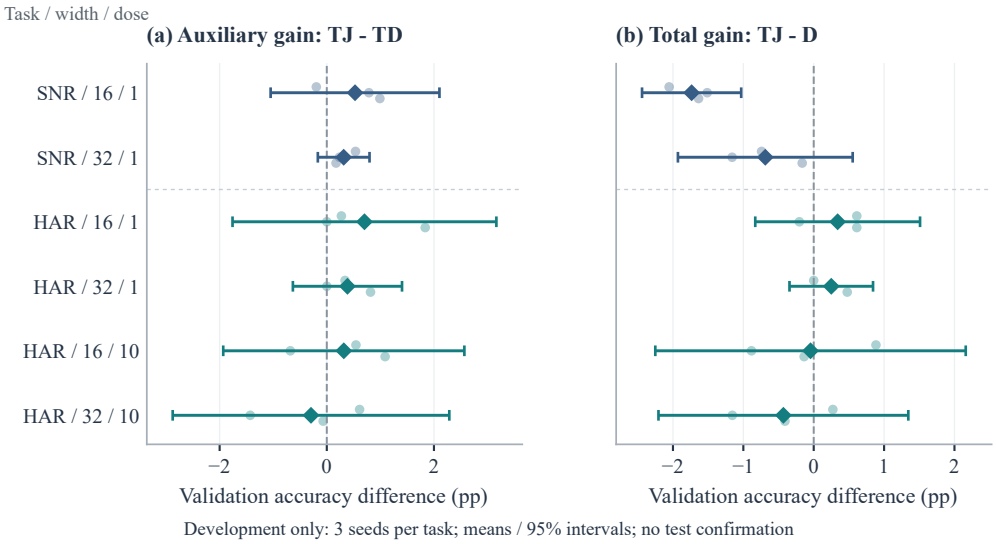



Figure 16: Neither task passes the nonlinear-channel development screen. Auxiliary and total gains for all six candidates, with three seed points and descriptive paired 95% intervals per row. Labels give task, side width, and dose. The screen additionally requires positive J–D and the TJ–J margin (Table 40). No candidate proceeds to test confirmation or full-data latency evaluation.

contrast to zero. Both serializations and the correction audit are retained; the grid and decision rule stay fixed. Because the screen fails, the planned test, auxiliary-MSE, and CPU-budget conjunction is not evaluated.

All 165 saved state files are locally checked: six healthy states, 87 endpoint model–Adam states and 72 deployed side models. A separate tensor-only process reconstructs all 87 validation consumers; accuracy, NLL and ECE replay exactly, with auxiliary errors also checked. All 36 constrained-side consumers retain their old validation predictions/NLL/ECE; maximum logit error is $2.5 \times 10^{-14}$. Fused/unfused feature error is at most $1.1\times10^{-14}$. The nonlinear prefix changes in every constrained branch, and frozen base weights and Adam states remain unchanged. Structural tests include arbitrary nonlinear changes, native-precision fusion, full-rank/no-nullspace and unconstrained/base-drift negative controls.

The construction supports nonlinear updates with retained affine-consumer outputs, but none of the tested candidates closes the utility and cost target. Isolating the prefix's contribution would need a matched side with its prefix frozen. A pre-registered vision screen did not reproduce the accuracy failure at the tested doses, although the frozen readout changed 42–56% of its decisions (Appendix H.17). Stronger vision auxiliaries and non-affine consumers are untested.

Table 41: Post-hoc single-encoder chart, ten independent SNR confirmation seeds. Means on the full fixed test corpus. Current logits are algebraically preserved by head rebasing. Old-reader metrics can improve despite extensive prediction changes. The five-point disagreement budget certifies none of the ten repaired joint consumers.

| Consumer | accuracy | NLL | ECE | changed predictions |
|---|---|---|---|---|
| Healthy old | 0.251 | 2.044 | 0.043 | 0 |
| Native J, old readout | 0.260 | 2.168 | 0.166 | 56.69% |
| Chart J, old readout | 0.264 | 1.984 | 0.019 | 44.07% |
| Native J, current head | 0.281 | 1.980 | 0.053 | — |
| Chart J, rebased current | 0.281 | 1.980 | 0.053 | 0 (vs. native J) |
| Supervised refit | 0.285 | 1.993 | 0.070 | — |

### H.15 Post-hoc Coordinates after Nonlinear Learning

The frozen-trunk control removes reference inference but also restricts learning. Here both nonlinear layers remain trainable throughout continuation. Only after training do we fit an affine chart from new features into the old coordinates, using established linear alignment.

With row-vector features, write the current logits as $hW^\top+\beta$. An invertible chart emits $\widetilde{h} = hM+b$. Rebasing the current head by

$$W'^\top = M^{-1}W^\top, \qquad \beta' = \beta - bM^{-1}W^\top \tag{30}$$

preserves these logits in real arithmetic. Composing the chart with the encoder's final affine layer leaves its shapes unchanged. This elementary reparameterization preserves the current head; recovery of the old consumer depends on how well the fitted chart describes its drift.

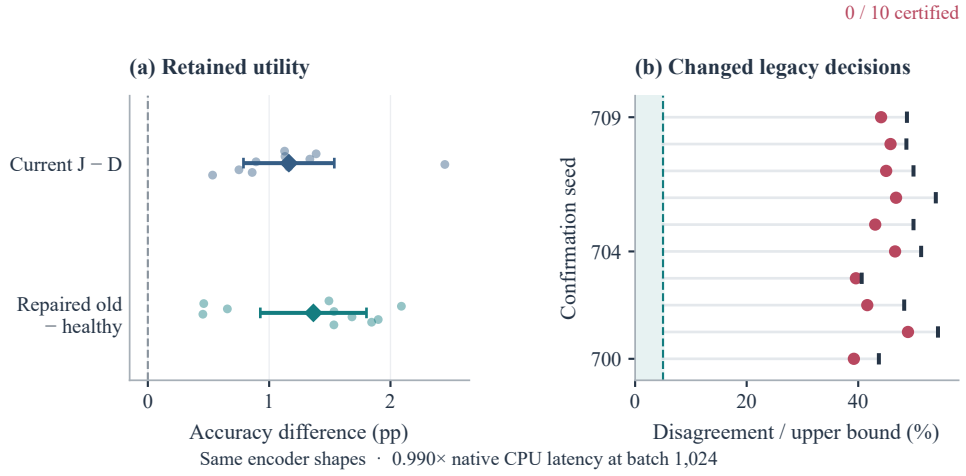
The committed protocol uses three development seeds 600–602 per task. HAR reconstructs inputs with $10\%$ of fitting primary labels, dose 10 and $60 + 30$ epochs; RadioML predicts SNR with $20\%$, dose 1 and $25 + 25$ epochs. Both use width 64, Adam $10^{-3}$, batch 256, fp32 nonlinear layers and a float64 final affine map. D/J clone the healthy model and complete optimizer state and share future minibatches. All nonlinear trunks change. Development evaluates validation only. Each chart uses $1{,}024$ uniformly sampled fitting inputs, with no labels. Old anchor features are cached offline.

Two fixed candidates are checked in order: diagonal statistics, then paired ridge with standardized coefficient $10^{-3}$. Singular values of the raw coordinate map are floored at $10^{-3}$ of the largest before rebasing. The development gate requires positive mean J–D, old accuracy within $0.5$ point of healthy, NLL increase at most $0.05$, ECE increase at most $0.02$, unchanged current predictions with logit error below $10^{-8}$, identical encoder shapes, and mean latency ratios at most $1.20/1.10$ for batches $1/1{,}024$. HAR fails the auxiliary-benefit gate. SNR diagonal repair fails ECE; paired ridge passes and alone opens confirmation.

Ten fixed new seeds 700–709 receive the unchanged SNR recipe. A queue interruption stopped one seed; only that seed was rerun, and the provenance is retained. Current J–D is $1.16$ points, paired descriptive $95\%$ interval $[0.79, 1.54]$. Rebasing preserves every current prediction; maximum logit error is $1.44\times10^{-12}$. The old consumer improves from $0.251$ healthy accuracy to $0.264$, a difference of $1.36$ points, $[0.93, 1.80]$. Mean NLL and ECE decrease by $0.0604$ and $0.0242$. Native J reaches $0.281$ current accuracy; the supervised refit reaches $0.285$. These are low-label results on one fixed modulation split.

The complete target fails. The repaired consumers change $44.07\%$ of healthy predictions on average. A separate forty-consumer family uses total $\delta = 0.05$, $1{,}024$ uniform monitor inputs and a $0.05$ error budget. None of the ten repaired joint consumers certifies; their upper remainder disagreement bounds range from $0.406$ to $0.543$. The check abstains; it does not report a loss increase, since the paired accuracy differences are positive. An accuracy repair is not a decision repair.

On one CPU thread, alternating 31 timed repetitions after five warm-ups give mean median-latency ratios $0.900$ at batch 1 and $0.990$ at batch $1{,}024$. Both branches evaluate final-affine features and primary logits in float64. The rebased head stores float64 parameters; the native fp32 head is cast for logit evaluation. We interpret the result as near-native encoder cost, not an algorithmic speedup. Auxiliary heads are not included in these primary inference timings. The chart and current heads are




(a) All seed differences and paired 95% CI. (b) Dots: observed; ticks: finite-corpus bounds.

Figure 17: A folded chart retains utility at native encoder cost but changes many legacy decisions. (a) All ten seed differences and paired 95% intervals. (b) Repaired-J disagreement with healthy predictions and simultaneous finite-corpus upper bounds. Every bound exceeds the five-point budget. Healthy aggregate accuracy is 0.251; a higher aggregate score does not imply decision compatibility.

composed offline; inference does not require the reference encoder or a separate projection. Fitting still requires anchor feature extraction and matrix estimation, not constant time.

All native model/Adam states, fitted charts, deployed models, source hashes and timing samples are retained locally. Drivers are `posthoc_chart_cpu.py` and its frozen confirmation wrapper; analysis is `summarize_posthoc_chart.py`.

### H.16 A Decision-Preserving Training Constraint

Each arm above fails one axis. A training-time constraint asks whether all three are reachable together. Continuation minimises $L_{\text{primary}} + \alpha L_{\text{aux}} + \gamma R$ with

$$R = \frac{1}{n} \sum_i \max\Big(0,\ \max_{j \neq p_{0i}} z_i[j] - z_i[p_{0i}]\Big), \tag{31}$$

where $z_i$ is the frozen consumer's logit vector on the current features and $p_{0i}$ its prediction on the healthy features. $R$ vanishes at initialisation, so it penalises decisions that would flip, not drift. The healthy encoder and readout enter fitting only; inference is the single shared encoder.

The protocol is the ten-seed benefit confirmation unchanged: RadioML, 20% primary labels, width 64, Adam $10^{-3}$, batch 256, dose 1, 25 warm-up and 25 continuation epochs, the same splits, minibatch order, healthy readout and monitor partition. On development seeds 800–802 the penalty trades disagreement for gain smoothly: at $\gamma = 0.1, 0.3, 1.0, 3.0$ the primary gain is $1.35, 1.26, 1.11, 0.08$ points and the validation disagreement is $38.4, 33.4, 24.3, 14.0\%$, against 1.06 points and 53.3% for joint training. The fixed rule selected the smallest $\gamma$ whose development disagreement is at most 5%; none reaches it, so it selected $\gamma = 3$.

Ten confirmation seeds give 0.72 points $[0.31, 1.12]$ at $\gamma = 1$ with 29.2% disagreement and $-0.04$ $[-0.46, 0.38]$ at $\gamma = 3$ with 17.8%; both use one encoder at inference. Neither is certified: no seed meets the five-point budget and the largest exact upper bound is 0.26 at $\gamma = 3$. The frozen consumer's accuracy is 0.262 at $\gamma = 3$, above the healthy 0.258, while 17.8% of its decisions differ. The $\gamma = 1$ row is a post-hoc knee confirmation run after the fixed rule selected $\gamma = 3$ and the primary arm failed; it is reported separately and does not change the primary decision.

At no inference cost, the constraint trades 0.25 of the 0.97-point gain for roughly half the decision changes, and still misses the budget. A second corpus tests whether the frontier belongs to the

Table 42: What each compatibility arm buys and costs, all on the same ten SNR confirmation seeds, splits and minibatch order. Gain is primary accuracy against the detached control, with paired descriptive 95% intervals; *changed* is the fraction of the healthy consumer's predictions that differ on the fixed test corpus; *certified* counts seeds whose exact finite-population bound is at most 0.05. Only the reference-anchored control certifies every seed, and it needs the reference encoder at inference. The decision-preserving arms share one encoder and still miss the budget. The post-hoc chart reaches 1.16 points at 44.07% on its own ten confirmation seeds.

| Arm | gain (pp) | changed | certified | encoders |
|---|---|---|---|---|
| Joint training | 0.97 [0.63, 1.31] | 58.9% | 0/10 | 1 |
| Decision, $\gamma = 1$ | 0.72 [0.31, 1.12] | 29.2% | 0/10 | 1 |
| Decision, $\gamma = 3$ | −0.04 [−0.46, 0.38] | 17.8% | 0/10 | 1 |
| Reference-anchored | 0.89 [0.60, 1.18] | 0.0% | 10/10 | 2 |

Table 43: The same decision-preserving constraint on the second corpus: HAR with the reconstruction auxiliary at dose 10, ten confirmation seeds, and the same splits and minibatch order. Gain is primary accuracy against the detached control with paired descriptive 95% intervals; reconstruction is held-out MSE as a ratio to that control; *changed* and *certified* are as in Table 42. The fixed rule selected $\gamma = 0.1$ because every development value already sat below the budget. The constraint certifies more seeds than the detached control it is measured against, joint training is the only arm whose mean changes more than five percent, and the gain is flat.

| Arm | gain (pp) | reconstruction | changed | certified | encoders |
|---|---|---|---|---|---|
| Detached control | — | 1.00 | 2.2% | 6/10 | 1 |
| Joint training | −0.08 [−0.86, 0.69] | 0.81 | 5.3% | 1/10 | 1 |
| Decision, $\gamma = 0.1$ | 0.22 [−0.15, 0.60] | 0.78 | 2.3% | 8/10 | 1 |

method or to the shift. An addendum, fixed before any HAR outcome, reruns the same grid, rule and seeds on HAR with its reconstruction auxiliary at dose 10, the highest-scoring HAR candidate in the earlier benefit development, where the auxiliary's primary gain is near zero. Every development value already meets the budget: disagreement is $1.6, 1.2, 1.1, 2.5\%$ at $\gamma = 0.1, 0.3, 1.0, 3.0$, against 2.6% for joint training, so the rule selected the smallest, $\gamma = 0.1$. Ten confirmation seeds at that value give $+0.22$ points $[-0.15, 0.60]$ over the detached control and $+0.31$ $[-0.68, 1.29]$ over joint training, with reconstruction error at 0.78 of the control against 0.81 for joint training; the gain axis is flat in both comparisons. Mean disagreement is 2.3%, between 1.5 and 3.1% per seed, and eight of ten seeds certify. The two remaining seeds have 2.7 and 3.0% point disagreement with exact upper bounds of 5.4 and 5.6%, so they abstain rather than contradict the budget. Unconstrained joint training on the same seeds changes 5.3% of legacy decisions, misses the budget on three seeds, certifies one, and loses 1.1 points of frozen accuracy, including a 5.6-point loss on one seed that the constrained arm avoids (Table 43). The budget is reachable where the shift is small and out of reach where it is large. Exact compatibility costs a second encoder at inference, approximate compatibility costs gain, and the five-point budget sits beyond the strength at which the gain vanishes.

### H.17 A Vision Boundary Screen

The failure needs an update that changes the scale of the features. A mild auxiliary on a pretrained vision backbone did not supply one in our screens. We ran two pre-registered development screens and did not open the official test split. The model is an ImageNet-pretrained ResNet-18 on CIFAR-100 at $64 \times 64$ input, continued for three epochs with SGD at $10^{-3}$ for the encoder and $10^{-2}$ for the heads, momentum 0.9, weight decay $5 \times 10^{-4}$, batch 128, and frozen batch-normalization running statistics. A linear readout fitted on the healthy features reaches 0.521 validation accuracy across the three replication seeds (0.513–0.528). Each branch clones the pretrained model and shares minibatch order, and 2,048 paired anchors are isolated from fitting.

The first screen used a natural rotation auxiliary at dose 1 and dose 10. The readout did not degrade: it reached 0.606 in the primary-only branch and 0.588 at dose 1, while the corresponding current heads reached 0.620 and 0.615; the auxiliary gave no measurable primary benefit (−0.0006). The

Table 44: Vision boundary screens on ImageNet-pretrained ResNet-18/CIFAR-100 at $64 \times 64$. All branches clone the same healthy model, share minibatch order, and train three epochs. Trace ratio and scale are measured on the fixed $5{,}000$-example validation split against healthy features; *changed* is the fraction of the frozen readout's predictions that differ from healthy. The primary-only and $\lambda = 0.5$ rows are three-seed means (seeds 900–902, healthy $0.521$); their largest SDs are $0.037$ (trace ratio), $0.027$ (scale), $0.006$ (frozen), $0.013$ (current), $0.018$ (refit) and $1.1$ points (changed). The dose rows are single runs, and the dose-1 row is an independent bf16 run. Frozen accuracy moves by $-2.9$ to $+8.5$ points while 42–56% of decisions change. The official test split is never loaded.

| Branch | trace ratio | scale | frozen | current | refit | changed |
|---|---|---|---|---|---|---|
| Healthy | 1.000 | 1.000 | 0.521 | — | — | 0 |
| Primary only | 0.474 | 0.688 | 0.606 | 0.620 | 0.637 | 42.1% |
| Rotation, dose 1 | 0.321 | 0.567 | 0.588 | 0.615 | 0.632 | 44.8% |
| Rotation, dose 3 | 0.134 | 0.367 | 0.484 | 0.583 | 0.594 | 55.8% |
| Scale, $\lambda = 0.5$ | 0.099 | 0.315 | 0.557 | 0.643 | 0.637 | 47.0% |
| Rotation, dose 5 | 0.005 | 0.067 | 0.013 | 0.023 | 0.079 | constant |

dose-10 branch produced a non-finite loss in its first epoch and was not retuned or rerun. The pre-registered failure criterion was a loss of at least five points. The screen therefore cannot test repair: the fixed diagonal-statistics family stayed near 0.61–0.62, and the paired-ridge family was unstable across anchor budgets (0.17–0.58).

The second screen crossed the affine boundary deliberately. The dose was fixed before outcomes from the feature trace alone: of two one-epoch scale instruments, $\lambda = 0.2$ gave a trace ratio of $0.180$ and $\lambda = 0.5$ gave $0.116$, so the rule selected $0.5$ for a three-epoch branch; readout accuracy was not used. Its endpoint trace ratio is $0.099 \pm 0.003$, a scale of $0.315 \pm 0.005$, below the median top-two crossing scale $0.411$ of the healthy readout. The predeclared crossing criterion still fails. Frozen accuracy is $0.557$, 3.6 points above healthy, and the standardized refit reaches $0.637$. A rotation auxiliary at dose 3 reaches scale $0.367$ and frozen $0.484$ ($-2.9$ points), refit $0.594$. At dose 5 the scale is $0.067$; the frozen readout collapses to the constant predictor ($0.013$, equal to the largest healthy class frequency) and the current head collapses with it ($0.023$). No branch combines a crossed scale with a silent readout loss.

What the scale does change is the decisions. A post-hoc replay evaluates the healthy readout on $c\,h$ over a fixed grid of $c$ and isolates the scale component from the direction change learned in continuation. Accuracy falls by only 3.5 points at $c = 0.432$, the seed-900 median crossing scale, where 21.1% of predictions change, and by 7.0 points at $c = 0.318$, where 33.9% change. Only at $c = 0.15$, where accuracy falls to $0.169$, do 77.7% of decisions change. The healthy readout is 52% accurate, so a changed prediction is nearly as likely to correct an error as to create one, and an incompatible interface hides behind stable accuracy.

The trained branches show the same separation. The primary-only branch raises frozen accuracy by 8.5 points and changes 42.1% of predictions; dose 1 raises it by 7.4 points and changes 44.8%, reproduced within one point in an independent bf16 run (42.5% at 8.5 points); the selected scale branch gains 3.6 points and changes 47.0%; dose 3 loses 2.9 points and changes 55.8% (Table 44). The standardized refit ranks the two auxiliary branches above the primary-only branch, while the frozen readout ranks them below it; the largest accuracy gain of the four still arrives with a 42.1% change of legacy decisions.

Standard unlabeled baselines fail the same way. Recomputing per-feature input statistics, the AdaBN operation, raises the old readout from $0.606$ to $0.622$ on the primary-only branch and from $0.557$ to $0.630$ on the selected scale branch, yet 40.4% and 41.7% of its decisions still differ. The paired linear map, the label-free old-to-new map of the LDC family, reaches $0.585$ and $0.580$ with 42.2% and 44.0% changed. Alignment recovers accuracy; it does not recover the interface.

Both screens bound the accuracy failure, not the compatibility failure. The affine law governs pure contraction, and the replay shows decisions accumulating as the scale falls. In a trained branch the same trace ratio arrives with a direction change the law does not model: that change raises the refit probe and offsets the scale damage to accuracy, but not to decisions. The silent accuracy failure

needs contraction strong enough to reach the bias-only predictor while the features stay decodable. The HAR variance auxiliary contracts to $0.015$ of healthy against a median crossing scale of $0.054$ and lands there, whereas the vision branches that contract that far destroy the refit probe as well. The compatibility failure is the more robust of the two: it appears in every branch whose features move, including the ones that improve accuracy.

### H.18 A Fixed-Corpus Unlabeled Consumer Check

For a present checkpoint, we complement the variance audit with unlabeled disagreement against the healthy consumer. The check combines the relative-loss inequality with exact finite-population sampling, extending unlabeled performance estimation (Garg et al., 2022) and update certificates (Usama & Qadir, 2026) with reference predictions and an option to abstain.

For a fixed corpus of $N$ inputs, let $f_0$ be the healthy consumer, $f$ a fixed candidate, and $R_T$ the mean 0–1 error on subset $T$. For every assignment of labels,

$$R_T(f) - R_T(f_0) \le D_T(f, f_0), \qquad D_T = |T|^{-1} \sum_{i \in T} \mathbf{1}\{f(x_i) \neq f_0(x_i)\}. \tag{32}$$

The inequality follows pointwise: error can increase only where predictions change. It is sharp when the labels equal the healthy predictions. Conversely, changing predictions can improve accuracy, so large disagreement does not establish harm.

Uniformly sample $n$ monitor inputs without replacement, independently of the fixed candidate family. If $c$ predictions disagree on this monitor and the unknown full-corpus disagreement count is $K$, then $c$ has the hypergeometric distribution with parameters $(N, K, n)$. With family size $M$ and total tail probability $\delta$, define

$$K^+(c) = \max\{K \in \{c, \dots, N - n + c\} : F_{N,K,n}(c) \ge \delta/M\}, \qquad U = \frac{K^+(c) - c}{N - n}. \tag{33}$$

Here $F$ is the exact hypergeometric CDF. Inverting its lower tail bounds the full-corpus disagreement count, hence $D_T \le U$ on the unmonitored remainder. For each candidate the failure probability is at most $\delta/M$; a union bound gives simultaneous coverage for all $M$ candidates, allowing selection from that fixed family. The statement needs no independence between adjacent windows, and it covers the fixed corpus with classification error, not a new population, future updates, NLL, ECE, or conformal coverage. Monitor-adaptive training and unconstrained construction of new candidates would need a different analysis.

We predeclare $n = 1{,}024$, $\delta = 0.05$, and budgets $0.05$ and $0.10$ for relative loss increase. The bank contains all 300 endpoints from the 30 earlier paired runs plus 35 CNN endpoints, each evaluated with the same nine fixed repair arms: $M = 3015$. All maps use the full fitting feature set; a restricted anchor budget would need its own evaluation. A seeded uniform monitor partition is shared across candidates in a run, and old/new reference predictions on those monitor inputs are required. Healthy reference predictions can be cached for a fixed corpus or obtained from the retained healthy encoder; this access is additional to paired training anchors for richer repairs. The first repair in the predeclared hierarchy satisfying $U \le \varepsilon$ is selected, or the audit abstains. Monitor labels never enter this choice. Candidate remainder accuracy and loss increase are scored only after choices are fixed; archived healthy/frozen metrics are also checked independently for reproducibility.

Across $105$ joint endpoints, the 5-point budget selects $18$ candidates and abstains on $87$; the 10-point budget selects $49$ and abstains on $56$. The latter includes $12$ unchanged frozen consumers and $37$ repaired consumers. Observed selected-consumer loss increases exceed budget in $0$ and $0$ cases, respectively. The healthy-reference scatter heuristic makes $16$ and $11$ false-safe declarations. This comparison uses scatter relative to healthy features on the same monitor, which is available without a continued primary-only counterfactual; it is distinct from retrospective comparison with a fully trained single-task endpoint. The fixed unsigned probe's safe declarations and primary-only controls are also retained in the numerical summaries.

At the 10-point budget, HAR supplies $49$ selections; RadioML supplies $0$ of $45$, including its near-null continuation cases. Disagreement can leave an accuracy-preserving or improving candidate uncertified when many decisions change. The check bounds relative 0–1 error on the fixed corpus; absolute accuracy and warning lead time require other evidence. All candidates, decisions,

Table 45: Fixed-family consumer audit. Each corpus supplies 1,024 uniformly sampled unlabeled monitor inputs; remaining labels score the decisions afterward. The 3,015 fixed endpoint/repair candidates share total tail probability 0.05. *Scatter false safe* uses a scatter ratio $\geq 0.3$ relative to healthy features on the same monitor, despite frozen-consumer loss increase above budget. Selection uses the first certified candidate, including identity/frozen; uncertified cases abstain. Counts describe dependent checkpoints, not independent trials.

| Case | budget | joint | scatter false safe | frozen certified | candidate selected | selected false safe |
|---|---|---|---|---|---|---|
| HAR isotropic | 0.05 | 30 | 0 | 0 | 6 | 0 |
| HAR isotropic | 0.10 | 30 | 0 | 3 | 24 | 0 |
| HAR projected | 0.05 | 15 | 10 | 0 | 3 | 0 |
| HAR projected | 0.10 | 15 | 8 | 0 | 10 | 0 |
| HAR recon. | 0.05 | 15 | 3 | 1 | 9 | 0 |
| HAR recon. | 0.10 | 15 | 3 | 9 | 15 | 0 |
| RadioML SNR | 0.05 | 15 | 3 | 0 | 0 | 0 |
| RadioML SNR | 0.10 | 15 | 0 | 0 | 0 | 0 |
| RadioML I/Q MLP | 0.05 | 15 | 0 | 0 | 0 | 0 |
| RadioML I/Q MLP | 0.10 | 15 | 0 | 0 | 0 | 0 |
| RadioML I/Q CNN | 0.05 | 15 | 0 | 0 | 0 | 0 |
| RadioML I/Q CNN | 0.10 | 15 | 0 | 0 | 0 | 0 |

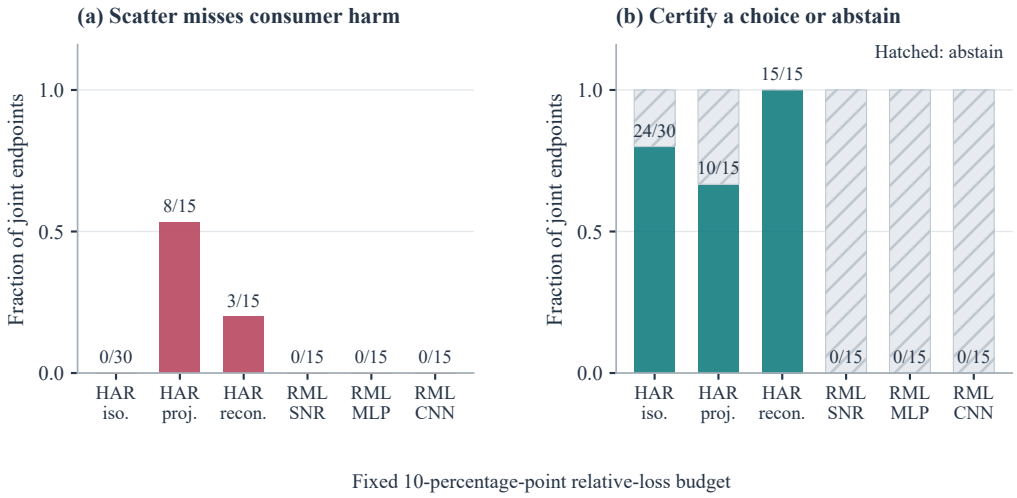



Figure 18: Unlabeled disagreement supports conservative repair selection. Fixed 10-percentage-point relative-error budget. Left: a monitor-scatter rule can declare a harmful frozen consumer safe. Right: the first certified repair is selected from the fixed hierarchy; otherwise the rule abstains. The final two groups use I/Q reconstruction. Bars show fractions of joint endpoints, with counts annotated. Within-seed branches are dependent. Simultaneous coverage applies to the fixed candidate family on the unmonitored corpus remainder.

bounds, remainder scores, map hashes, and source links are archived. Checks cover finite-population coverage, boundary cases, monotonicity, sharpness, selection, abstention, and temperature-only NLL drift. Drivers: `consumer_risk.py` and `checkpoint_consumer_audit.py`; analysis: `summarize_consumer_frontier.py`.

### H.19 Dissociation Between Contraction and Utility

Three quantities are often conflated: variance contraction, scale-invariant information, and readout utility. Table 46 separates them in six cases, each using the protocol cited in its section.

Table 46: Six cases separating contraction, decodability, and readout utility. Relative $S$ is the same-seed scatter ratio. The scale check is the standardized probe where available; PR measures spectral concentration in the battery row. Utility names the primary or frozen-readout metric. SNR and subject-ID readout studies use five seeds, I/Q three; the battery row uses the corrected MATR application.

| Case | Rel. $S$ | Scale check | Utility | Diagnosis |
|---|---|---|---|---|
| HAR toxifier (100) | 0.016 | probe 0.955 (healthy 0.947) | frozen 0.241; input recal 0.952; retrained 0.955 | contraction; information intact; readout broken |
| MATR battery | 3.55 | PR 2.49 | SOH 5.57 vs 2.10 detached | no contraction; negative transfer |
| RadioML SNR (3000) | 0.137 | n/a | own 0.205; frozen 0.099; retrained 0.229; detached readout 0.315 | relative collapse; readout near chance; sampled flow flux mostly negative |
| RadioML I/Q (3000) | 1.05 | n/a | own 0.344; frozen 0.195; retrained 0.351; detached readout 0.312 | no contraction; frozen readout broken by drift |
| HAR subject-ID (10) | 0.93 | n/a | own 0.887; frozen 0.384; retrained 0.901 | no contraction; frozen readout broken by drift |
| HAR reconstruction | 1.00 | n/a | 0.942 | benign head |

The six cases show why no single scalar orders representation quality. In HAR, scatter contracts by a factor of 60 while task information remains linearly available and input recalibration repairs the readout. MATR has negative transfer without contraction. SNR combines relative collapse and readout failure, but mostly negative sampled flow flux leaves its Adam attribution unresolved. I/Q retains scatter 1.05 and refitted accuracy 0.351 versus 0.312 detached, while the frozen readout falls to 0.195. Subject-ID similarly gives joint accuracy 0.887 at scatter 0.93, yet frozen accuracy 0.384 versus 0.901 refitted. The last row is benign at its tested dose.

### H.20 RadioML Dose–Response and Natural Scale

Table 47 varies the auxiliary dose for two real RadioML heads at $m = 64$, 25 epochs and three seeds, fully labeled: SNR regression (the head that collapses at the reported dose) and I/Q reconstruction (a benign head). The detached reference is 0.306 accuracy with $0/3$ collapses for SNR and 0.307 with $0/3$ for reconstruction; its relative scatter is 1.031 and 1.101, respectively.

Table 47: RadioML dose–response, fully labeled. Joint training, three seeds, $m = 64$. "Coll." is the scatter-collapse count; $\alpha\rho$ is the nominal dose times the measured per-example radial ratio. The SNR onset lies between $\alpha = 3$ and 10, and the head collapses in every seed from $\alpha = 30$ onward; the reconstruction head never collapses at any tested dose.

| $\alpha$ | SNR acc. | SNR coll. | SNR rel. $S$ | SNR $\alpha\rho$ | I/Q acc. | I/Q coll. | I/Q rel. $S$ |
|---|---|---|---|---|---|---|---|
| 1 | 0.329 | 0/3 | 0.932 | 0.28 | 0.305 | 0/3 | 1.054 |
| 3 | 0.318 | 0/3 | 0.600 | 1.28 | 0.308 | 0/3 | 1.038 |
| 10 | 0.277 | 2/3 | 0.291 | 5.84 | 0.312 | 0/3 | 1.096 |
| 30 | 0.248 | 3/3 | 0.211 | 22.6 | 0.303 | 0/3 | 1.005 |
| 100 | 0.229 | 3/3 | 0.163 | 70.3 | 0.312 | 0/3 | 1.057 |
| 300 | 0.214 | 3/3 | 0.146 | 187 | 0.321 | 0/3 | 1.252 |
| 1000 | 0.207 | 3/3 | 0.144 | 557 | 0.338 | 0/3 | 1.267 |
| 10000 | 0.204 | 3/3 | 0.140 | 4138 | 0.320 | 0/3 | 0.641 |

SNR collapse begins between $\alpha = 3$ and 10, with all seeds below the criterion by 30; the historical 3000 comparison is deep within this observed regime. I/Q reconstruction stays near or above reference scatter through 1000 and reaches 0.641 at 10000, above 0.3. Thus nominal dose alone does not order the two heads. Their measured radial loadings differ sharply: at 30, SNR has $\alpha\rho = 22.6$, while I/Q loading is near zero. The signed-flux audit below checks what those unsigned measurements can explain.

Table 48 adds STF-cal at the three doses around and above the SNR onset. At $\alpha = 30$, calibrated filtering reaches 0.303 against joint 0.248 and detached 0.306, with no collapse; at $\alpha = 100$ and 300 the recovery shrinks to 0.278 and 0.236 even though the gate opens in every seed. The reconstruction gate stays shut (gate rate 0.00) at all three doses and STF-cal stays at 0.308. Filtering helps near the onset, not at every dose.

Table 48: STF-cal at the SNR onset (RadioML, fully labeled, three seeds, $m = 64$). Gate rate and mean margin are for the calibrated arm; joint and STF-0 show the reference endpoints.

| $\alpha$ | joint acc. | STF-0 acc. | STF-cal acc. | gate / margin |
|---|---|---|---|---|
| 30 | 0.248 | 0.306 | 0.303 | 1.00 / 14.1 |
| 100 | 0.229 | 0.306 | 0.278 | 1.00 / 47.1 |
| 300 | 0.214 | 0.306 | 0.236 | 1.00 / 141.3 |

### H.21 Readout Dose Scan and Signed Flux

Table 49 repeats the healthy-readout transfer across the SNR dose scan at $m = 64$, 25 epochs, and five seeds. The frozen readout is 0.176 at $\alpha = 3$ (relative scatter 0.558, above the 0.3 criterion), 0.121 at $\alpha = 10$ (relative scatter 0.274, at the criterion), and near chance from $\alpha = 30$ onward. Retraining recovers 0.322 at $\alpha = 3$ and 0.229 at $\alpha = 3000$, and exceeds the joint encoder's own accuracy at every dose, so the representation retains linearly recoverable information. The readout therefore degrades before the operational scatter criterion fires; a scatter threshold alone is not a deployment certificate. Detachment preserves 0.315 at every dose. Driver: `repro/scripts/real_head_readout_transfer.py`; records: `results/READOUT/rml_snr_readout_a*.json`.

Table 49: Historical RadioML SNR readout dose scan (five seeds, $m = 64$, 25 epochs). Own uses the joint primary head; frozen and recalibrated reuse the healthy weights; retrained fits new weights; detached uses the healthy readout on the STF-0 encoder. Relative $S$ is the same-seed scatter ratio.

| $\alpha$ | rel. $S$ | healthy | frozen | recal. | retrained | detached | own |
|---|---|---|---|---|---|---|---|
| 3 | 0.558 | 0.315 | 0.176 | 0.146 | 0.322 | 0.315 | 0.315 |
| 10 | 0.274 | 0.315 | 0.121 | 0.114 | 0.283 | 0.315 | 0.273 |
| 30 | 0.198 | 0.315 | 0.102 | 0.099 | 0.256 | 0.315 | 0.247 |
| 100 | 0.160 | 0.315 | 0.106 | 0.090 | 0.240 | 0.315 | 0.229 |
| 300 | 0.145 | 0.315 | 0.103 | 0.097 | 0.234 | 0.315 | 0.214 |
| 3000 | 0.137 | 0.315 | 0.099 | 0.094 | 0.229 | 0.315 | 0.205 |

The signed flux audit uses the same driver with `--flux-trace`. On a fixed diagnostic batch of 384 test examples we compute $V = \|C_B H\|_F^2/(2B)$, $v = \nabla_\theta V$, $r = -v^\top G_p$, and $c = v^\top G_a$, and record them at every epoch end; inside the first two epochs we also sample every 10 minibatches. At $\alpha = 3000$ the seed-median $c_t$ is $-3.97$ at epoch 1, $-49.3$ at epoch 5, $-66.1$ at epoch 10, and $-107.5$ at epoch 25; the aggregate signed cosine is below $-0.90$ at the endpoint. Of 75 epoch-end checkpoints, 74 have negative $c_t$; the exception is $+0.76$. Epoch-end sampling alone cannot rule out an early effect from that checkpoint. I/Q reconstruction gives $|c_t| < 10^{-5}$ at every epoch end, consistent with no variance flux. The SNR head thus shows relative collapse without consistently positive sampled gradient-flow flux. These observations alone do not identify its Adam-step contributions; Appendix H.7 audits actual updates separately. Records: `results/READOUT/rml_snr_flux*.json` and `results/READOUT/rml_iq_flux.json`.

At $\alpha = 3000$, every calibrated SNR gate opens and the selected deletion reaches its cap of 3 (calibrated $\rho = 0.44$–$0.49$, margins $1.3$–$1.5 \times 10^3$). Relative scatter remains $0.25$–$0.36$; frozen accuracy rises only from $0.094$ (joint seed median) to $0.136$, with recalibration $0.124$ and refitting $0.212$. The selected rank and reliably open gate therefore fail to restore compatibility at this dose. Records: `results/READOUT/rml_snr_readout_stfcal.json`.

# I Anti-Collapse Regularizers at Matched Doses

Table 25 evaluates two regularizer families at the doses and seeds of Table 4: VICReg's variance hinge plus off-diagonal covariance penalty, and Barlow Twins' redundancy-reduction penalty. Each is applied to the shared representation with weight $\lambda$. VICReg's invariance term is omitted because this experiment has no augmented-view pair. All rows use the same joint-training baseline; these are adaptations of the regularizers to this setting, not evaluations of the complete original pipelines.

Regularization can clear the operational collapse threshold without restoring the single-task reference or improving the task. In the coarse grid, VICReg first avoids collapse in all seeds at $\lambda = 10$ for HAR, 300 for battery, and 10000 for RadioML. Barlow Twins does not do so at any tested weight.

On HAR, VICReg scatter stays near $0.36$ from $\lambda = 10$ to $10^4$, while accuracy falls from $0.944$ to $0.821$. At the lowest clearing weight, accuracy is close to joint training ($0.945$) and detachment. On battery, scatter increases through the reference level, but the best tested RMSE is $0.339$ at $\lambda = 300$ (relative scatter $0.59$), compared with $0.297$ for detachment. Larger weights worsen RMSE to $0.390$ at $10^3$ and $0.423$ at $10^4$; detachment is better in nine of ten seeds. On RadioML, $\lambda = 10000$ gives zero collapse and scatter $1.29$, but accuracy drops to $0.215$, below both collapsed joint training ($0.256$) and detachment ($0.341$). These cases separate the scatter criterion from predictive utility.

The Barlow Twins adaptation yields absolute scatter as low as $0.011$, $0.012$, and $0.013$ for HAR, battery, and RadioML, respectively, or factors of 172, 6, and 24 below their references. It fails to clear the collapse threshold throughout the tested grid, with HAR accuracy reaching approximately $0.34$–$0.44$ and RadioML approximately $0.11$ in the low-accuracy configurations.

Additional ten-seed sweeps use $\lambda \in \{2, 5, 20, 50, 200, 500\}$ on HAR and $\lambda \in \{200, 500, 700, 1500, 2000, 5000\}$ on battery. The recorded battery transition is bracketed by 150 and 200, and the lowest measured battery RMSE remains $0.339$ at $\lambda = 300$, above detachment. HAR scatter remains near $0.36$ above its crossing. RadioML was not refined: the sampled collapse rate is 1.0 at 1000, 0.10 at 3000, and 0.0 at 10000. The experiments therefore locate an operational transition over a finite grid rather than a precise critical weight.